SECTION 26 05 00 - COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes:
1. Basic electrical methods.
2. Grounding and bonding.
3. Hangers and supports.
4. Electrical identification.
5. Motor starters, controls, and connections to mechanical equipment.
6. Electrical system testing and inspection.

B. Related Documents: The contract documents, as defined in Section 011000 - Summary of Work, apply to the work of this section. Additional requirements and information necessary to complete the work of this section may be found in other documents.

C. Related Sections:
1. Section 019113 - General Commissioning Requirements
2. Section 024113 - Selective Site Demolition
3. Section 078400 - Firestopping
4. Section 220500 - Common Work Results for Plumbing
5. Section 230500 - Common Work Results for HVAC
6. Section 251104 - Metering Devices
7. Section 251304 - Facility System Integration into EEMS
8. Section 260513 - Medium Voltage Cables
9. Section 260519 - Low-Voltage Electrical Power Conductors and Cables
10. Section 260533 - Raceway and Boxes for Electrical Systems
11. Section 260623 - Lighting Control Devices
12. Section 260800 - Commissioning of Electrical Systems
13. Section 262200 - Low Voltage Transformers
14. Section 262416 - Panelboards
15. Section 262726 - Wiring Devices
16. Section 262816 - Enclosed Switches and Circuit Breakers
17. Section 264128 - Surge Protective Devices (SPD's)
18. Section 265100 - Interior Lighting (LED - Solid State)
19. Section 265600 - Exterior Lighting
20. Section 270500 - Common Work Results for Communications
21. Section 271100 - Communications Equipment Room Fittings
22. Section 271300 - Communications Backbone Cabling
23. Section 271500 - Communications Horizontal Cabling
24. Section 272133 - Data Communications - Wireless Access Points
25. Section 272134 - Video Communications - Outdoor Wireless Access Points
26. Section 275116 - IP Integrated, Public Address Zone Paging System
27. Section 275313 - Wireless, Synchronized, GPS Clock System
28. Section 275319 - Distributed Antenna System (DAS)
29. Section 281304 - Enterprise Physical Access Control System (ePACS)
30. Section 281524 - IP Video Intercom and Exterior Gate Control System
31. Section 281600 - Intrusion Detection System
32. Section 283100 - Fire Emergency Voice/Alarm Communication System (EVACS)

1.02 REFERENCES

A. National Electrical Contractors Association (NECA):

1. NECA SI - Standard of Installation.

B. National Electrical Manufacturers Association (NEMA):
1. NEMA KS 1 - Enclosed Switches.

C. National Electrical Testing Association (NETA):
1. NETA ATS - Acceptance Testing Specifications for Electrical Power Distribution Equipment and Systems.

D. National Fire Protection Association (NFPA):
1. NFPA 70 - National Electrical Code.

1.03 SUBMITTALS

A. Section 013300 - Submittal Procedures: Procedures for submittals.
1. Product Data:
a. Grounding electrodes and connections.
b. Starter electrical characteristics and connection requirements.
2. Assurance/Control Submittals:
a. Electrical System Test Reports: Submit report including the following directly to the USPS Project Manager from Testing Laboratory, with copy to Contractor. Prepare reports in conformance with Section 014000 - Quality Requirements.
b. Summary of project.
c. Description of equipment tested.
d. Description of test.
e. Test results.
f. Conclusions and recommendations.
g. Appendix, including appropriate test forms.
h. List of test equipment used and calibration date.
i. Signature of responsible Testing Laboratory Officer.
j. Certificates: Manufacturer's certificate that each Product specified meet or exceed specified requirements.
k. Qualification Documentation: Submit documentation of experience indication compliance with specified qualification requirements.

B. Section 017704 - Closeout Procedures and Training: Procedures for closeout submittals.
1. Project Record Documents: Accurately record the following.
a. Locations of components and grounding electrodes.

1.04 QUALITY ASSURANCE

A. Installer Qualifications: Company specializing in performing Work of this Section with minimum 5 years documented experience.

B. Regulatory Requirements:
1. Products: Listed and classified by Underwriters Laboratories, Incorporated as suitable for the purpose specified and indicated.
2. Work herein shall conform to all applicable laws, ordinances and regulations in accordance with the latest applicable requirements of:
a. The National Electrical Code (NFPA 70).
b. National Electrical Manufacturer's Associates.
c. Standards of National Fire Protection Association (NFPA 72, 90A and 101).
d. Underwriter's Laboratories.
e. Occupational Safety and Health Agency Standards.
f. Illuminating Engineering Society Handbook.
g. The International Existing Building Code.
h. The International Electrical Code.

i. ASHRAE Standard 90.1.
j. The International Energy Conservation Code.

1.05 BASIC ELECTRICAL METHODS

A. Drawings are schematic and diagrammatic. Use judgment and care to install electrical Work to function properly and fit within building construction and finishes. Electrical conductors, conduit, components, not shown or specified, which are required for any device or system to produce a complete and operative system are required to be furnished and installed.

B. Exact location of outlets is determined from dimension on Drawings, manufacturer's shop drawings, or as may be determined at Project Site. Do not scale Drawings for exact location of any item. Verify item mounting heights as required by project conditions prior to rough-in.

C. Route conduits and wiring associated with new equipment and systems above ceilings, in existing chases, and concealed within building structure.

D. Surface mounted raceways or conduit permitted only at locations indicated on Drawings.

E. Circuit grouping, conduit or cable runs and home runs are indicated with number of conductors shown in each raceway to clarify operation and function of various systems. Provide proper number of conductors and conduits or cables to provide operative system as indicated on Contract Documents. Do not regroup any feeder circuits, branch circuits, home runs, and zone alarms at any point, from that shown on Contract Documents. Each conduit run shall contain no more than (6) current carrying conductors.

F. Branch and home run circuits are indicated as 2, 3, or 4 wire circuits unless otherwise noted. Do not connect two ungrounded conductors to same circuit breaker/fused switch in any panel. Circuit runs consist of a maximum of five conductors; 3 phase conductors, 1 neutral conductor, and 1 equipment ground conductor, unless otherwise noted. Do not splice branch circuit conductors in any panels, safety switches, or circuit breakers in separate enclosures.

G. The sharing of neutral conductors for multiwire branch circuits is prohibited. All branch circuits shall contain individual neutrals.

H. Proposed equipment, switches or devices, shown mounted on and/or adjacent to equipment, which if installed, would impair proper operation of existing or new equipment, shall be removed and relocated by Contractor as required so equipment will function properly. Notify Contracting Officer immediately if any such condition exists.

I. Seal and make permanently watertight penetrations by electrical raceways or equipment through ceilings, walls or floors.
   1. Seal penetrations in non-fire rated ceilings, walls or floors material specified in Section 079200 - Joint Sealants.
   2. Seal penetrations in fire rated walls with material specified in Section 078400 - Firestopping.

J. Tighten electrical connectors and terminals, including screws and bolts, in accordance with equipment manufacturer's published torque tightening values for equipment connectors. Where manufacturer's torqueing requirements are not indicated, tighten connectors and terminals to comply with tightening torques specified in UL 486A, and NFPA 70.

K. Install equipment and materials to provide required maintenance and code working clearance for servicing and maintenance. Coordinate final location of concealed equipment and devices requiring access with final location of required access panels and doors. Allow required space for removal of parts that require replacement or servicing.

L. Remove existing equipment, lighting fixtures, switches, and receptacles as required to facilitate proposed installation and as specified in Section 024119 - Selective Structure Demolition. Remove existing wiring and conduit serving items to be removed. Conduit in inaccessible areas shall be cut off below finished surfaces and existing surface patched to match existing. Provide blank plates on existing flush mounted outlet boxes that will be abandoned. Remove all abandoned conductors from raceways.

PART 2 - PRODUCTS

2.01 GROUNDING AND BONDING

A. Grounding System Resistance: Five ohm.

B. Rod Electrodes:
   1. Material: Copper.
   2. Diameter: 3/4 inch.
   3. Length: 10 feet.

C. Mechanical Connectors: Bronze.

D. Electrode Conductor:
   1. Material: Bare stranded copper.

2.02 HANGERS AND SUPPORTS

A. Product Requirements: Furnish and install approved materials, sizes, and types of anchors, fasteners, and supports to carry loads of equipment and conduit, including weight of wire in conduit plus 300 pounds.

B. Materials and Finishes: Corrosion resistive.

C. Anchors and Fasteners:
   1. Steel Structural Elements: Beam clamps and welded fasteners.
   2. Concrete Surfaces: Self-drilling anchors and expansion anchors.
   3. Hollow Masonry, Plaster, and Gypsum Board Partitions: Toggle bolts and hollow wall fasteners.
   4. Solid Masonry Walls: Expansion anchors.
   5. Sheet Metal: Sheet metal screws.
   6. Wood: Wood screws.

2.03 ELECTRICAL IDENTIFICATION

A. Nameplates:
   1. Engraved three-layer laminated phenolic plastic, white letters on black background.
   2. Locations:
      a. Each electrical distribution and control equipment enclosure.
      b. Communication cabinets.
      c. Terminal Cabinets.
      d. Individual motor starter.
      e. Separately enclosed circuit breakers.
      f. Panelboards
      g. Transformers.
      h. Pull boxes.
      i. Lighting contactor/control panel enclosure.
      j. Relays.
      k. Switches and disconnects.
   3. Letter Size:
      a. Use 1/8 inch letters for identifying individual equipment and loads.

b. Use 1/4 inch letters for identifying grouped equipment and loads.
4. Nameplates shall indicate voltage of the equipment and the source of power from the upstream device or equipment.

B. Wire and Cable Markers:
1. Description: Cloth tape or tubing type wire markers.
2. Locations: Each conductor at panelboard gutters, pull boxes, outlet and junction boxes, and each load connection.
3. Identification:
a. Power and Lighting Circuits: Branch circuit or feeder number indicated on Drawings.
b. Control Circuits: Control wire number indicated on schematic and interconnection diagrams on Drawings.
c. Communications Cable: Per section 270500.

C. Conduit Markers:
1. Underground conduit routings shall be marked utilizing magnetic marker tape set atop of the entire conduit run.
a. Underground-Type Plastic Line Marker: Manufacturer's standard detectable permanent, bright colored, continuous-printed plastic tape, intended for direct-burial service; not less than 6 inches wide by 4 mils thick. Provide tape with printing which most accurately indicates type of service of buried cable. Locate tape 12 inches above top of conduit.
2. Detectable marker tape cannot be installed atop of directionally bored underground conduits. Directionally bored conduits containing only fiber cabling shall therefore be equipped with a #12/AWG copper conductor to enhance detection.

D. Arc Flash Warning Signs: Furnish signs in accordance with NEC Article 110.16, warning of potential arc flash hazard and requiring suitable Personal protective equipment. Locate and install signs per INSTALLATION Section of this specification.

E. Confined space markings: Work within electrical manholes and underground vaults must comply with "confined space" OSHA requirements. Manhole covers and the entrance to underground vaults shall be stamped or marked as "CONFINED SPACE - PERMIT REQUIRED".

F. Receptacles and Switches: All coverplates for receptacles and switches shall be labeled with the branch circuit number. Label shall be machine generated and permanently affixed to the outside of the coverplate.

2.04 MOTOR STARTERS, CONTROLS, AND CONNECTIONS TO MECHANICAL EQUIPMENT

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
1. ABB/G.E. Industrial Solutions (ABB/GEIS), Mebane, NC (800) 431-7867.
2. Allen-Bradley Company, Milwaukee, WI (414) 382-2000.
3. Cutler-Hammer Eaton Corp, Milwaukee, WI (800) 833-3927.
4. Square D Company, Palatine, IL (847) 397-2600.
5.
6. Siemens Energy and Automation, Alpharetta, GA (800) 964-4114.
7. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Motor Starters
1. Provide manual, single phase, 120/277V, toggle type, motor rated switches with thermal overload element (sized at 115 percent of full load current) for fractional horsepower motors not requiring automatic control interfaces.

2. Provide across-the-line, AC magnetic motor starters in applications where controls other than manual on and off are involved. Motor starters shall be UL labeled. Provide starters with the following features:
   a. Rating for the voltage and current imposed.
   b. Enclosure for the application usage: NEMA 1 for dry, indoors, NEMA 3R for outdoors, etc.
   c. Control circuit voltage and amperage to match coil voltage and ratings of control apparatus.
   d. Control transformers with primary and secondary fusing for control circuits, as required.
   e. Overload elements for every conductor leg above ground. Elements are to be "thermal alloy" type, resettable and properly sized to motor nameplate rating. Elements located near boilers, heat strips, duct heaters or other heat sources or where heating by conduction or radiation can occur, shall be ambient temperature compensated types.
   f. Adjustable phase loss/phase reversal protection (0-15 seconds), factory set at 7 seconds and a minimum of two field convertible auxiliary contacts.
   g. Cover-mounted control switch is to be a "start-stop" or "hand-off-auto" type with "running" and "auto" pilot lights, as required by the control sequence. A suitable reset device for manually resetting overcurrent trip shall be provided.
3. Starters for motors 10 hp or less shall be connected to automatically return the motor to service after a power interruption. Starters for motors over 10 hp shall be equipped with time delay relays so that after a power resumption and after a preset delay of 0-30 seconds, the motor shall automatically be returned to service.
4. Combination magnetic motor starter/fused disconnect unit shall be utilized wherever possible.

C. Furnish and Install the Following:
   1. Conduit, wiring and electrical connections to motors, safety switches, starters, relays, electrical interlock circuits, valves, unit heaters, fan coil units, air handling units, and other similar equipment, required for complete and ready for operation. Coordinate with and review other sections of the specifications describing electrical equipment in order to fully understand the wiring requirements.
   2. Starters as indicated on Drawings except factory provided starters such as those physically mounted on the unit or any piece of equipment where starter is furnished as an integral part of the equipment.
   3. Electrical line voltage control components and installation as specified in Division 26 Sections.
   4. Furnish and install low voltage (below 50 volts) control wiring as indicated on Drawings using metallic conduit and No. 12 type THHN wire, minimum.
   5. Thermostat and special wire other than building wire.

D. Refer to Drawings for quantity and size of motor starters.

E. Individual and group mounted motor starters within motor control centers and those starters factory provided integral with the equipment shall be furnished in accordance with paragraph 2.4 B.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.

B. Verification of Conditions: Verify that field measurements, surfaces, substrates and conditions are as required, and ready to receive Work.

C. Report in writing to the USPS Project Manager prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.
D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the United States Postal Service.

3.02 INSTALLATION - GROUNDING AND BONDING

A. Install rod electrodes at locations indicated. Install additional rod electrodes as required to achieve specified resistance to ground.

B. Provide grounding well pipe with cover at each rod location. Install well pipe top flush with finished grade or surface.
C. Provide grounding electrode conductor and connect to reinforcing steel in foundation footing, building steel above grade and metallic cold water pipe.
D. Provide bonding and grounding in conformance with NFPA 70.
E. Equipment Grounding Conductor: Provide separate, insulated conductor within all lighting and power raceways. Terminate each end on suitable lug, bus, or bushing.
F. Testing and Inspection:
   1. Inspect and test in accordance with NETA ATS, where applicable.
   2. Perform inspections and tests listed in NETA ATS, Section 7.13.
   3. Test ground resistance of system with clamp-on ground resistance tester. The resistance of the grounding system shall not exceed 5 ohms. Where tests show resistance-to-ground is over 5 ohms, take appropriate action to reduce resistance to 5 ohms, or less, by driving additional ground rods, lengthening the rods or installing ground enhancing materials; then retest to demonstrate compliance. Install rods at least 8 feet apart.

3.03 INSTALLATION - HANGERS AND SUPPORTS

A. Install products in accordance with manufacturer's published instructions.

B. Furnish and install anchors, fasteners, and supports in accordance with NECA SI.
C. Do not fasten supports to pipes, ducts, mechanical equipment, and conduit.
D. Do not use spring steel clips and clamps.
E. Do not use powder-actuated anchors.
F. Obtain permission from structural engineer before drilling or cutting structural members.
G. Fabricate supports from structural steel angle or structural steel channel. Rigidly weld members or use hexagon head bolts to present neat appearance with adequate strength and rigidity. Use spring lock washers under all nuts.
H. Install surface-mounted cabinets and panelboards with minimum of four anchors.
I. In wet and damp locations use structural steel channel supports to stand cabinets and panelboards one inch off wall.
J. Use sheet metal channel to bridge studs above and below cabinets and panelboards recessed in hollow partitions.

3.04 INSTALLATION - ELECTRICAL IDENTIFICATION

A. Install nameplate parallel to equipment lines.

B. Secure nameplate to equipment front using stainless steel screws. Use minimum two screws at each end of nameplate.
C. Secure nameplate to outside surface of door on panelboards and switchboards.
D. Install Arc Flash Warning Signs on switchboards, panelboards, control panels, meter socket enclosures, and motor control centers likely to require examination, adjustment, servicing, or maintenance while energized. Locate sign so as to be clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

3.05 INSTALLATION - MOTOR STARTERS. CONTROLS, AND CONNECTIONS TO MECHANICAL EQUIPMENT

A. Verify and check equipment manufacturer's nameplate and installation instructions to obtain exact location of outlets for equipment before installation.

B. Wire and connect line voltage controls in accordance with approved wiring diagrams. Provide line voltage interlock and control wiring as indicated on Drawings using conduit and No. 12 type THHN wire.

3.06 FIELD QUALITY CONTROL - ELECTRICAL TESTING AND INSPECTION

A. Section 014000 - Quality Requirements: Field testing and inspection.

B. Section 260800 - Commissioning of Electrical Systems: Requirements related to Division 26 Commissioning
C. Conduct testing to Determine that Electrical Equipment and Systems:
   1. Are in conformance with Contract Documents and applicable reference standards.
   2. Is properly installed without damage due either to installation or shipment.
   3. Operate correctly, meet design intent, and are performing at optimum level, in safe manner.

D. Provide a complete written record of operational values to be used as a baseline for future operational testing.

E. Instrumentation:
   1. Provide calibration program that assures applicable test instrumentation is maintained within rated accuracy and directly traceable to National Bureau of Standards.
   2. Calibrate instruments in accordance with following frequency schedule:
      a. Field Instruments:
      b. Analog - 6 months maximum.
      c. Digital - 12 months maximum.
      d. Leased Specialty Equipment: 12 months. (Where accuracy is guaranteed by lessor.)
   3. Dated Calibration Labels: Visible on test equipment.
   4. Keep records current; Show date and result of instruments calibrated or tested.
   5. Maintain current instrument calibration instruction and procedure for each test instrument.
   6. Calibrating Standard: Higher accuracy than that of instrument being calibrated.

F. Regulatory Requirements:
   1. Safety Practices: Include, but not limited to, the following requirements:
      a. Occupational Safety and Health Act of 1970 - OSHA.
      b. Accident Prevention Manual for Industrial Operations, Seventh Edition, National Safety Council, Chapter 4.
      c. Applicable State and Local Safety Operating Procedures.
      d. NETA Safety/Accident Prevention Program.
      e. United States Postal Service Safety Practices.
      f. NFPA 70E - Electrical Safety Requirements for Employee Workplace.
      g. American National Standards for Personnel Protection, ANSI Z244.1.

2. Perform tests with apparatus de-energized except where otherwise specifically required herein.
3. Testing Laboratory: Provide a designated safety representative present at Project Site and supervise safety operations.
4. Power Circuits: Conductors shorted to ground by a hot line grounded device approved for the purpose.
5. Do not proceed until safety representative has determined that it is safe to do so.
6. Testing Laboratory: Provide sufficient protective barriers and warning signs to conduct specified tests safely.

G. Tests and inspections include, but are not limited to the following:
1. Proper operation of lights and equipment.
2. Continuity of raceway system.
3. Insulation leakage and impedances.
4. Ground system resistance.
5. Elimination of reverse rotation and single-phasing of motors.
6. Sub-system tests indicated in other Sections.
7. Proper operation of communications systems specified in Section 270500.
8. Proper operation of intrusion detection systems specified in Section 281600.
9. Proper operation of video surveillance system specified in Section 282305.
10. Proper operation of fire alarm system specified in Section 283100.

H. Load balance all electrical phases, at device, panels, and switchboards.

I. Perform electrical system testing and inspection as specified in each related Section and as specified in this Section.

END OF SECTION 26 05 00

SECTION 26 05 01 - WORK IN EXISTING BUILDING – ELECTRICAL

PART 1 - GENERAL

1.01 DESCRIPTION

A. Furnish all labor, materials, services, equipment, and appliances required in conjunction with the work in existing building as indicated in the Contract Documents.

PART 2 - PRODUCTS

2.01 MATERIALS

A. Use materials to match existing construction unless specified elsewhere in these Contract Documents. Materials shall comply to local codes, UL, and be properly applied for their intended function.

PART 3 - EXECUTION

3.01 EXISTING CONDITIONS

A. Inspect the jobsite prior to bidding and be familiar with all existing conditions. Include the cost of the work required to accommodate the existing conditions in the bid proposal.

B. Obtain data related to existing facilities from existing documents, measurements, notations, photographs, surveys and other observations at the site.

C. Relocate existing items as required to accommodate the new construction. Remove, relocate and reconnect equipment and accessories that are to be reused.

D. Coordinate the Work with other divisions of the specifications. Determine which items and equipment are to remain, to be relocated or be removed, and perform all work consistent with the Scope of Work.

E. Loads, which are existing and are to remain shall be connected to the new distribution system as shown on the Drawings or as required to maintain their proper operation.

F. Refer to other divisions of the specifications and determine equipment which requires power to be disconnected, power to be relocated and disconnect power and relocate power to this equipment.

3.02 DISRUPTION OF EXISTING FUNCTIONS

A. Access: Access to and use of the existing facilities and site will be restricted and shall be under the direction and control of the Owner.

B. Disruptions: Maintain existing electrical, communications, alarm, and other existing systems, and maintain existing functions in service except for schedule disruptions. Where existing functions to remain in use are disrupted, they shall be fully restored after disruption, in full compliance with this division of the specifications for new work.

C. Scheduling of Disruptions: Seek and obtain approval two weeks in advance of event of date, starting, time, and duration of each required disruption.

D. Notice of Disruption: Date, time and duration of each disruption shall be subject to the Owner's prior approval, and shall include the following information in the form of a memorandum submitted by the Contractor to the Architect for approval by the Owner:

Facility/System	Starting Date	Starting Time	Duration

E. Emergency Disruptions: When circumstances preclude obtaining advance approval as specified above, make request immediately on knowledge of the requirement, and perform work so as to cause the minimum amount of disruption, for the minimum duration.
F. Notification: Notify the Architect and the Owner immediately by telephone and then in writing, as changes and additions to the scheduled disruption requirements become known.
G. Duration: Complete as large a portion of the work as possible before initiating disruption and perform only that work necessary so as to minimize duration of disruption. Maintain adequate personnel, supplies, materials, equipment, tools, and other resources at job site to avoid unnecessary delay in resumption of normal service.

3.03 SALVAGE, DEMOLITION AND RELOCATION

A. General:
   1. Modify, remove, or relocate materials and items indicated in the Contract Documents and required by the installation of new facilities.
   2. Working jointly with the work under other divisions of the specifications establish and mark salvage and demolition items before commencing work; report items scheduled for relocation, reinstallation or reuse, which are found to be in damaged condition: await further instructions from the Owner's Representative and/or the Architect before commencing with work.
   3. New materials and items of similar design and quality may be substituted for materials and items indicated to be relocated upon approval of shop drawings, product data, and samples.
   4. Remove carefully, in reverse order to original assembly or placement, items, which are to be relocated.
   5. Protect items until relocation is complete.
   6. Clean and repair items to be relocated, and provide new materials, fittings, and appurtenances required to complete the relocation and to restore to good operating order.
   7. Perform the relocation work in accordance with applicable sections of these specifications, utilizing skilled workers.
B. Relocating Devices: Remove and reinstall in locations designated by the Owner's Representative and the Architect temperature control system devices, relays, wire, conduit, fixtures, equipment and other devices required for the operation of the various systems that are installed in existing-to-be-removed construction.

3.04 EXISTING RACEWAYS

A. Reuse raceways where possible and where permitted by local codes. Rework raceways to meet code requirements. Secure all raceways, which are not properly supported. Paint raceways when exposed to view to match surroundings if existing finish is damaged or soiled.
B. Fasten existing boxes and raceways securely to provide proper support.

3.05 NEW RACEWAYS

A. Provide new raceways where required to provide wiring as indicated in the Contract Documents.
B. Where raceways must be exposed to view, use wiremold, trench duct or wireway securely fastened, and painted to match surroundings. Provide number of coats of paint as required to cover prime coat of original finish of wiremold.

3.06 EXISTING LIGHTING FIXTURES

A. Service existing lighting fixtures as follows:
   1. Clean reflective, lenses, and sight exposed portions.
   2. Relamp with new lamps of the same types removed.

3. Repair or replace lampholders, ballasts, wiring, and door latching and hinging mechanisms.
4. Reconnect to circuit wiring, tighten connections.

B. Existing lighting fixtures may be replaced with new fixtures in lieu of the actions required by Paragraph 3.08 A above, if, in the Contractor's opinion, costs to the Owner would be lower.

3.07 EXISTING CEILINGS

A. Provide a typewritten list of existing damaged ceilings and ceiling tiles. Disregard rooms in which ceilings are to be repaired and replaced. Correlate list to room numbers indicated on drawings.

B. Mark damaged ceilings and ceiling tiles with easily removable red "stick-on" labels, minimum size two square inches.

C. Submit list prior to commencing work. Do not start work until list is reviewed by the Architect and Owner; otherwise repair and replace damaged ceilings and ceiling tiles.

3.08 EXISTING PANELBOARDS

Existing panelboards to be reused:

A. Clean interiors and exteriors.

B. Inspect for components damage and notify the architect if repair or replacement may be required.

C. Tighten conduit and wire termination's.

D. Verify panelboards and panelboard feeders are of adequate capacity for loads to be served.
1. Activate loads connected to panelboards to simulate 100 per cent demand.
2. Measure and record amperage readings of phase and neutral conductors of panelboards feeders.
3. Provide typewritten record of recorded measurements to the Architect for review.

E. Rebalance loads as specified in other sections of the specifications to provide for evenly-balanced phases.

F. Provide new typewritten circuit directories.

G. Provide new panelboard identification labels if panelboards designation changes or if no labels exist.

3.09 EXISTING WIRING

A. Inspect existing wiring for damage. Repair or replace damaged wiring.

B. Secure and label existing wiring which is to be disturbed.

C. Tighten existing wiring terminations and connections.

END OF SECTION 26 05 01

SECTION 26 05 19 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Building wire and cable.
   2. Branch-circuit cable.
   3. Wiring connectors and connections.
   4. Drop cords.
   5. Busways.

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other documents.

C. Related Sections:
   1. As specified in Section 260500 - Common Work Results for Electrical: Basic electrical methods.

1.02 REFERENCES

A. As specified in Section 260500 - Common Work Results for Electrical.

1.03 SUBMITTALS

A. As specified in Section 260500 - Common Work Results for Electrical.

1.04 QUALITY ASSURANCE

A. As specified in Section 260500 - Common Work Results for Electrical.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Section 016000 - Product Requirements: Transport, handle, store, and protect products.

B. Deliver in accordance with NEMA WC 26.

PART 2 - PRODUCTS

2.01 BUILDING WIRE AND CABLE

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Alcan Cable, Atlanta, GA (770) 392-2376.
   2. Anixter, Inc., Skokie, IL (800) ANIXTER.
   3. General Cable, Highland Heights, KY (800) 526-4391.
   4. General Electric, Plainville, CT (860) 747-7111.
   5. Okonite, Ramsey, NJ (201) 825-0300.
   6. Southwire Company, Carrollton, GA (800) 444-1700.
   7. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Description: Single conductor insulated wire.

C. Conductor: Copper, except conductors #1/0 AWG and larger may be compact stranded aluminum if equipped with compression lugs and installed per manufacturer's recommendations and the National Electrical Code.
D. Insulation Voltage Rating: 600 Volts.
E. Insulation: NFPA 70, Type THHN/THWN or Type XHHW-2.
F. Multiconductor cable: Metal clad cable, Type MC with ground wire.
   1. Type "MC" cable shall be permitted for use in exposed or accessible ceiling spaces only. Type "MC" cable shall not be utilized above inaccessible hard ceilings or in damp locations. Cable shall be supported and secured where such support does not exceed 3 ft. intervals and shall be properly color coded to identify phase, neutral, ground and switch legs.

2.02 WIRING CONNECTORS

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Buchanan Construction Products, Hackettstown, NJ (800) 610-5201.
   2. Thomas and Betts, Memphis, TN (800) 695-1901.
   3. 3M, St. Paul, MN (800) 364-3577.
   4. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Compression Connectors; Conductor sizes #12 through #6 AWG:
   1. Buchanan: 2006S or 2011S.
   2. Thomas and Betts
   3. 3M

2.03 DROP CORDS

A. Description: Continuous length of cable with 20 Amp, 120 Volt locking blade type connector body at one end as indicated on Drawings. Secure cable at both ends with wire type stainless steel cable grips to prevent transmission of tension directly to conductors or terminal screws.

B. Junction Box: Furnished and installed flush with ceiling anchored to building structure for fastening of upper cord grip.
C. Cable: Type SO 600 Volt flexible cord with three #12 stranded wires.
D. Connector Body: Single 20 Amp, 120 Volt, grounding receptacle of twist-lock type at one end and straight blade type at other end that grips on cable insulation and is manufactured for use with wire cable grips. Furnish and install drop cords in length required for a receptacle height of 6 feet 8 inches above finished floor.

2.04 BUSWAYS

A. Basis of Design: General Electric "Spectra" series.

B. Manufacturers: Subject to compliance with project requirements, manufacturers offering products which may be incorporated in the Work include the following:
   1. Eaton Corporation, Cutler-Hammer Products, Pittsburg, PA (800) 525-2000.
   2. General Electric Company (800) 626-2000.
   3. Siemens Energy and Automation, Alpharetta, GA (800) 964-4114.
   4. Square D Company, Palatine, IL (800) 392-8781.
   5. No substitutions permitted.

C. Provide factory shop drawing submittals for each type of busway.

1. Show fabrication and installation details of busway, including plans, elevations and sections of components and attachments to other construction elements. Designate components and accessories, including clamps, brackets, hanger rods, splice-plate connectors, expansion-joint assemblies, straight lengths and fittings.
2. Seismic-Restraint Details: Signed and sealed by a qualified Professional Engineer, licensed in the state where Project is located, who is responsible for their preparation.
   a. Design Calculations: Calculate requirements for selecting seismic restraints.
   b. Detail fabrication, including anchorages and attachments to structure and to supported busways.

D. Furnish and install a totally enclosed, low-impedance 5 wire, copper, busway system of the indicated ratings with all necessary fittings, power takeoffs, hanging devices and accessories.
1. Materials and installation shall comply with applicable codes, recommended practices and standards of ANSI, IEEE, NEMA and UL. All components of the busway shall be UL listed. Arrangements, details and locations shall be as shown on the drawings and specified herein. The housing shall be of extruded aluminum and all hardware shall be painted to prevent corrosion.
2. Joints shall be of the one-bolt removable/isolatable type with through-bolts that can be checked for tightness without de-energizing the system. The means of visual indication shall be a color change in the head of the bolt. It shall be possible to make up a joint from one side in the event the busway is installed against a wall or ceiling. The joint shall be so designed as to allow removal of any length without disturbing adjacent lengths. Belleville springs shall be provided to give positive pressure over complete contact area.
3. Plug-in and feeder busway shall use identical parts and all multi-stacks shall be phase collected.
4. The maximum hot-spot temperature rise at any point in the busway at continuous rated load shall not exceed 55 degrees C above a maximum ambient temperature of 40 degrees C in any position.

E. Furnish and install busway plugs of the types and ratings indicated. Plugs shall be UL labeled. Housing shall completely enclose the switching device and shall be of sheet steel furnished in ASNI-61 grey enamel over a rust inhibitor. Provide stab shields that protect ground plug body to busway housing before stabs make power contact. Provide grounding terminal inside plug body and adequate shielding to prevent access to live parts when cover is open. A ground stab to engage grounding tab on busway and internal ground bus shall be provided. Provide means for padlocking cover and operating handle in "off" position. The operating handle shall be easily moved from end to side of vice versa so that it will be in the correct position to operate from the floor.
1. Circuit breaker type plugs shall have an interrupting rating as indicated on the drawings. They shall have a releasable cover interlock that prevents opening of cover except with the breaker in "off" position. An interlock to prevent insertion or removal from busway when in "on" position shall be provided, as well as an interlock (releasable) to prevent closing circuit breaker with cover open.
2. Plug assists shall be furnished on all plugs over 100 Amps that will mechanically engage or disengage the plug from the busway, but only when the plug is in the "off" position.

PART 3 - EXECUTION

3.01 EXAMINATION

A. As specified in Section 260500 - Common Work Results for Electrical.

3.02 PREPARATION

A. Completely and thoroughly swab raceway before installing wire.

3.03 INSTALLATION - CONDUCTORS

A. Wiring methods:
   1. Concealed Dry Interior Locations: Use only building wire, Type THHN/THWN or Type XHHW-2 insulation in metallic raceway or MC multiconductor cable.
   2. Exposed Dry Interior Locations: Use only building wire, Type THHN/THWN or Type XHHW-2 insulation in metallic raceway or MC multiconductor cable.
   3. Above Accessible Ceilings: Use only building wire, Type THHN/THWN or Type XHHW-2 insulation in metallic raceway or MC multiconductor cable.
   4. Wet or Damp Interior Locations: Use only building wire, Type THW or THWN or Type XHHW-2 insulation in raceway.

B. Install products in accordance with manufacturers published instructions and NECA SI.

C. Use solid conductor for feeders and branch circuits 10 AWG and smaller.
D. Use stranded conductors for control circuits and final connections to all vibration equipment.
E. Use conductor not smaller than 12 AWG for power and lighting circuits.
F. Use conductor not smaller than 14 AWG for control circuits.
G. Use 10 AWG conductors for 20 ampere, 120 Volt branch circuits longer than 75 feet.
H. Use 10 AWG conductors for 20 ampere, 277 Volt branch circuits longer than 200 feet.
I. Pull all conductors into raceway at same time.
J. Use approved wire pulling lubricant for all building wire.
K. Protect exposed cable from damage.
L. Neatly train and lace wiring inside boxes, equipment, and panelboards in accordance with NECA Standards.
M. Clean conductor surfaces before installing lugs and connectors.
N. Make splices, taps, and terminations to carry full ampacity of conductors with no perceptible temperature rise.
O. For splices and taps, use only compression connectors for copper or aluminum conductors, 6 AWG and larger. Tape uninsulated conductors and connector with electrical tape to 150 percent of insulation rating of conductor.
   1. Splicing of copper feeder conductors #3 AWG and larger is prohibited.
   2. Splicing of aluminum feeder conductors #1 AWG and larger is prohibited.
   3. Splices within branch circuit or feeder conductors located underground or below grade shall not be provided. All splices shall be terminated above grade.

P. Use solderless pressure compression connectors with insulating covers for copper conductor splices and taps, 8 AWG and smaller.

Q. Use conductors rated 90 degrees C, inside a ballast compartment or within 6 inches of any ballast.
R. Conductor Sizes #8 and Larger: Class B stranding.
S. Install Drop Cords to building structure at locations indicated on Drawings as indicated on Drawings.
T. The sharing of neutral conductors for multiwire branch circuits is prohibited. All branch circuits shall contain individual neutral conductors.

3.04 INSTALLATION - BUSWAYS

A. Horizontal runs of busway shall be UL listed for hanging on 10-foot centers in any position. Vertical riser runs of busway shall be supported with rigid and/or spring hangers. (Max. 16 ft. centers).

B. Final field measurements shall be made by the contractor prior to release for manufacture to assure coordination with other trades. Contractor shall coordinate routing of busways with field conditions.

C. Contractor shall provide all necessary mounting hardware as recommended by the manufacturer. Utilize trapeze hangers, spring isolators, and ½ inch all-thread on 10-foot centers. Installation shall comply with local seismic zone requirements.
   1. Provide approved manufacturer's standard clamps, hangers, brackets, splice plates, reducer plates, blind ends, expansion joints and connectors. Obtain busway components from a single manufacturer.

D. Engraved nameplates: ½ inch high black letters on yellow laminated plastic nameplate, engraved with the following wording: WARNING! DO NOT USE BUSWAY AS WALKWAY, LADDER OR SUPPORT.

3.05 CONSTRUCTION

A. Interface with Other Work:
   1. Identify wire and cable using Thomas and Betts type WM vinyl markers.
   2. Identify each conductor with its circuit number or other designation indicated on Drawings in all junction, pull, terminal boxes and cabinets. Identify neutrals with common circuit numbers in all junction, pull and terminal boxes, panels and cabinets.

3.06 WIRING COLOR CODE

A. Comply with the following color code for each voltage system.

B. 208Y/120 Volt System:
   1. Phase A - Black.
   2. Phase A Switch Leg - Black with "S" tag.
   3. Phase B - Red.
   4. Phase B Switch Leg - Red with "S" tag.
   5. Phase C - Blue.
   6. Phase C - Switch Leg - Blue with "S" tag.
   7. Travelers - Yellow.
   8. Neutral - White.
   9. Equipment Ground - Green.

C. 480Y/277 Volt System:
   1. Phase A - Brown.
   2. Phase A Switch Leg - Brown with "S" Tag.
   3. Phase B - Orange.
   4. Phase B Switch Leg - Orange with "S" Tag.
   5. Phase C - Yellow.
   6. Phase C Switch -Leg- Yellow with "S" Tag.
   7. Travelers - Yellow with "T" Tag.
   8. Neutral - Grey.
   9. Equipment Ground - Green with Yellow stripe.

D. Use same color for same phase throughout. Use same colors for switch legs. Travelers shall be yellow. Phase rotation shall be same in all panels. Identify large cables with colored tape.

E. Provide identification tags on each conductor entering panel, switch, junction box and pull box to identify conductor.

3.07 FIELD QUALITY CONTROL

A. As specified in Section 260500 - Common Work Results for Electrical.

B. Cables, 600 Volt or less and size no. 3 or larger, shall be meggered using an industry-approved "megger with a minimum of 500 Volt internal generating voltage. All inspection, cleaning and testing procedures shall be in compliance with the recommendations and standards outlined in the "maintenance testing specifications for electrical power distribution equipment and systems", latest edition, published by International Electrical Testing Association (NETA). Insulation resistance test values shall be no less than 250 megaohms. A typewritten report of all readings shall be prepared and submitted.

END OF SECTION 26 05 19

SECTION 26 05 33 – RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Metal conduit.
   2. Flexible metal conduit.
   3. Liquidtight metal conduit.
   4. Electrical metallic tubing.
   5. Fittings and conduit bodies.
   6. Wall and ceiling outlet boxes.
   7. Pull and junction boxes.
   8. Cable trays.
   9. Floor boxes with covers (other uses.)

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other documents.

C. Related Sections:
   1. Section 230500 - Common Work Results for HVAC.
   2. Section 260500 - Common Work Results for Electrical.
   3. Section 262726 - Wiring Devices.
   4. Section 270500 - Common Work Results for Communication.
   5. Section 281304 - Enterprise Physical Access Control System (ePACS).
   6. Section 281600 - Intrusion Detection.
   7. Section 282305 - Integrated Security and Investigative Platform (ISIP) CCTV System.
   8. Section 283100 - Fire Emergency Voice/ Alarm Communication System (EVACS).
   9. Section 312000 - Earth Moving.
   10. Section 312300 - Excavation and Fill.

1.02 REFERENCES

A. American Society for Testing and Materials (ASTM):
   1. ASTM A 123 - Specification for Zinc (Hot-Galvanized) Coatings on Iron and Steel Products.

B. American National Standards Institute (ANSI):
   1. ANSI C80.1 - Rigid Steel Conduit, Zinc Coated.
   2. ANSI C80.3 - Electrical Metallic Tubing, Zinc Coated.
   3. ANSI C80.5 - Rigid Aluminum Conduit.

C. National Electrical Contractors Association (NECA):
   1. NECA "Standard of Installation."

D. National Electrical Manufacturers Association (NEMA):
   1. NEMA FB 1 - Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit and Cable Assemblies.
   2. NEMA RN 1 - Polyvinyl Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit.
   3. NEMA TC 2 - Electrical Plastic Tubing (EPT) and Conduit (EPC-40 and EPC-80).
   4. NEMA TC 3 - PVC Fittings for Use with Rigid PVC Conduit and Tubing.
   5. NEMA VE 1 - Metallic Cable Tray Systems.

E. National Fire Protection Association (NFPA):
   1. NFPA 70 - National Electrical Code.

1.03 SYSTEM DESCRIPTION

A. Design Requirements
   1. Conduit Size: NFPA 70, unless indicated otherwise on Drawings.

1.04 QUALITY ASSURANCE

A. Regulatory Requirements:
   1. Conform to requirements of NFPA 70.
   2. Provide products listed and classified by Underwriters Laboratories, Incorporated.

1.05 PROJECT OR SITE CONDITIONS

A. Existing Utilities: Contact local utility companies and make arrangements to obtain utility company location and marking service prior to start of Work.
   1. Locate existing underground utilities in areas of Work using "Ground Penetrating Radar (GPR)" detection. If utilities are to remain in place, provide means of support and protection during trenching and excavation operations.
      a. Pothole and locate existing underground utilities at locations to assure that no conflict with Work of this Contract will occur and required clearance is available to prevent damage to existing utilities.
      b. Perform potholing minimum 10 days before start of excavation or underground work.
   2. Should uncharted, or incorrectly charted, piping or other utilities be encountered during excavation, consult utility company and Contracting Officer immediately for directions.
   3. Coordinate with Contracting Officer and utility companies to keep existing utility services and facilities in operation.
   4. Repair damaged utilities to satisfaction of utility company, at no additional cost to U.S. Postal Service.
   5. Do not interrupt existing utilities serving facilities occupied and used by U.S. Postal Service or others, during occupied hours, except when permitted in writing by Contracting Officer and then only after acceptable temporary utility services have been provided and approved by Contracting Officer.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Section 016000 - Product Requirements: Transport, handle, store, and protect products.

B. Accept conduit on site. Contractor inspect for damage prior to acceptance.
C. Protect conduit from corrosion and entrance of debris by storing above grade. Provide appropriate covering.
D. Protect PVC conduit from sunlight.

PART 2 - PRODUCTS

2.01 CONDUIT REQUIREMENTS

A. Where conduit is required by standards, codes, or required elsewhere, minimum size shall be as follows:
   1. 1/2 inch for power and branch circuit wiring, unless indicated otherwise. All homerun conduits shall be 3/4 inch, minimum.
   2. 3/4 inch for communications cable, unless indicated otherwise.
   3. 3/4 inch for low voltage, control, intercom, security and communications unless indicated otherwise.
   4. Underground conduits shall be sized 1 inch, minimum.

2.02 METAL CONDUIT

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Allied Tube & Conduit, Harvey, IL (800) 882-5543.
   2. Wheatland Tube Co., Collinswood, NJ (800) 257-8182.
   3. Republic Wire & Cable, Rocky Mount, NC (800) 533-8198.
   4. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Rigid Galvanized Steel Conduit (GRC): ANSI C80.1, UL6.

C. Intermediate Metal Conduit (IMC): UL1242.

D. Fittings and Conduit Bodies: NEMA FB1 Material to match conduit.

2.03 FLEXIBLE METAL CONDUIT (FMC)

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Hubbell, Millford, CT (203) 882-4800.
   2. Electriflex, Roselle, IL (800) 323-6174.
   3. 0-Z/Gedney, Farmington, CT (860) 677-5541.
   4. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Description: Interlocked steel and aluminum construction.

C. Fittings: NEMA FB 1.

2.04 LIQUID-TIGHT FLEXIBLE METAL CONDUIT

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Hubbell, Millford, CT (203) 882-4800.
   2. Electriflex, Roselle, IL (800) 323-6174.
   3. Anixter, Inc., Skokie, IL (800) ANIXTER.
   4. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Description: Interlocked steel and aluminum construction with PVC jacket.

C. Fittings: NEMA FB 1.

2.05 ELECTRICAL METALLIC TUBING (EMT)

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Allied Tube & Conduit, Harvey, IL (800) 882-5543.
   2. Wheatland Tube Co., Collinswood, NJ (800) 257-8182.
   3. Republic Wire & Cable, Rocky Mount, NC (800) 533-8198.
   4. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Description: ANSI C80.3; galvanized tubing.

C. Fittings and Conduit Bodies: NEMA FB 1; steel set-screw type. Die-cut Zinc not permitted.

2.06 NONMETALLIC CONDUIT

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Carlon, Cleveland, OH (800) 322-7566.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Description: NEMA TC 2; Schedule 40 PVC.

C. Fittings and Conduit Bodies: NEMA TC 3.

2.07 FITTINGS

A. Manufacturer: Raco, Inc., South Bend, IN (219) 234-7151.
   1. Subject to compliance with project requirements, one of the following manufacturers may also be provided:
      a. Steel City.
      b. 0-Z/Gedney.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Conduits 1/2 inch thru 1 inch enter junction boxes, pull boxes, panels, cabinets, and gutters, provide the following:
   1. Rigid Conduit: Raco 1222, 1223, 1224.
   2. Flexible Metal Conduit: Raco 3302, 3303, 3304, 3305, 3306, 3308.
   3. Liquidtight Flexible Metal Conduit: Raco 3511, 3512, 3513, 3541, 3542, 3543.

C. Conduits 1-1/4 inch and larger entering junction boxes, pull boxes, panels, cabinets, and gutters, provide Insulated throat type bushings; Raco 1225, 1226, 1228, 1230, 1232, 1234, 1236.

D. Provide threaded joint connectors and malleable iron no thread compression box connectors on rigid conduit. Do not provide fittings requiring set screws or indentor type applications including BM connectors.

E. Provide only steel set-screw couplings and connectors on EMT conduit.

2.08 CONDUIT STRAPS AND HANGERS

A. Strap Manufacturer: Raco, Inc., South Bend, IN (219) 234-7151
   1. Subject to compliance with project requirements, one of the following manufacturers may also be provided:
      a. Steel City.
      b. Unistrut.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Hanger Manufacturer: Steel City/Thomas & Betts, Memphis, TN (800) 888-0211.
   1. Subject to compliance with project requirements, one of the following manufacturers may also be provided:
      a. Unistrut.
      b. Raco.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

C. Straps: Two-hole push on stamped steel straps on surface areas such as concrete, masonry, wide flange beams, columns, and wood.
   1. Rigid Conduit: Raco 2232, 2233, 2234, 2235, 2336, 2238.

2. Electrical Metallic Tubing: Raco 2092, 2093, 2094.

D. Hangers: Lay-in pipe hanger.
   1. Conduits 1-1/4 Inch and Larger: Steel-City C-149.

E. Trapeze Hangers for Conduits Grouped Together: Hangers consisting of all thread rods sized as required and Kingdorff channel.
   1. Steel City B-909, 1/2 inch x 1-7/8 inch (12 gauge) with single bolt channel pipe straps.
   2. Steel City C-105, C-105-AL, or C-106, (no wire permitted for anchoring conduit).

2.09 SEAL-OFF AND EXPANSION FITTINGS

A. Seal-Off Fitting Manufacturer: Crouse-Hinds, Syracuse, NY (315) 477-5531.
   1. Subject to compliance with project requirements, one of the following manufacturers may also be provided:
      a. Killark.
      b. Appleton.
      c. O-Z/Gedney.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Expansion Fitting Manufacturer: OZ/Gedney, Farmington, CT (860) 677-5541
   1. Subject to compliance with project requirements, one of the following manufacturers may also be provided:
      a. Crouse-Hinds.
      b. Killark.
      c. Appleton.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

C. Provide seal-off fittings where required by governing authority, code, or as indicated on Drawings.
   1. Vertical Runs: Crouse-Hinds Type EYS.
   2. Horizontal and Vertical Runs: Crouse-Hinds Type EZS.
   3. Elbows: Crouse-Hinds Type EYS.
   4. Sealing Compound: "Chico X" fiber and "Chico A".

D. Provide expansion fittings in conduits where indicated on Drawings or where required to pass through expansion joints embedded in concrete.
   1. O-Z/Gedney Type AX.

2.10 OUTLET BOXES

A. Sheet Metal Outlet Boxes: NEMA OS 1, galvanized steel.
   1. Luminaire and Equipment Supporting Boxes: Rated for weight of equipment supported; include 1/2 inch male fixture studs where required.
   2. Receptacle and Device Boxes - 4 inch square x 2-1/8 inch deep with raised, single gang, plaster ring unless indicated otherwise.
   3. Switch Boxes: 2 inch x 4 inch x 2-1/8 inch deep, unless indicated otherwise.
   4. Communication Boxes: 4 inch square x 3 inch deep with raised gang plaster ring unless indicated otherwise.

B. Cast Boxes: NEMA FB 1, Type FD, aluminum. Provide gasketed cover by box manufacturer. Provide threaded hubs.

C. Wall Plates for Finished Areas: Specified in Section 262726.

2.11 PULL AND JUNCTION BOXES

A. Sheet Metal Boxes: NEMA OS 1, galvanized steel.

2.12 CABLE TRAY

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Chalfant Cable Trays, Cleveland, OH (216) 521-7922.
   2. Cable Management Solutions, Incorporated, Deer Park, NY (800) 308-6788.
   3. GS Metals Corporation, Pinckneyville, IL (800) 851-9341.
   4. Southwire Co., Carrollton, GA (800) 444-1700.
   5. Mono-Systems, Inc., Rye Brook, N.Y. (914) 934-2075.
   6. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Provide factory shop drawing submittals for each type of cable tray.
   1. Show fabrication and installation details of cable tray, including plans, elevations and sections of components and attachments to other construction elements. Designate components and accessories, including clamps, brackets, hanger rods, splice-plate connectors, expansion-joint assemblies, straight lengths and fittings.
   2. Seismic-Restraint Details: Signed and sealed by a qualified Professional Engineer, licensed in the state where Project is located, who is responsible for their preparation.
      a. Design Calculations: Calculate requirements for selecting seismic restraints.
      b. Detail fabrication, including anchorages and attachments to structure and to supported cable trays.

C. Description: NEMA VE 1, ladder tray, wire mesh tray or solid bottom tray as indicated on drawings.

D. Material: Steel or aluminum.
E. NEMA Load/Span Class: 20C
F. Finish: ASTM A 525, pre-galvanized or clear aluminum.
G. Inside Width and Depth: Indicated on Drawings. Inside Radius of Fittings: 24 inches (minimum).
H. Provide with compartment dividers as indicated on drawings. Same materials and finish as tray.
I. Straight Section Rung Spacing: 9 inches on center (ladder tray only).
J. Provide approved manufacturer's standard clamps, hangers, brackets, splice plates, reducer plates, blind ends, barrier strips, connectors, and grounding straps. Obtain cable tray components from a single manufacturer.
K. Engraved Nameplates: 1/2 inch high black letters on yellow laminated plastic nameplate, engraved with the following wording:

WARNING! DO NOT USE CABLE TRAY AS WALKWAY, LADDER, OR SUPPORT. USE ONLY AS MECHANICAL SUPPORT FOR CABLES AND TUBING!

2.13 FLOOR BOXES

A. Type: Modular, flush-type dual-service units suitable for wiring method used. Provide dual-service units within carpeted areas only.

B. Compartmentation: Barrier separates power and signal compartments.
C. Housing Material: Die-cast aluminum, satin-finished.

D. Power Receptacle: NEMA WD 6, Configuration 5-20R, ivory finish, unless otherwise indicated.
E. Signal Outlet: Blank cover with brushed cable opening, unless otherwise indicated.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.

B. Verification of Existing Conditions: Verify that field measurements, surfaces, substrates and conditions are as required, and ready to receive Work.
   1. Verify routing and termination locations of proposed conduit prior to rough-in.

C. Existing Utilities, Conduits and Piping:
   1. Locate the routings of the existing underground utilities, conduits and piping in the areas of the Work prior to the installation of the proposed conduit. Trace the locations of existing underground utilities, conduits, and piping using "Ground Penetrating Radar (GPR)" detection.
      a. The routings of these existing underground utilities, conduits and piping shall be identified and marked to avoid any trenching conflicts. These routings shall be recorded on the As-Built drawings prepared by the Contractor for future reference.
   2. Existing utilities, conduits and piping that are to remain shall be supported and protected during the trenching process.

D. Report in writing to the USPS Project Manager prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.

E. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the United States Postal Service.

3.02 INSTALLATION - RACEWAYS

A. Install in accordance with the following schedule, unless indicated otherwise on Drawings: Plastic flexible PVC conduit shall not be permitted. Flexible metal conduit shall be permitted for electrical power and security wiring only and not permitted for fire alarm cables.
   1. Above suspended ceilings: Galvanized or sherardised thick wall rigid steel (GRC) or electrical metallic tubing (EMT).
   2. Metal stud walls: Galvanized or sherardised thick wall rigid steel (GRC) or electrical metallic tubing (EMT).
   3. Exposed interior areas: Galvanized or sherardised thick wall rigid steel (GRC) or electrical metallic tubing (EMT).
   4. Exposed exterior areas: Galvanized or sherardised thick wall rigid steel (GRC).
   5. Underground or below slab areas: Rigid polyvinyl chloride conduit (PVC-Sched. 40).

B. Install conduit in accordance with NECA "Standard of Installation."

C. Install nonmetallic conduit in accordance with manufacturer's instructions. Nonmetallic conduit shall only be used under slabs or direct buried in earth. Conduit penetrations through slab including elbows shall be galvanized rigid conduit.
D. Conduit routing indicated on Drawings are approximate locations unless dimensioned. Route parallel and perpendicular to building construction for complete wiring system regardless whether exposed or concealed.
E. Arrange supports to prevent misalignment during wiring installation.

F. Support conduit using coated steel or malleable iron straps, lay-in adjustable hangers, clevis hangers, and split hangers.
G. Group related conduits; support using conduit rack. Construct rack using approved steel channel and provide space on each rack for 25 percent additional conduits.
H. Fasten conduit supports to building structure and surfaces under provisions of this section.
I. Do not support conduit with wire or perforated pipe straps in any type structure. Remove wire used for temporary supports. Steel tie wire may be used to anchor conduit down to reinforcing rods in concrete encasement only.
J. Do not attach conduit or boxes to ceiling support wires. Boxes shall be independently supported.
K. Arrange conduit to maintain headroom and present neat appearance. Maintain required clearance between conduit and piping.
L. Route all conduit, whether exposed or concealed, parallel and perpendicular to walls, ceilings, building structures, etc.
M. Maintain 12 inch clearance between conduit and surfaces with temperatures exceeding 104 degrees F.
N. Cut EMT conduit square using saw or pipe cutter; de-burr cut ends and ream. Bring conduit to shoulder of fittings; fasten securely.
O. Join nonmetallic conduit using cement as recommended by manufacturer. Wipe nonmetallic conduit dry and clean before joining. Apply full even coat of cement to entire area inserted in fitting. Allow joint to cure for 20 minutes, minimum.
P. Use conduit hubs or sealing locknuts to fasten conduit to sheet metal boxes in damp and wet locations and to cast boxes. Use Myers hub connectors on all conduit entering top or sides of all junction boxes, pull boxes, wiring gutters, exposed to weather.
Q. The number of conduit bends per box shall comply with NFPA 70, Article 360. Conduit bends for "SCS" installation shall not exceed two 90 degree bends or exceed a total of 180 degrees of bend between pull boxes or conduit ends. Pull boxes shall be sized per NEC codes per conduit installed. Use conduit bodies to make sharp changes in direction, as around beams. Use hydraulic one-shot bender to fabricate or use factory elbows for bends in metal conduit larger than 2 inch size.
R. Avoid moisture traps; provide junction box with drain fitting at low points in conduit system.
S. Provide suitable fittings to accommodate expansion and deflection where conduit crosses control and expansion joints.
T. Provide suitable nylon pull string or #14 AWG steel wire in each conduit excluding sleeves and nipples.
U. Ground and bond conduit per NFPA 70.
V. Coat all metallic conduit with "General Electric" RTV silicone sealer where conduit is installed in exterior areas or in contact with concrete or earth.
W. Conduits shall be sized as indicated on Drawings. Where sizes are not indicated, conduit shall be sized per NFPA 70.
X. Cap all upturned conduits during construction rough-in to prevent moisture or debris from entering. Pull through each and every conduit a dry swab of sufficient size to remove any and all moisture.

Y. Maximum length of flexible metal conduit (Greenfield), or flexible liquid-tight shall be 5 feet.
Z. Assure ground continuity on all branch circuitry conduits with two locknuts, one inside and one outside of all boxes, cabinets and gutters for rigid conduit. One locknut inside of all boxes, cabinets, and gutters for EMT.
AA. Provide conduit supports as follows:
   1. Galvanized rigid thick wall conduit (GRC), intermediate grade rigid conduit (IMC) and electrical metallic conduit (EMT) within three feet of all outlet boxes, junction boxes, cabinets, gutters, or fittings. Horizontally anchored at 10 foot maximum intervals. Other spacings indicated on Drawings.
   2. Flexible metal conduit (Greenfield) and liquid-tight flexible metal conduit, within 12 inches of all outlet boxes, junction boxes, cabinets, gutters, or fittings and bends or turns. Horizontally anchored at 4-1/2 foot intervals. 1/2 inch minimum size permitted.

BB. Overhead conduits shall be routed below the roof decking. Raceways shall not be routed between the decking flutes.

3.03 INSTALLATION - BOXES

A. Install boxes in accordance with NECA "Standard of Installation."

B. Install in locations as shown on Drawings, and as required for splices, taps, wire pulling, equipment connections and compliance with NFPA 70.
C. Set wall mounted boxes at elevations to accommodate mounting heights indicated or as required for specific project requirements. Orient boxes to accommodate wiring devices as specified in Section 262726.
D. Electrical boxes are indicated on Drawings in approximate locations unless dimensioned. Adjust box location up to 10 feet if required to accommodate intended purpose with no additional cost to contract. Coordinate mounting heights and locations of outlets mounted above counters, benches, and backsplashes.
E. Maintain headroom and present neat mechanical appearance.
F. Install pull boxes and junction boxes above accessible ceilings and in unfinished areas only. Install pull boxes in freezer and dock area above bottom chord of structural joist. Pullboxes sized in excess of 12 inches shall be equipped with hinged and hasped covers.
G. Install outlet and junction boxes within inaccessible ceiling areas, no more than 6 inches from ceiling access panel or from removable recessed luminaire.
H. Locate outlet boxes to allow luminaires positioned as indicated on Drawings.
I. Align adjacent wall mounted outlet boxes for switches, thermostats, and similar devices.
J. Locate flush mounted box in masonry wall to require cutting of masonry unit corner only. Coordinate masonry cutting to achieve neat opening. Use approved raised gang covers in masonry and stud walls.
K. Flush mounted boxes shall not be mounted back-to-back in walls; provide minimum 6 inches separation. Provide minimum 24 inches separation in acoustic rated walls.
L. Secure flush mounted box to interior wall and partition studs. Accurately position to allow for surface finish thickness. Use approved stamped steel bridges to fasten box between studs.
M. Install flush mounting box without damaging wall insulation or reducing its effectiveness.
N. Use approved adjustable steel channel fasteners spanning joist for hung ceiling outlet box.

O. Provide factory sectioned multi-gang boxes where more than one adjacent device is to be mounted. Sectional boxes shall not be permitted.

3.04 INSTALLATION - CABLE TRAYS

A. Install trays level and plumb in accordance with manufacturer's published instructions.

B. Install metallic cable tray in accordance with NEMA VE 2.

C. Support cable trays as follows:
   1. Use anchors and fasteners as specified in Section 260500.
   2. Provide supports at each connection point and at the end of each run.
   3. Design supports including attachment to structure to carry the greater of calculated load multiplied by a factor of four or the calculated load plus 200 lb.

D. Locate cable tray with sufficient space to permit access for installing cables.

E. Make changes in directions and elevations using standard fittings. Use expansion connectors where required.

F. Ground and bond cable tray under provisions of Section 260500.

G. Provide continuity between tray components.

H. Use antioxidant compound to prepare aluminum contact surfaces before assembly.

I. Provide #2 AWG bare copper equipment grounding conductor through entire length of tray; bond to each section.

J. Connections to tray may be made using mechanical connectors.

K. Install warning signs at 50 feet on center along cable tray, located to be visible.

3.05 FIELD QUALITY CONTROL

A. Section 014000 - Quality Requirements: Field inspection.

B. Inspect conduit installation, types, sizes, fittings and attachment to structure.

C. Inspect box installation, locations, connection to conduit, and attachment to structure.

D. Inspect cable tray installation, locations, connection to conduit, and attachment to structure.

3.06 ADJUSTING

A. Adjust flush-mounting outlets to make front flush with finished wall material.

B. Install knockout closures in unused box openings.

3.07 CLEANING

A. Clean interior of boxes to remove dust, debris, and other material.

B. Clean exposed surfaces and restore finish like new.

END OF SECTION 26 05 33

SECTION 26 27 26 - WIRING DEVICES

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Wall switches.
   2. Receptacles.
   3. Device plates and box covers.
   4. Multi-Outlet surface raceway.
   5. TelePower Poles.
   6. Digital Interval Countdown Timer Switch

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other documents.

C. Related Sections:
   1. Section 260500 - Common Work Results for Electrical.

1.02 REFERENCES

A. National Electrical Contractors Association (NECA):
   1. NECA "Standard of Installation."

B. National Electrical Manufacturers Association (NEMA):
   1. NEMA WD 1 - General Requirements for Wiring Devices.
   2. NEMA WD 6 - Wiring Devices - Dimensional Requirements.

C. National Fire Protection Association (NFPA):
   1. NFPA 70 - National Electrical Code.

1.03 QUALITY ASSURANCE

A. Qualifications:
   1. Manufacturer: Company specializing in manufacturing the Products specified in this section with minimum 5 years documented experience.

B. Regulatory Requirements:
   1. Conform to requirements of NFPA 70.
   2. Provide Products listed and classified by Underwriters Laboratories, Incorporated.

1.04 SUBMITTALS

A. Product data required.

PART 2 - PRODUCTS

2.01 WALL SWITCHES

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Hubbell, Inc, Milford, CT (203) 882-4800.
   2. Leviton Manufacturing, Company, Inc., Little Neck, NY (800) 824-3005.
   3. Pass & Seymour, Syracuse, NY (800) 776-4035.

B. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

C. Provide 20Amp, 120/277Volt, specification grade, flush single pole toggle switches with side and back wired screw terminals. All switches shall be equipped with grounding screws.

D. Single Pole Switch:
   1. Leviton Cat. No.1221-2.
   2. P&S Cat. No. PS20AC1I.
   3. Hubbell Cat. No. HBL1221.

E. Double Pole Switch:
   1. Leviton Cat. No. 1222-2.
   2. P&S Cat. No. PS20AC2.
   3. Hubbell, Cat. No. HBL1222.

F. Three-way Switch:
   1. Leviton, Cat. No. 1223-2.
   2. P&S Cat. No. PS20AC-3.
   3. Hubbell Cat. No. HBL1223.

G. Indicator Switch:
   1. Leviton Cat. No. 1221-PLR (Red).
   2. P&S Cat. No. PS20AC1-RPL (Red).
   3. Hubbell Cat. No. HBL1221PL (Red).

H. Locator Switch:
   1. Leviton Cat. No. 1221-LHC (Clear).
   2. P&S Cat. No. PS20AC1-CSL (Clear).
   3. Hubbell Cat. No. HBL1221IL (Clear).

I. Locking Switch:
   1. Leviton Cat. No. 1221-2L.
   2. P&S Cat. No. PS20AC1-L.
   3. Hubbell Cat. No. HBL1221L.

J. Color: Switches located within the Retail Area to be mounted in "blue" or "red" painted walls shall be black. All other switches shall be white unless indicated otherwise.

2.02 RECEPTACLES

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Leviton Manufacturing, Company, Inc., Little Neck, NY (800) 824-3005.
   2. Pass & Seymour, Syracuse, NY (800) 776-4035.
   3. Hubbell, Inc, Milford, CT (203) 882-4800.
   4. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Provide duplex, specification grade, 20Amp, 125Volt, 2 pole, 3 wire receptacles with grounding screw.

C. Duplex Convenience Receptacle:
   1. Leviton Cat. No. 5362.
   2. P&S Cat. No. 5362.
   3. Hubbell Cat. No. HBL5352.

D. Tamper and Weather Resistant GFCI Receptacle (Side Wired Feed-Thru):
   1. Hubbell Cat. No. GFR5362SG.

E. Color: Receptacles located within the Retail Area to be mounted in "blue" or "red" painted walls shall be black. All other receptacles shall be white unless indicated otherwise.

2.03 WALL PLATES

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. P&S Sierra.
   2. Hubbell.
   3. Leviton.
   4. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Coverplate: Coverplates to be installed within the Retail Areas on "blue" or "red" painted walls shall be black smooth thermoplastic. All other coverplates shall be white smooth thermoplastic unless otherwise noted.
   1. Sierra TP8-W.

C. Weatherproof Coverplate: Gasketed cast metal with hinged gasketed device.
   1. Sierra 4510 cast aluminum.

D. Integral locking and pad-lockable coverplates:
   1. Duplex receptacles shall be equipped with Decora style, stainless steel, single gang, locking coverplates: Pass & Seymour/Legrand #WR26L.
   2. Quadraplex receptacles shall be equipped die-cast metal, low profile, two gang, flip type, pad-lockable coverplates: Hubbell/TayMac #MX2050S.
      a. Provide two (2) keyed, padlocks for each quadraplex coverplate: Master Lock #4120KA.
      b. All quadraplex receptacles to be keyed alike.

2.04 MULTI-OUTLET SURFACE RACEWAY

A. Manufacturer: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Legrand/Wiremold, West Hartford, CT (800) 621-0049.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Raceway Material: Anodized aluminum with manufacturer's standard hardware and fittings. Length as indicated on drawings.

C. Wire: Factory pre-wired with No. 12 AWG minimum. Provide equipment grounding conductor.

D. Wiring Devices: NEMA5-20R duplex receptacles and/or telecommunication outlets. Quantity as indicated on drawings.

E. Provide single channel raceway for applications requiring power receptacles only. Provide dual channel raceway for applications requiring power receptacles and telecommunications outlets.

F. Single channel, single cover raceway.
   1. Wiremold #AL3000 Series.

G. Dual channel, single cover raceway
   1. Wiremold #AL4000 Series.

2.05 TELE/POWER POLE

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following:
   1. Legrand/Wiremold, West Hartford, CT (800) 621-0049.

2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Raceway Material: Anodized aluminum with manufacturer's standard hardware and fittings. Length as indicated on drawings.

C. Wire: Factory pre-wired with No. 12 AWG minimum. Provide equipment grounding conductor.

D. Wiring Devices: NEMA5-20R duplex receptacles and/or telecommunication outlets. Quantity as indicated on drawings.

E. Basis of Design: Wiremold NP620 Series.

2.06 DIGITAL INTERVAL COUNTDOWN TIMER SWITCH

A. Provide flush wall mounted line voltage, digital, countdown timer switch with the following features:
   1. The timer switch shall be preset to turn loads "off" after a preset interval time of 60 minutes maximum. Switch shall be equipped with manual on/off pushbutton.
   2. Timer switch shall mount in a standard single gang wall box and shall fit behind a decorator style face plate. The control switches shall not protrude more than 1/8 inch from the wall.
   3. Timer switch shall have no minimum load requirement from 0 to 800 Watt @ 120 VAC - 60 Hz.
   4. Optional flash and beep warnings shall notify occupants when the interval countdown reaches one minute.
   5. The switch shall not require a neutral, simplifying installation and shall feature terminal style wiring, which makes installation easier.
   6. Basis of Design:
      a. Sensor Switch #PTS-60 (60 minute max.)
      b. Intermatic #EI215 (1800W @ 120 VAC).

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.

B. Verification of Conditions: Verify that field measurements, surfaces, substrates and conditions are as required, and ready to receive Work.
   1. Verify that outlet boxes are installed at proper height.
   2. Verify that wall openings are neatly cut and will be completely covered by wall plates.
   3. Verify that branch circuit wiring installation is completed, tested, and ready for connection to wiring devices.

3.02 PREPARATION

A. Provide extension rings to bring outlet boxes flush with finished surface.

B. Clean debris from outlet boxes.

3.03 INSTALLATION

A. Install wiring devices as indicated, in accordance with manufacturer's written instruction, applicable requirements of the NEC and NECA "Standard of Installation", and in accordance with recognized industry practices to fulfill project requirements.

B. Install devices plumb and level.

C. Install switches with OFF position down.

D. Install receptacles with grounding pole on bottom.
E. Connect wiring device grounding terminal to branch circuit equipment grounding conductor.
F. Connect wiring devices by wrapping conductor 2/3 of screw diameter in clockwise direction around screw terminal. Tighten screw to 12 pound-inches. Do not use spring pressure devices for wire connections.
G. Install galvanized steel plates on outlet boxes and junction boxes in unfinished areas, above accessible ceilings, and on surface mounted outlets.
H. Provide coverplates on switch, receptacle, and blank outlets.
I. Receptacles mounted within 8 feet of a fire extinguisher shall be equipped with integral locking or pad-lockable coverplates as specified in paragraph 2.3 D.

3.04 LABELING

A. All coverplates for receptacles and switches shall be labeled with the branch circuit number. Label shall be machine generated and permanently affixed to the outside of the coverplate.

3.05 CONSTRUCTION

A. Interface with other work:
   1. Coordinate locations of outlet boxes provided under Section 260533 to obtain mounting heights indicated on Drawings.

3.06 FIELD QUALITY CONTROL

A. Section 014000 - Quality Requirements: Field inspection.

B. Prior to energizing circuitry, test wiring for electrical continuity, and for short circuits. Ensure proper polarity of connections is maintained. Subsequent to energization, test wiring devices to demonstrate compliance with requirements.
C. Inspect each wiring device for defects.
D. Operate each wall switch with circuit energized and verify proper operation.
E. Verify that each receptacle device is energized.
F. Test each receptacle device for proper polarity.
G. Test each GFCI receptacle device for proper operation.

3.07 ADJUSTING

A. Adjust devices and wall plates to be flush, level and plumb with wall.

3.08 CLEANING

A. Section 017300 - Execution: Cleaning installed work.

B. Clean exposed surfaces to remove splatters and restore finish.

END OF SECTION 26 27 26

SECTION 26 41 28 - SURGE PROTECTIVE DEVICES (SPD)

PART 1 - GENERAL

1.01 SUMMARY

A. This section describes the materials and installation requirements for surge protective devices (SPD) for the protection of all power and communications circuits. Provide and install materials, labor and auxiliaries required to furnish and install complete surge suppression for the protection of building electrical and electronics systems from the effects of induced transient voltage surge and lightning discharge as indicated on drawings.
   1. Provide surge suppression devices for the following equipment:
      a. Each main electrical service switchboard as indicated for on drawings.
      b. Distribution and branch panels as indicated for on drawings.
      c. All electronic communications equipment installed including but not limited to: fire alarm, intrusion, ePACS, CCTV, and paging systems.
   2. Provide surge suppression protection on all exterior lighting and communications systems wiring if the "expected lightning stroke frequency" exceeds the "tolerable lightning frequency" of the structure.

B. Related documents: The contract documents, as defined in Section 011000-Summary of Work, apply to work of this section. Additional requirements and information necessary to complete the work of this section may be found in other documents.

C. Related sections:
   1. Section 260500 - Common Work Results for Electrical.
   2. Section 262413 - Switchboards.
   3. Section 264100 - Facility Lightning Protection.
   4. Section 265600 - Exterior Lighting.
   5. Section 275116 - IP Integrated, Public Address Zone Paging System.
   6. Section 275117 - Video Intercom and Exterior Gate Control System.
   7. Section 281304 - Enterprise Physical Access Control System.
   8. Section 281600 - Intrusion Detection System.
   9. Section 282305 - Integrated Security and Investigative Platform (ISIP) CCTV System.
   10. Section 283100 - Fire Emergency Voice/Alarm Communication System (EVACS).

1.02 REFERENCES

A. IEEE C62.41.1, IEEE Guide on the Surge Environment in Low-Voltage (1000 V and Less) AC Power Circuits,

B. IEEE C62.41.2, IEEE Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits,

C. IEEE C62.45, IEEE Recommended Practice on Surge Testing for Equipment Connected to Low-Voltage (1000 V and Less) AC Power Circuits.

D. National Electrical Code: Article 285.

E. UL 1283 - Electromagnetic Interference Filters.

F. UL 1449, 4th Edition, effective December 30, 2014 - Surge Protective Devices.

1.03 SUBMITTALS

A. Section 013300 - Submittal Procedures: Procedures for submittals.

1. Shop Drawings: Indicate outline and support point dimensions, voltage, main bus ampacity, integrated short circuit ampere rating, circuit breaker and fusible switch arrangement and sizes.
2. Manufacturer's Installation Instructions: Indicate application conditions and limitations of use stipulated by Product testing agency. Include instructions for storage, handling, protection, examination, preparation, installation, and starting of Product.
3. Certification submitted SPDs are manufactured in the United Sates.
4. Shall include UL 1449 Listing documentation verifying the following:
   a. Short Circuit Current Rating (SCCR)
   b. Voltage Protection Ratings (VPRs) for all modes
   c. Maximum Continuous Operating Voltage rating (MCOV)
   d. I-nominal rating (I-n)
   e. Type 1 Device Listing
      1) VPR, MCOV, I-n, and Type 1 information is posted at www.UL.com, under Certifications, searching using UL Category Code: VZCA. SCCRs are posted in manufacturer's UL docs.
      2) UL data and visual inspection takes precedence over manufacturer's published documentation.

B. Section 017704 - Closeout Procedures and Training: Procedures for closeout submittals:
   1. Project Record Documents: Record actual locations of Products; indicate actual branch circuit arrangement.
   2. Operation and Maintenance Data: Include spare parts data listing; source and current prices of replacement parts and supplies; and recommended maintenance procedures and intervals.
   3. Submit data showing compliance with UL 1449 4th edition.

1.04 QUALITY ASSURANCE

A. SPDs must be manufactured in the United Sates.

B. Manufacturer Qualifications: Engage a firm with at least 10 years experience in manufacturing transient voltage surge suppressors.

C. Manufacturer shall be ISO 9001 or 9002 certified.

D. The manufacturer of this equipment shall have produced similar electrical equipment for a minimum period of 5 years. When requested by the Engineer, an acceptable list of installations with similar equipment shall be provided demonstrating compliance with this requirement.

E. The SPD shall be compliant with the Restriction of Hazardous Substances (RoHS) Directive 2002/95/EC.

1.05 DELIVERY, STORAGE AND HANDLING

A. Handle and store equipment in accordance with manufacturer's Installation and Maintenance Manuals. One copy of this document to be provided with the equipment at time of shipment.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Subject to compliance with project requirements, manufacturers offering specified items which may be incorporated in the Work include the following.
   1. ASCO/Advanced Protection Technologies, Incorporated, Clearwater, FL (800) 237-4567
   2. Emerson/Liebert Corporation, Columbus, OH, (800) 877-9222
   3. Atlantic Scientific Corporation, Melbourne, FL, (800) 544-4737
   4. Current Technology Inc., Irving, TX, (800) 238-5000

5. Ditek Surge Protection, Largo, FL, (800) 753-2345

B. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

2.02 SERVICE ENTRANCE SURGE PROTECTIVE DEVICES (SPDS)

A. Models:
   1. Basis of Design: Advanced Protection Technologies: "TEXAS" Series.

B. Surge Protective Device Description: Replaceable module type complying with UL 1283 and UL 1449 4th Edition Listed. Provide unit with the following features and accessories:
   1. LED indicator lights for power and protection status.
   2. Audible alarm, with silencing switch, to indicate when protection has failed.
   3. One set of dry contacts rated at 5.0 amperes, 240 volts ac, for remote monitoring of protection status.

C. Short Circuit Current Rating: SPD shall be UL labeled with 200kA Short Circuit Current Rating (SCCR). Fuse ratings shall not be considered in lieu of demonstrated withstand testing of SPD, per NEC 285.6.

D. SPD Type: SPD shall be UL labeled as Type 1 (verifiable at UL.com), intended for use without need for external or supplemental overcurrent controls. Every suppression component of every mode, including N-G, shall be protected by internal overcurrent and thermal overtemperature controls. SPDs relying upon external or supplementary installed safety disconnectors do not meet the intent of this specification.

E. In Rating: SPD shall be UL labeled with 20kA Inominal (I-n) (verifiable at UL.com) for compliance to UL 96A Lightning Protection Master Label and NFPA 780.

F. SPD shall provide surge current diversion paths for all modes of protection; L-N, L-G, N-G, and L-L in WYE systems, and L-L, L-G in DELTA Systems.

G. Minimum Single Impulse Surge Current Capability (single pulse rated) per phase shall be.
   1. Single Impulse Surge Current Capacity is to be 300 kA.

H. Connection Means: Permanently wired via internal disconnect. The device shall have a NEMA designed and certified safety interlocked integral disconnect switch. The switch shall be located within the unit with an externally mounted metal manual operator.

I. Protection modes and UL 1449 4th Edition Voltage Protection Rating for grounded WYE circuits with voltages of 480Y/277, 3-phase, 4-wire shall be as follows:

VOLTAGE	L-N	L-G	N-G
208Y/120V	700V	700V	700V
480Y/277V	1500V	1500V	1500V

J. Install devices at service entrance at load side, with ground lead bonded to service entrance ground.

K. Test unit in accordance with manufacturer's written instructions.

2.03 DISTRIBUTION SURGE PROTECTIVE DEVICES (SPDS)

A. Models:
   1. Basis of Design: Advanced Protection Technologies: "TEXDS" Series.

B. Surge Protective Device Description: Non-modular type complying with UL 1283 and UL 1449 4th Edition Listed. Provide unit with the following features and accessories:
   1. LED indicator lights for power and protection status.

C. Short Circuit Current Rating: SPD shall be UL labeled with 200kA Short Circuit Current Rating (SCCR). Fuse ratings shall not be considered in lieu of demonstrated withstand testing of SPD, per NEC 285.6.

D. SPD Type: SPD shall be UL labeled as Type 1 (verifiable at UL.com), intended for use without need for external or supplemental overcurrent controls. Every suppression component of every mode, including N-G, shall be protected by internal overcurrent and thermal overtemperature controls. SPDs relying upon external or supplementary installed safety disconnectors do not meet the intent of this specification.

E. In Rating: SPD shall be UL labeled with 20kA Inominal (I-n) (verifiable at UL.com).

F. SPD shall provide surge current diversion paths for all modes of protection; L-N, L-G, N-G, and L-L in WYE systems, and L-L, L-G in DELTA Systems.

G. Minimum Single Impulse Surge Current Capability (single pulse rated) per phase shall be.
   1. Single Impulse Surge Current Capacity is to be 150 kA.

H. Connection Means: Permanently wired via internal disconnect. The device shall have a NEMA designed and certified safety interlocked integral disconnect switch. The switch shall be located within the unit with an externally mounted metal manual operator.

I. Protection modes and UL 1449 4th Edition Voltage Protection Rating for grounded WYE circuits with voltages of 480Y/277, 3-phase, 4-wire shall be as follows:

VOLTAGE	L-N	L-G	N-G
208Y/120V	700V	700V	700V
480Y/277V	1500V	1500V	1500V

J. Install devices as close as possible to distribution or branch panelboards.

K. Test unit in accordance with manufacturer's written instructions.

2.04 FIRE ALARM AND SECURITY SYSTEM SURGE PROTECTIVE DEVICES (SPDS)

A. Power Surge Protection
   1. SPD shall be listed or recognized in accordance with UL 1449 4th Edition verifiable by visiting UL.com.
   2. SPD shall provide surge current L-N or L-G mode of protection.
   3. Every mode of protection shall be protected by internal overcurrent and thermal overtemperature controls.
   4. SPD shall meet or exceed the following criteria:
      a. Minimum surge current capability (single pulse rated) per phase shall be:
         1) 120/240 Panel Application 50kA per phase
      b. UL 1449 4th Edition listed Voltage Protection Ratings for shall not exceed the following:

VOLTAGE	L-N/L-G	MCOV
120V or 240/120V	600V	150V

   5. SPD shall have a warranty for a period of five years, incorporating unlimited replacements of suppressor parts if they are destroyed by transients during the warranty period.

B. Signal line protection
   1. SPD shall be solid state, silicon avalanche diode circuitry for protection from overvoltages on 2 or 4 wire signal lines such as balanced pair telephone, metallic pair telephone, buried and overhead field cable, remote radio equipment, and control systems. Unit shall have an external ground lug or wire. Connect ground lug or wire to protected equipment grounding system with a No. 12 green insulated stranded ground wire as short as possible.
      a. Pins Protected: Pins 4, 5 on the RJ- 45 Interface; Pins 3, 4 on the RJ- 11 Interface.
      b. Clamping Voltage: 310 Volts in 500 nsec.

c. Surge Capacity: 1500 Watts for 1 msec.
d. Protection Mode:
   1) Common Mode Pins 4, 5 to shielding braid.
   2) Differential Mode Pins 4,5.
e. Shall be listed to UL497A Telco Specification.

C. Video 75 ohm coaxial cable
1. Solid state, silicon avalanche diode circuitry for non-interrupting overvoltage protection of RG- 59/U coaxial cable. Unit shall be provided with one female input connector for "F" series male connector, one output RG-59/U coax cable terminated with an "F" series male cable end connector and A #16 stranded, 18 inch long grounding wire on output end of unit or similar arrangement. Securely mount adjacent to protection equipment and ground to equipment or local building ground if an equipment ground is not available.
   a. Normal Operating Characteristics
      1) Voltage .....5.8V max
      2) Current.....500ma max
      3) Frequency.....DC to 10 Mhz
      4) Insertion Loss.....3.5db @ 4Mhz
   b. Protection Requirements
      1) Transient suppression level.....7.5v Voltage Protection Level
      2) Transient response.....<5 nanoseconds
      3) Operating temp.....-20o C to +50o C
      4) Energy dissipation.....15,000 watts (10X1000 Test Wave)

PART 3 - EXECUTION

3.01 EXAMINATION

A. As specified in Section 260500 - Common Work Results for Electrical.

3.02 INSTALLATION

A. The installation shall meet the following criteria:
1. Install per manufacturer's recommendations and contract documents.
2. Install units plumb, level and rigid without distortion.
3. One primary lightning arrestor shall be installed external to the service entrance in accordance with manufacturer instructions.
4. Service Entrance SPD shall be installed on the load side of the main service disconnect.
5. Service Entrance SPD ground shall be bonded to the service entrance ground.
6. At Service Entrance, a UL approved disconnect switch shall be provided as a means of servicing if a 60A breaker is not available.
7. One SPD shall be installed external to each designated distribution panelboard.
8. At Distribution and Branch, SPD shall have an independent means of disconnect such that the protected panel remains energized. A 40A breaker (or larger) may serve this function.
9. SPD shall be installed per manufacturer's installation instructions with lead lengths as short (less than 24") and straight as possible. Gently twist conductors together.
10. Before energizing, installer shall verify service and separately derived system Neutral to Ground bonding jumpers per NEC.

3.03 ADJUSTMENTS AND CLEANING

A. Remove debris from SPD and wipe dust and dirt from all components.

B. Repaint marred and scratched surfaces with touch up paint to match original finish.

3.04 TESTING

A. Check tightness of all accessible mechanical and electrical connections to assure they are torqued to the minimum acceptable manufacture's recommendations.

B. Check all installed panels for proper grounding, fastening and alignment.

END OF SECTION 26 41 28

SECTION 26 51 00 - INTERIOR LIGHTING (LED - SOLID STATE)

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Interior luminaires and accessories.
   2. Emergency lighting units.
   3. Exit signs.
   4. Ballast/Light emitting diode (LED) drivers.
   5. Light Sources.
   6. Luminaire accessories.

B. Substitutions:
   1. Section 016000 - Product Requirements: Product substitutions permitted by manufacturers listed in Paragraph 2.1A.

C. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other Documents.

D. Related Sections:
   1. Section 260500 - Common Work Results for Electrical: Basic electrical methods.
   2. Section 260623 – Lighting Control Devices.

1.02 REFERENCES

A. As specified in Section 260500 – Common Work Results for Electrical.

B. Illuminating Engineering Society (IES):
   1. IES LM-79 - (2008) Electrical and Photometric Measurements of Solid-State Lighting Products.
   2. IES LM-80 - (2015) Measuring Lumen Maintenance of LED Light Sources.
   3. IES TM-21 - (2011; Addendum B 2015) Projecting Long Term Lumen Maintenance of LED Light Sources.

C. National Fire Protection Association (NFPA):
   1. NFPA 70 - National Electrical Code.
   2. NFPA 101 - Life Safety Code.

D. National Electrical Manufacturers Association (NEMA):
   1. NEMA ANSILG C78.377 – (2017) Electric Lamps— Specifications for the Chromaticity of Solid State Lighting Products.
   2. NEMA SSL 1 – (2010) Electronic Drivers for Led Devices, Arrays, or Systems.
   3. NEMA SSL 3 - (2011) High-Power White LED Binning for General Illumination.

E. Federal Communications Commission Parts 18.305, 18.307 (EMI RFI).

F. American Society of Heating, Refrigerating and Air Conditioning, Inc.
   1. ANSI/ ASHRAE/ IES Standard 90.1.

G. Underwriters Laboratories (UL)
   1. UL 1472 – (2015) UL Standard for Safety Solid-State Dimming Controls.
   2. UL 1598 – (2008; Reprint Oct 2012) Luminaires.
   3. UL 844 – (2012; Reprint Mar 2016) UL Standard for Safety Luminaires for Use in Hazardous (Classified) Locations.
   4. UL 8750 – (2015; Reprint Feb 2018) UL Standard for Safety Light Emitting Diode (LED) Equipment for Use in Lighting Products.

5. UL 924 – (2016; Reprint Nov 2017) UL Standard for Safety Emergency Lighting and Power Equipment.

1.03 SUBMITTALS

A. As specified in Section 260500 - Common Work Results for Electrical.
   1. Product Data: Submit catalog cuts, drawings, descriptive matter and lighting performance characteristics as required to completely define the materials and construction details employed, finishes applied, dimensions, hinging, latching and relamping provisions, and electrical characteristics.
   2. Assurance/Control Submittals:
      a. Manufacturer's Instructions: Indicate application conditions and limitations of use stipulated by Product testing agency. Include instructions for storage, handling, protection, examination, preparation, and installation of Product.

B. Section 017704 - Closeout Procedures and Training: Procedures for closeout submittals:
   1. Operation and Maintenance Data: Submit manufacturer's operation and maintenance instructions for each type of luminaire.

1.04 DEFINITIONS

A. For LED luminaire light sources, “Useful Life” is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent “Rated Lumen Maintenance Life” as defined in IES LM-80.

B. For LED luminaires, “Luminaire Efficacy” (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.

C. Total harmonic distortion (THD) is the root mean square (RMS) of all the harmonic components divided by the total fundamental current.

1.05 QUALITY ASSURANCE

A. As specified in section 260500 – Common Work Results for Electrical.

B. Products shall be tested, approved and labeled/listed by Underwriters Laboratories, Inc., or by a nationally recognized testing laboratory (NRTL).

C. Electrical equipment and materials shall be new and within one year of manufacture, complying with the latest codes and standards. Re-built, refurbished and/or re-manufactured electrical equipment and materials shall not be furnished on this project.

1.06 MAINTENANCE

A. Section 017704 - Closeout Procedures and Training: Procedures for closeout submittals.

B. Extra Products: At completion of installation, deliver to the USPS Project Manager.
   1. Two of each luminaire lens type.
   2. Each component type: Provide quantity for each unique ballast/driver, and LED array equal to 2 percent of luminaire total, but not less than two of each type.

PART 2 - PRODUCTS

2.01 LUMINAIRE TYPES

A. Type A1: Lithonia #2BLT4-XXX-ADP-EZ1-LP840.
   1. Description: Recessed, 2 feet W x 4 feet L x 3 inch D LED type troffer with side reflectors and dropped acrylic center lens non-air handling.

2. Lens: High performance extruded acrylic diffuser with curved linear prisms.
3. Housing:
   a. 22 gauge steel body, flush steel door with mitered corners.
   b. Frame and housing finished with baked white enamel or powder coated finish.
4. Ballast/Driver: LED high efficiency – 32W at 4000 Lumen. Wattage based on lumen package selected.
5. Mounting:
   a. Recessed in Inverted T suspended ceiling.
   b. Recessed in gypsum board ceiling; provide flanged frame-in kit.
6. Lamps: 4000 Lumen LED array; 4000K rated 60,000 hours at LLD = 0.8.
7. Marking: Luminaires are to be labeled on the interior side with lumen package used.
8. Alternate Manufacturers:
   a. Columbia #LCAT24-40-XX-G-ED1-U.
   b. Metalux #24CZ2-XXX-HE-UNV-L840-HCD-1.

B. Type R5E: Gotham #EVO6-40/XXX-WR-MD-MVOLT-EZ1-EM.
1. Description: Recessed 6 inch dia. aperture LED downlight with 90 min. battery backup.
2. Reflector: Low brightness white painted, self-flanged reflector.
3. Ballast/Driver: 20 Watt/2000 Lumen LED light engine with remote phosphor technology; Wattage based on lumen package selected; 5-year factory warranty.
4. Housing:
   a. Frame to be 18 gauge galvanized steel ring.
   b. Mounting ring shall be secured to grid ceiling bar hangers (supplied with luminaire).
   c. Luminaire frame to be supported from the structure by at least two opposing corners.
5. Junction Box:
   a. Junction box to be code approved for through wiring.
   b. Junction box to be secured to the mounting ring and accessible from two sides.
   c. Junction box to be pre-wired and accessible per code through the ceiling trim opening.
6. Mounting:
   a. 24 inch grid ceiling bar hangers shall be supplied by manufacturer and securely fastened to grid.
   b. Provide 28 inch 'C' channel mounting bars and flange kit for drywall ceilings.
7. Voltage: multi-volt.
8. Lamp: 1000 Lumen, 1500 Lumen or 2000 Lumen, 4000K, remote phosphor enclosed LED array; 60,000 hours at LLD = 0.7.
9. Marking: Luminaires are to be labeled on the interior side with lumen package used.
10. Alternate Manufacturers:
    a. Portfolio #LD6B-XX-D010-EU6B-1020-80-40-6LB-M-1-MW-EM.
    b. Prescolite Lighting LTR-6RD-H-XX-XXL-DM1 LTR-6RD-T-XX-40K-8-MD-WC-EM.
    c. Intense Lighting: #SS6G4DRR-XX-408-ED10V1-IC630-W-SF-EM.

2.02 LUMINAIRES

A. Provide luminaires as indicated in luminaire schedule and details on project plans. Provide luminaires complete with light sources of quantity, type and wattage indicated. Provide all luminaires of the same type by the same manufacturer. Luminaires must be specifically designed for use with the driver or ballast and light source provided.

B. LED Luminaires:
1. Install ballast/drivers, LED arrays and specified accessories at the factory.
2. Luminaires must have a minimum 5 year manufacturer's warranty.

3. Luminaires must have a minimum L70 lumen maintenance value of 50,000 hours as calculated by IES TM-21, with data obtained per IES LM-80 requirements.
4. Luminaire drive current value must be identical to that provided by test data for luminaire in question.
5. Luminaires must be listed with the DesignLights Consortium 'Qualified Products List' when falling into category of "General Application" luminaires, i.e. Interior Directional, Display Case, Troffer, Linear Ambient, or Low/High Bay. Requirements are shown in the DesignLights Consortium "Technical Requirements Table" at https://data.energystar.gov/dataset/EPA-Recognized-Laboratories-For- Lighting-Products/jgwf-7qrr.
6. Provide Department of Energy 'Lighting Facts' label for each luminaire.

C. Luminaires for hazardous locations:
1. In addition to requirements stated herein, provide LED luminaires for hazardous locations which conform to UL 844 or which have Factory Mutual certification for the class and division indicated.

2.03 LED DRIVERS

A. NEMA SSL 1, UL 8750. LED drivers must be electronic, UL Class 1, constant-current type and comply with the following requirements:
1. Output power (watts)and luminous flux (lumens) as shown in luminaire schedule for each luminaire type to meet minimum luminaire efficacy (LE) value provided.
2. Power Factor (PF) greater than or equal to 0.9 over the full dimming range when provided.
3. Current draw Total Harmonic Distortion (THD) of less than 20 percent.
4. Class A sound rating.
5. Operable at input voltage of 120-277 volts at 60 hertz.
6. Minimum 5 year manufacturer's warranty.
7. RoHS compliant.
8. Integral thermal protection that reduces or eliminates the output power if case temperature exceeds a value detrimental to the driver.
9. UL listed for dry or damp locations typical of interior installations.
10. LED driver shall tolerate sustained open circuit and short circuit output conditions without damage.
11. LED driver shall comply with the requirements of the FCC rules and regulations, Title 47 CFR Part 15 Non-Consumer (Class A).

2.04 LIGHT SOURCES

A. NEMA ANSLG C78.377, NEMA SSL 3. Provide type and wattage as indicated in luminaire schedule on project plans.
B. LED arrays shall have a correlated color temperature (CCT) of 4000K; minimum color rendering index (CRI) value of 80.
C. High power, white light output utilizing phosphor conversion (PC) process or mixed system of colored LEDs, typically red, green and blue (RGB).
D. Provide light source color consistency by utilizing a binning tolerance within a 4 step McAdam ellipse.
E. Luminaire shall have door frame and lens compatible for use with LED arrays and integral airflow ventilation system.

2.05 LED EMERGENCY DRIVERS

A. Provide LED emergency driver with automatic power failure detection, test switch and LED indicator (or combination switch/indicator) located on luminaire exterior and provide self-diagnostic function integral to emergency driver. Integral nickel cadmium or lithium iron phosphate battery is required to supply a minimum of 90 minutes of emergency power at 700 Lumens (at 100 Lumens/Watt). Driver must be RoHS compliant, rated for installation in plenum-rated spaces and damp locations, and be warranted for a minimum of five years.

2.06 LUMINAIRE SUPPORT HARDWARE

A. Wire:
   1. ASTM A641/A641M; Galvanized, soft tempered steel, minimum 0.11 inches in diameter, or galvanized, braided steel, minimum 0.08 inches in diameter.

B. Threaded Rods:
   1. Threaded steel rods, 3/16 inch diameter, zinc or cadmium coated.

C. Straps:
   1. Galvanized steel, one inch by 3/16 inch, conforming to ASTM A653/A653M, with a light commercial zinc coating or ASTM A1008/A1008M with an electrodeposited zinc coating conforming to ASTM B633, Type RS.

2.07 EQUIPMENT IDENTIFICATION

A. Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

B. Provide labeled luminaires in accordance with UL 1598 requirements. All luminaires must be clearly marked for operation of specific light sources and ballasts or drivers. Note the following light source characteristics in the format "Use Only _____":
   1. Correlated color temperature (CCT) and color rendering index (CRI) for all luminaires.
   2. All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. Ballasts or drivers must have clear markings indicating multi-level outputs and indicate proper terminals for the various outputs.

2.08 FACTORY APPLIED FINISH

A. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum, meets requirements of NEMA 250 corrosion-resistance test.

2.09 RECESS- AND FLUSH-MOUNTED LUMINAIRES

A. Provide access to lamp and ballast from bottom of luminaire. Provide trim for the exposed surface of flush-mounted luminaires as indicated on project drawings and specifications.

2.10 SUSPENDED LUMINAIRES

A. Provide hangers capable of supporting twice the combined weight of luminaires supported by hangers. Provide with swivel hangers to ensure a plumb installation. Provide cadmium-plated steel with a swivel-ball tapped for the conduit size indicated. Hangers must allow fixtures to swing within an angle of 45 degrees. Brace pendants 4 feet or longer to limit swinging. Single-unit suspended luminaires must have twin-stem hangers. Multiple-unit or continuous row luminaires must have a tubing or stem for wiring at one point and a tubing or rod suspension provided for each unit length of chassis, including one at each end. Provide rods in minimum 0.25 inch diameter.

PART 3 - EXECUTION

3.01 EXAMINATION

A. As specified in Section 260500 – Common Work Results for Electrical.

3.02 INSTALLATION

A. Electrical installations must conform to IEEE C2, NFPA 70, and to the requirements specified herein. Install luminaires to meet the requirements of ASHRAE 90.1 and ASHRAE 189.1. To encourage consistency and uniformity, install luminaires of the same manufacture and model number when residing in the same facility or building.

B. Luminaires:
   1. Set luminaires plumb, square, and level with ceiling and walls, in alignment with adjacent luminaires and secure in accordance with manufacturers' directions and approved drawings. Installation must meet requirements of NFPA 70. Obtain approval of the exact mounting height on the job before commencing installation and, where applicable, after coordinating with the type, style, and pattern of the ceiling being installed.
   2. Recessed and semi-recessed luminaires must be independently supported from the building structure by a minimum of four wires, straps or rods per luminaire and located near each corner of the luminaire. Ceiling grid clips are not allowed as an alternative to independently supported luminaires.
   3. Round luminaires or luminaires smaller in size than the ceiling grid must be independently supported from the building structure by a minimum of two wires, straps or rods per luminaire, spaced approximately equidistant around. Do not support luminaires by acoustical tile ceiling panels.
      a. Where luminaires of sizes less than the ceiling grid are indicated to be centered in the acoustical panel, support each independently and provide at least two 3/4 inch metal channels spanning, and secured to, the ceiling tees for centering and aligning the luminaire. Provide wires, straps, or rods for luminaire support in this section.

C. Suspended Luminaires:
   1. Provide suspended luminaires with swivel hangers so that they hang plumb and level. The stem, canopy and luminaire must be capable of 45 degree swing. Pendants, rods, or chains, 4 feet or longer excluding luminaire, must be braced to prevent swaying using three cables at 120 degree separation.
   2. Suspended luminaires in continuous rows must have internal wireway systems for end to end wiring and must be properly aligned to provide a straight and continuous row without bends, gaps, light leaks or filler pieces.
   3. Match supporting pendants with supported luminaire. Aircraft cable must be stainless steel. Canopies must be finished to match the ceiling and must be low profile unless otherwise shown.
   4. Maximum distance between suspension points must be 10 feet or as recommended by the manufacturer, whichever is less.

D. Locate ceiling luminaires as indicated on reflected ceiling plan.

E. Install surface mounted luminaires and exit luminaire signs plumb and adjust to align with building lines and with each other. Secure to prevent movement. Mount exit signs to outlet box mounted flush in wall or ceilings. Outlet box for ceiling mounted exit signs: Connect to rigid conduit system.

F. Install recessed luminaires using accessories and firestopping materials to meet regulatory requirements for fire rating. In fire rated ceilings, recessed luminaires must carry 1 hour UL fire rating classification.

G. Install earthquake clips to secure recessed grid-supported luminaires in place.

H. Install wall mounted luminaires, emergency lighting units and exit luminaire signs at height as scheduled.
I. Install accessories furnished with each luminaire.
J. Bond products and metal accessories to branch circuit equipment grounding conductor.
K. Install specified light sources in each emergency lighting unit, exit luminaire sign, and luminaire.
L. Wire exit signs and emergency lighting units ahead of the local switch, to the normal lighting circuit located in the same room or area.
M. Luminaire whips shall be steel or aluminum. M/C cable shall be permissible for luminaire whips/connections. Luminaire whips/connections shall be made with a minimum of #12 AWG copper conductors. Equipment grounding conductors shall be provided in all luminaire whips and/or connections.
    1. All luminaire whips shall be supported to luminaire support wire/cable with an approved fastener equal to an Erico "KX" flexible conduit hanger or other UL listed supports and fasteners.
N. Luminaires are not to be used as a raceway unless stamped for use as raceway by manufacturer. Single luminaire in lay-in ceilings shall not be used for raceway and shall be connected to an outlet box located within six feet (6') of fixture with flexible conduit or luminaire whips.

3.03 FIELD QUALITY CONTROL

A. Section 014000 - Quality Requirements: Field testing and inspection.
B. Operate each luminaire after installation and connection. Inspect for proper connection and operation.
C. Final acceptance will be based on measurement of initial lighting levels after required hours of burn in as specified in USPS Mail Processing Facility Design Criteria, not maintained lighting levels.

3.04 WARRANTY

A. Provide a written 5 year on-site replacement warranty for material, luminaire finish, and workmanship. On-site replacement includes transportation, removal, and installation of new products.
    1. Include finish warranty to include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.
    2. Material warranty must include:
        a. All drivers.
        b. Replacement when more than 10 percent of LED sources in any lightbar or subassembly are defective or non-starting.
B. Warranty period must begin on date of beneficial occupancy. Provide the USPS Project Manager with signed warranty certificates prior to final payment.

3.05 ADJUSTING

A. Aim and adjust luminaires as directed by the USPS Project Manager.
B. Position exit luminaire sign directional arrows as indicated.

3.06 CLEANING

A. Section 017300 - Execution: Cleaning installed work.
B. Clean electrical parts to remove conductive and deleterious materials.
C. Remove dirt and debris from enclosures.

D. Clean photometric control surfaces as recommended by manufacturer.
E. Clean finishes and touch up damage.

END OF SECTION 26 51 00

SECTION 26 56 00 - EXTERIOR LIGHTING

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Exterior luminaires and accessories.
   2. Poles.
   3. Ballast/Drivers.

B. Substitutions:
   1. Section 016000 – Product Requirements: Product substitutions permitted by manufacturers listed in Paragraph 2.1A.

C. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the work of this section. Additional requirements and information necessary to complete the work of this section may be found in other documents.

D. Related Sections
   1. Section 033000 - Cast-in-Place Concrete: Concrete for pole foundation.
   2. Section 260500 - Common Work Results for Electrical: Basic electrical methods.
   3. Section 260623 - Lighting Control Devices.

1.02 REFERENCES

A. As referenced in Section 260500 – Common Work Results for Electrical.

B. Illuminating Engineering Society North America (IESNA):
   1. IESNA RP-8 - Recommended Practice for Roadway Lighting.
   2. IESNA RP-20 – Recommended Practice for Lighting for Parking Facilities.
   3. IESNA RP-33 - Recommended Practice for Lighting for Exterior Environments.

C. Federal Communications Commission Parts 18.305, 18.307 (EMI RFI).

D. American Society of Heating, Refrigerating and Air Conditioning, Inc.
   1. ANSI/ ASHRAE/ IES Standard 90.1.

1.03 SUBMITTALS

A. As specified in Section 260500 – Common Work Results for Electrical.
   1. Product Data:
      a. Luminaire dimensions, ratings, and performance data.
      b. Complete computer data printout of illumination levels based on a 5 by 5 foot grid pattern.
   2. Shop Drawings:
      a. Indicate dimensions and components for each luminaire which is not a standard Product of the manufacturer.
      b. Indicate illumination levels in accordance with layout and scheduled luminaires indicated on Drawings.

1.04 QUALITY ASSURANCE

A. As specified in Section 260500 – Common Work Results for Electrical.

1.05 DELIVERY, STORAGE, AND HANDLING

A. As specified in Section 260500 – Common Work Results for Electrical.

1.06 MAINTENANCE

A. Section 017704 – Closeout Procedures and Training: Procedures for closeout submittals.
B. Extra Products: At completion of installation, deliver to the USPS Project Manager.
   1. Each component type: Provide quantity for each unique ballast/driver, surge protector and LED array equal to 2 percent of luminaire total, but not less than two of each type.

PART 2 - PRODUCTS

2.01 LUMINAIRE TYPES

A. Type SP1: Lithonia #DSX0LED-PX-40K-80CRI-TXX-XVOLT-SPA.
   1. Description: 14 inch W x 26 inch L x 3 inch H, low profile, rectilinear architectural arm-mounted sharp cut-off, solid state, LED luminaire.
   2. Reflector: Anodized segmented reflectors, beam distribution as required.
   3. Housing: Rugged aluminum rectilinear housing with all seams continuously welded for integrity. Corrosion-resistant polyester powder coat. Finish by the USPS Project Manager.
   4. Ballast/Driver: 38W at 4700 Lumen thru 166W at 18,000+ Lumen. Wattage based on lumen package selected.
   5. Mounting: 20 to 25 feet high square, tapered aluminum pole. (match existing)
   6. Voltage: 277.
   7. Lamp: 4700 Lumen thru 18,000+ Lumen LED array; 4000K, 60,000 hours @ LLD = 0.7.
   8. Quantity of luminaires per pole as shown on the design drawings.
   9. Published Life: 60,000 hours at 70 percent lumen depreciation.
   10. Label: UL listed for wet locations.
   11. Warranty: Full five (5) year factory replacement warranty (internal components).
   12. Alternate Manufacturers:
      a. Current Lighting #VP-MICROSTRIKE-1-160LXXX-4K8-XX-XXX-A.
      b. McGraw-Edison #GLEON-SAXX-740 Series.

2.02 LUMINAIRES

A. Provide luminaires as indicated in luminaire schedule and details on project plans. Provide luminaires complete with light sources of quantity, type and wattage indicated. Provide all luminaires of the same type by the same manufacturer. Luminaires must be specifically designed for use with the driver or ballast and light source provided.
B. LED Luminaires:
   1. Install ballast/drivers, LED arrays and specified accessories at the factory.
   2. Luminaires must have a minimum 5 year manufacturer's warranty.
   3. Luminaires must have a minimum L70 lumen maintenance value of 50,000 hours as calculated by IES TM-21, with data obtained per IES LM-80 requirements.
   4. All luminaires shall be fused. Locate fuses within handhole of pole for pole mounted luminaires.
   5. Voltage: 277.
   6. Provide individual surge protectors within handhole of each pole mounted luminaire. Branch circuit breakers feeding pole mounted luminaires shall also be equipped with surge protection.

2.03 LED DRIVERS

A. NEMA SSL 1, UL 8750. LED drivers must be electronic, UL Class 1, constant-current type and comply with the following requirements:
   1. Output power (watts)and luminous flux (lumens) as shown in luminaire schedule for each luminaire type to meet minimum luminaire efficacy (LE) value provided.

2. Power Factor (PF) greater than or equal to 0.9 over the full dimming range when provided.
3. Current draw Total Harmonic Distortion (THD) of less than 20 percent.
4. Class A sound rating.
5. Operable at input voltage of 120-277 volts at 60 hertz.
6. Minimum 5-year manufacturer's warranty.
7. RoHS compliant.
8. Integral thermal protection that reduces or eliminates the output power if case temperature exceeds a value detrimental to the driver.
9. UL listed for wet locations typical of exterior installations.
10. LED driver shall tolerate sustained open circuit and short circuit output conditions without damage.
11. LED driver shall comply with the requirements of the FCC rules and regulations, Title 47 CFR Part 15 Non-Consumer (Class A).

2.04 LIGHT SOURCES

A. NEMA ANSLG C78.377, NEMA SSL 3. Provide type and wattage as indicated in luminaire schedule on project plans.
B. LED arrays shall have a correlated color temperature (CCT) of 4000K; minimum color rendering index (CRI) value of 70.
C. High power, white light output utilizing phosphor conversion (PC) process or mixed system of colored LEDs, typically red, green and blue (RGB).
D. Provide light source color consistency by utilizing a binning tolerance within a 4 step McAdam ellipse.

2.05 EQUIPMENT IDENTIFICATION

A. Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.
B. Provide labeled luminaires in accordance with UL 1598 requirements. All luminaires must be clearly marked for operation of specific light sources and ballasts or drivers. Note the following light source characteristics in the format "Use Only _____".
C. Correlated color temperature (CCT) and color rendering index (CRI) for all luminaires.
D. All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. Drivers must have clear markings indicating multi-level outputs and indicate proper terminals for the various outputs.

2.06 POLES

A. Manufacturers:
   1. Section 016000 - Product Requirements: Product substitutions permitted by manufacturers listed in Paragraph 2.1A.

B. Material and Finish: Aluminum. Finish by the USPS Project Manager.
C. Section Shape – match existing.
D. Height: - match existing.
E. Base: Nonbreakaway.

F. Accessories:
   1. Handhole.
   2. Anchor bolts.
   3. Base cover.
   4. Bolt covers.
   5. Ground rod and conductor.

G. Approximate Loading Capacity Ratings:
   1. Luminaire Weight: 16 pounds.
   2. Luminaire and Bracket Effective Projected Area: 1 square feet.
   3. Steady Wind: 90 miles per hour minimum, with gust factor of 1.3.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Provide 3000 PSI minimum concrete for lighting poles bases at locations indicated, in accordance with Section 033000 and details shown on drawings.
B. Install poles plumb. Provide double nuts to adjust plumb. Grout around each base and provide bolt covers.
C. Bond luminaires, metal accessories and metal poles to branch circuit equipment grounding conductor. Provide supplementary 3/4 inch x 10 foot copper clad rod with #2/AWG/copper grounding electrode conductor at each pole.

3.02 FIELD QUALITY CONTROL

A. Conform to Section 014000: Quality Requirements.
B. Operate each luminaire after installation and connection. Inspect for improper connections and operation.
C. Measure illumination levels to verify conformance with layout and performance requirements.
D. Take measurements during night sky, without moon or with heavy overcast clouds effectively obscuring moon.

3.03 ADJUSTING

A. Aim and adjust luminaires to provide illumination levels and distribution as directed.

3.04 CLEANING

A. Conform to Section 017300 -Execution: Cleaning and protecting installed work.
B. Clean electrical parts to remove conductive and deleterious materials.
C. Remove dirt and debris from enclosure, pole, and base.
D. Clean photometric control surfaces as recommended by manufacturer.
E. Clean finishes and touch up damage.

3.05 PROTECTION OF FINISHED WORK

A. Conform to Section 017300 – Execution: Protecting installed work.

END OF SECTION 26 56 00

SECTION 27 05 00 - COMMON WORK RESULTS FOR COMMUNICATIONS

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes the following structured cabling system provisions:
   1. Pre-Construction Design Review/Monthly Status Meetings.
   2. Pre-Work Submittals.
   3. Contractor RCDD/Installer requirements.
   4. Labeling.
   5. Post-Work Close-Out Submittals.

B. Related Documents:
   1. The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section.
   2. USPS Structured Cabling System Best Practices, 01 October, 2022.
   3. Cable Termination Schedule - Template.
   4. "Sample" Fiber Backbone Schematic.
   5. Additional requirements and information necessary to complete the Work of this Section may be found in other Documents.

C. Related Sections:
   1. Section 078400 - Fire stopping.
   2. Section 260500 - Common Work Results for Electrical.
   3. Section 260533 - Raceway and Boxes for Electrical Systems.
   4. Section 271100 - Communications Equipment Room Fittings.
   5. Section 271300 - Communications Backbone Cabling.
   6. Section 271500 - Communications Horizontal Cabling.
   7. Section 272133 - Data Communications - Wireless Access Points.
   8. Section 275319 - Distributed Antenna System (DAS).

1.02 REFERENCES

A. Conform to the current Edition of the following documents:
   1. TIA-568.0-*X* - Generic Telecommunications Cabling for Customer Premises.
   2. TIA-568.1-*X* - Commercial Building Telecommunications Infrastructure Standard.
   3. TIA-568-C.*X* - Twisted-Pair Copper Cabling and Components Standard.
   4. TIA-568.3-*X* - Optical Fiber Cabling and Components.
   5. TIA-568-C.*X* - Broadband Coaxial Cabling and Components.
   6. TIA-569 - Telecommunications Pathway and Spaces.
   7. TIA-570 - Residential Telecommunications Infrastructure Standard.
   8. TIA-598 - Fiber Optic Color Codes.
   9. TIA-607-*X* - Generic Telecommunications; Bonding and Grounding (Earthing) for Customer Premises.
   10. TIA-758 - Customer-Owned Outside Plant Telecommunications Infrastructure Standard.
   11. TIA-526-7 - Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant.
   12. TIA-526-14 - Optical Power Loss Measurement of Installed Multimode Fiber Cable Plant.
   13. BICSI Telecommunications Distribution Methods Manual (Latest Edition including all addendums).
   14. IEEE 1100 - Recommended Practice for Powering and Grounding Electronic Equipment (The Emerald Book).

B. National Electrical Manufacturer's Association (NEMA):
   1. NEMA WC 26 - Wire and Cable Packaging (Current Version adopted by the State, per NFPA website).

C. National Fire Protection Association (NFPA):
   1. NFPA 70 - National Electrical Code (Current Version).

D. Regulatory Requirements:
   1. Conform to requirements of Current NFPA 70.
   2. Products: Listed and classified by Underwriter's Laboratories Incorporated as suitable for the purpose specified and indicated.
   3. Perform Work that interfaces with Telephone Utility Company in accordance with Telephone Utility Company rules and regulations.
   4. Conform to current TIA standards and current BICSI TDMM for telecommunications installation.

E. Fire Stopping
   1. Fire stop penetrations of fire-resistive rated assemblies as specified in Section 078400 - Fire Stopping.

1.03 PRE-CONSTRUCTION DESIGN REVIEW/MONTHLY STATUS MEETINGS

A. Pre-Construction Review Meetings:
   1. Convene Issued for Construction (IFC) Review meeting with USPS IT Service Center representative.
   2. Require attendance of parties directly affecting Work of this Section. The USPS telecommunications system representative for Mail Processing Facilities projects will be the Raleigh Information Technology Support Center (RITSC) Subject Matter Expert, Area Maintenance Representative, Local Maintenance Manager, and the District IS Manager or his representative.
   3. Review conditions of operations, procedures, and coordination with related Work.
   4. Agenda:
      a. Tour, inspect, and discuss building conditions relating to communications cabling and equipment.
      b. Coordination with Telephone Utility Company (LEC) and the USPS telecommunications system representative will be by the Raleigh Service Center IT SME through the USPS Project Manager.
      c. Review exact location of each network related item within building construction, casework, and fixtures and their requirements.
      d. Review/Approve required Pre-Work Submittals.
      e. Review Drawings and Specifications.
      f. Review and finalize construction schedule related to voice and data installation, verify availability of materials, personnel, equipment, and facilities needed to complete project and avoid delays.
      g. Review required labeling process, inspections, and testing.
      h. Review cable routing and support.

B. Convene re-occurring Monthly Status Meetings at the construction site with Local Maintenance Manager, Raleigh IT Service Center SME representative and District IS Manager.

1.04 SUBMITTALS

A. Installer qualifications:
   1. Name and cell phone number of full-time BICSI RCDD on staff and copy of RCDD certification. Provide RCDD monthly site visit schedule during the construction period and copies of the reports.

2. Name of full time BICSI TECH on staff and copy of TECH certification for the Lead Installer to be onsite, 5 days a week minimum.
3. Name of full time BICSI Installers (INST1 minimum certified).

B. Product Data: Provide detailed data sheet clearly showing manufacturer Unit Price, Total Price, Model Number, Part Number, color, length, quantity for each material or equipment item specified. Including, but not limited to, backbone fiber, horizontal copper, patch panels, bonding busbars, wire baskets, ladder trays, wire managers (horizontal and vertical), equipment racks, patch cords, fiber interconnect panels, UPS's, rack mounted power strips.

C. Project Documents: The following documentation shall be provided before installation of cabling.
1. Proposed copper cable termination schedules identifying each cable terminated at the copper patch panel ports. Termination schedule shall be provided to USPS IT in printed form on CD-ROM.
2. Proposed fiber backbone schematic showing a visual representation of the fiber port connections for each distribution center within the CCR, TR's and TE's. Schematic shall be provided to USPS IT in a USPS compatible version of AutoCAD.
3. Proposed cable routings shall be provided with cable plant depicted on floor plans. The drawings must identify location of all T/Os (Telecommunications Outlets), TR/TE's (Telecommunication Rooms / Enclosures), Consolidated Computer Room (CCR) and any other installed component of the cabling solution. Show actual routing of the cable bundles (pathways) and backbone cables on the floor plans. Drawings shall be provided to USPS IT in a USPS compatible version of AutoCAD.
   a. Labeling shall conform to the USPS labeling guidelines. All 48 port Copper Patch Panels in the CCR, TR's or TE's shall be labeled 1 thru the end port number. For any questions, contact RITSC SME for clarification.
   b. A detailed cable termination record will be provided in sufficient detail, so that:
      1) Telephone Utility Company or telephone interconnect company can install cross connects.
      2) Postal Service users can install and maintain patch cords at patch panel fields.
      3) The location and size of the service entrance conduits are known.

1.05 INSTALLER QUALIFICATIONS

A. Installer: Minimum of one BICSI certified Technician on the job site at all times with documented formal training in the installation of Category 6, Category 6A and fiber optic cabling systems. 50% of onsite installers shall possess a certification for a total systems solution being installed from the manufacturer of the cabling and terminating hardware. The contractor must present these certifications to the Raleigh IT SME before beginning work.

B. Installer Company: Full time BICSI RCDD with current credentialing on staff. Company specializing in the installation of Category 6, Category 6A, and single-mode fiber optic Structured Cabling Systems with a minimum of 5 years documented experience. Installation certification - 50% of Low voltage installers must be trained by the manufacturer and currently certified to install manufactures product line of copper/fiber wiring. Provide current installer certifications before doing any copper or fiber installations. This certification is part of the 15 year warranty.

C. Lead Installer: Minimum of BICSI Technician Certification.

1.06 LABELING

A. Furnish and install machine generated labels.

B. Patch Panels, CP1 Enclosures, and Outlet Faceplates: Display outlet or cable identification number in uppercase lettering on permanent machine generated adhesive label stock. Each individual port requires a port number label. The faceplate cannot be labeled as a range.
C. Label the Consolidated Computer Room as CCR and TR/TE's as 1-01, 1-02, 1-03, etc. Example: 3rd TE for mail processing on the first floor: TE 1-03.
D. Label all copper patch panel ports in a horizontal fashion left to right in numerical sequence. Example: If there are seven 48 port copper patch panels in a rack, the ports are numbered consecutively from port 1 all the way through 336.
E. Label all (TE) Workroom Floor locations with a minimum of 18 inch high letters, contrasting color, block letters mounted on all three sides of the TE. Black letters for the metal sides and orange for the glass door is acceptable. Example: TE 1-02 on the three visible sides in contrasting colors.
F. Label all (WRF) Workroom Floor CP-1 NEMA-12 housing units with a minimum of six inch, black, block letters affixed to the outside of the enclosure. Label shall include the port numbers at the TE/TR/CCR.
   1. The total cable length to the TE, TR or CCR shall be permanently labeled on the inside cover of the enclosure.

G. All CP-1 type terminations will be labeled alphabetically. Example: First CP-1 would be labeled CP-1A, the second CP-1 would be labeled CP-1B, the third CP-1C, etc.

H. CP-1s will have the following Patch Panel Port assignments. Example: (CP-1A Ports 1-12), (CP-1B Ports 13-24) (CP-1C Ports 25-36), (CP-1D Ports 37-48), (CP-1E Ports 49-60) * Note CP-1E begins on the second patch panel numerically numbered ports 49 - 96.
I. Label Copper Patch Panel ports in the order the cables were terminated beginning with all CP-1 type terminations, followed then by T/O terminations in the order of six-plexes, quads and triplexes.
J. Label telecommunications outlet faceplate and CP-1 location in the same manner as the patch panel.
K. Display cable identification number in black uppercase lettering on machine generated permanent adhesive self-laminating label of contrasting color from cable sheath.
L. Place labels on each end of cable, maximum 6 inches from cable termination.

1.07 CLOSEOUT SUBMITTALS

A. Test Reports: Submit to the Raleigh Service Center IT SME through the USPS Project Manager from Testing Laboratory.

B. Prepare reports in conformance with Section 014000 - Quality Requirements.
C. Provide end-to-end tests.
D. Certificates: Manufacturer's certificate that Products meet or exceed specified requirements.
E. Section 017704 - Closeout Procedures and Training: Procedures for closeout submittals. Deliver prior to Final Acceptance.

F. Comprehensive test results for Category 6, Category 6A and fiber optic certification of cable plant per specifications of TIA/EIA-568-C, and all addendums. Immediately following new Category 6/6A copper and single-mode fiber installation, submit raw test results via e-mail to the Raleigh IT Service Center representative who will be performing copper and fiber site acceptance. All testing must be performed using an industry standard compliant test device. Test results must be furnished in format used by testing device. Vender generated spreadsheets or PDF's are not acceptable. Paper test results are not acceptable. There is a USPS 10MB attachment limit. USPS cannot access file sharing sites.

G. Project Record Documents: Accurately record the following as-built documentation:
   1. Final copper cable termination schedules, in printed form on CD-ROM.
   2. Final fiber backbone schematic showing a visual representation of the fiber port connections for each distribution center within the CCR, TR's and TE's. Schematic shall be provided to USPS IT in a USPS compatible version of AutoCAD.
   3. Cable routings (as-built drawings) shall be provided with cable plant depicted on floor plans prior to acceptance. The drawings must identify location of all T/Os (Telecommunications Outlets), TR/TE's (Telecommunication Rooms / Enclosures), Consolidated Computer Room (CCR) and any other installed component of the cabling solution. Show actual routing of the cable bundles (pathways) and backbone cables on the floor plans. Provide master overall set plus one set for each TR/TE which will detail T/O's and CP's served by that TR/TE. As-built drawings shall be provided to USPS IT electronically in a USPS compatible version of AutoCAD on a CD-ROM.

H. Operations and Maintenance Data: Data including wiring diagrams, parts lists, shop drawings, product data, manufacturer's instructions for cables and equipment and certifications identified above shall be provided.

I. Total Systems Solution Warranty: Minimum 15 year warranty from both manufacturer of cabling as well as connecting hardware when installed together according to predetermined manufacturers' specifications. Installer shall possess certifications from manufacturers of the components installed as a total systems solution and must present said certifications to the contracting officer through the USPS Project Manager in advance of beginning the Work.

PART 2 - PRODUCTS

2.01 CONDUITS, BOXES AND CABLE TRAYS

A. Specified in Section 260533 - Raceway and Boxes for Electrical Systems.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Verification of Conditions: Verify that field measurements, surfaces, substrates, and conditions are as required, and ready to receive Work.

3.02 INSTALLATION GUIDELINES

A. Special requirements for cable routing and installation:
   1. The majority of the structured cabling system wiring in this building will be installed above ceilings without conduit. All cabling used throughout this project shall comply with the requirements as outlined in the National Electric Code (NEC) article 725. All cabling shall bare CMP and/or appropriate markings for the environment in which they are installed. All interior cables shall be CMP rated.
   2. Where cables pass through partitions, walls and floors, provide properly bushed conduit sleeves to allow passage of the cabling.

3. Sealing of openings for cable pass through between floors and fire rated walls and partitions shall be provided in accordance with section 078400 - Firestopping.
    a. The USPS Project Manager shall be the final authority in approving firestopping materials and methods.

B. Support cables installed in ceiling spaces with cable tray and/or wide-base J-hooks Category 6/6A compliant suspension devices, anchored to building ceiling structural steel (red iron).
1. Maximum spacing between supports: Non-continuous supports shall be installed at intervals not exceeding 5 feet.
2. Maximum Number of 4 Pair Cables per support: 25.
3. Furnish and install additional supports as required.
4. Install complete cable support device system before starting installation of cable.
    a. Installation of cable before completion of support system not permitted.
    b. Unsupported cable not permitted.
5. Organize and group cables. Install cable group as single run through ceiling spaces following column and building lines. Do not install cable group runs diagonally across center of building.
6. Install armored fiber optic cabling in cable tray or approved support solution.
7. Cabling will not be suspended from any electrical conduits, HVAC ducts, sprinkler systems, gas, or water pipes, etc.
8. Cabling will not be attached to suspended ceiling grid system.
9. Cabling system shall be installed in approved suspension devices for telecommunications cabling.
10. Vertical runs of backbone and horizontal cabling (e.g.: cabling exiting thru-wall penetrations) shall be equipped with factory manufactured cable drop out fittings and kellums cord grips to properly support the cabling at the vertical bends.

C. Cable trays shall be required for areas of heavy cable concentration including but not limited to CCRs, TR/TE's, exposed workroom areas and large administrative hallways.
1. Maximum spacing between each cable tray support: Specified by manufacturer of cable tray.
2. Maximum number of cables supported by cable tray: Specified by manufacturer of cable tray not-to-exceed 40% fill ratio. Install additional cable tray or wire basket as needed to maintain 40% Maximum Fill Ratio.
3. Install complete cable tray system before starting installation of cable.
    a. Installation of cable before completion of tray system not permitted.
    b. Cabling shall not be bundled within cable tray.
    c. Provide factory manufactured cable drop-out fittings for transportation of cabling entering or exiting the cable tray.
4. Cable/Ladder trays, wire mesh tray or solid bottom cable tray shall be provided as specified in USPS MPF specification section 260533, paragraph 2.12.
    a. Provide equipment "drop off" fittings for cord drops and factory sweep corners for all changes in direction.

D. Cabling routed underground, or exterior of the building, or through inaccessible ceilings shall be contained in conduit. Provide flush boxes within finished areas and surface mounted, cast aluminum, "FD" factory boxes in unfinished areas. Provide 3/4 inch conduit risers with 90 degree bend and bushing for all T/O's.
1. Conduit/EMT, cable tray or wire basket shall be used in the ceiling of the workroom floor where a suspended ceiling system is not present. Cabling within exposed workroom areas, not routed within cable trays, must be contained within conduit raceways (3/4 inch minimum).
2. All conduit stubs must have a plastic bushing/collar installed at each end.
3. All conduit runs require an accessible pull-string in each conduit.

4. Interior conduits shall be a minimum of 3/4 inch in diameter. Conduits shall adhere to the 40 per cent fill ratio.
5. No conduit is to be buried directly in the slab.
6. There shall be no more than 180 degrees of bend in a conduit longer than 30 feet. All conduits that are comprised of more than two (2) ninety degree bends or a reverse bend shall have a properly installed pull box. Pull boxes shall be 12 x 12 x 6 inches for up to 1-inch EMT, 18 x18 x 8 inches for up to 1-1/2 inch EMT. Ninety degree bends in fiber runs shall be installed using dual forty-five degree bends.
7. Under no circumstances shall a pull box be used to change direction of a conduit. All conduits shall be installed in a manner so that cabling passes directly through the pull box without changing direction.
8. Underground service and interbuilding conduits shall be a minimum of four 4-inch diameter, buried minimum of 36 inch BFG, equipped with heavy wall rigid galvanized steel conduit elbows and risers and marked with red magnetic warning tape, refer to Module 1, 5-2.7.2. Conduits shall adhere to the 40 per cent fill ratio and each 4 inch conduit shall be provided with 3-cell mesh, fabric innerduct equipped with individual pull strings.
   a. Basis of Design: MaxCell "Edge"-Flexible, 3-cell, fabric innerduct.

E. Route cable for T/O (telecommunications outlets) as follows:
1. Wall Mounted: Through ceiling spaces to conduit stub-ups or junction boxes. Include drag lines.
2. Furniture System Cable Raceway: Point of entry to outlet.
3. Floor Outlet Box: Through under floor conduit to box requires prior approval from USPS.
4. Column Mounted-Workroom Floor: Through surface mount conduit stubs to junction box or cable tray.
5. Consolidated Computer Room: Along ladder rack from rack to locations to be run in ladder tray / basket tray.

F. Communications cabling and fixtures shall have minimum separation from the following devices:
1. Non-Shielded Electrical Cables: 12 inches.
2. Fluorescent Light Fixtures: 12 inches.

G. Cross electrical cables with communications cables at 90 degrees only. Data cables shall not run parallel with electrical cables, unless separated by 12 inch minimum.

H. Comply with cable manufacturers minimum bend radius requirements. For Category 6/6A, minimum bend radius shall be no less than 4 times diameter of outer sheath of cable. For Fiber optic cabling, minimum bend radius shall be no less than 10 times diameter of outer sheath of cable.
1. Do not stretch, stress, tightly coil, bend or crimp cables.
2. Replace cables that are severely stressed during installation at no additional cost to USPS.
3. Any armored cable that has had its armor sheathing broken shall be replaced in its entirety, end to end at no additional cost to USPS.

I. Cabling installed in plenum or non-plenum air returns.
1. All interior cabling installed shall be CMP plenum rated.

J. Cable Run Lengths: Route cables so that cable run length does not exceed recommended maximum distance.
1. UTP cabling from the back of the patch panel to the Telecommunications Outlet (T/O) is limited to a maximum total run of 295 feet and 230 feet for CP-1 Consolidation Points on the work room floor. Minimum cable run length between back of patch panel and type 1, consolidation points shall be 49 feet per TIA-568-C
2. Cable conductors shall be continuous ("homerun") from originating termination equipment to destination termination equipment.

K. Cables: Furnish and install communications cables as specified, in accordance with cable termination schedules, manufacturer's published instructions, TIA-568-C including all addendums and as indicated on Drawings.
   1. Dress cable to final location, remove sheath to point allowing splaying of conductor, and terminate. Make each termination uniform and precise. Hook and loop cable straps shall be used for bundling and dressing all cabling on manufacturer supplied Strain Relief Bars. No nylon zip ties shall be used for cable bundling or attachment. No wire managers will be used/substituted for Strain Relief Bars.
   2. Maintain sheath integrity. Remove minimum amount of sheath required for termination up to a maximum of 1/2-inch.
   3. Maintain manufacturer's twisting of wire pairs to termination point. Do not attempt to restore, modify, or add to manufacturer's twisting of cable. Do not untwist more than 1/2-inch of the stripped cable.
   4. Label each end with a machine generated, self-laminating label.
   5. Mechanical couplers or splices not permitted in fiber or copper cabling.
   6. A fiber optic service loop of sheathed fiber no less than 20 feet at each end of a fiber optic cable shall be installed at each termination point in/on the wire basket/ladder tray. All service loops shall be installed so that the minimum bend radius (10 times the outside diameter of the fiber) is maintained. All service loops shall be installed outside of the fiber optic termination housing. Once the fiber reaches the entrance point of the fiber optic enclosure, there shall be no less than 3 feet of unsheathed fiber installed neatly in the fiber optic storage tray prior to terminations being installed. Unsheathed fiber shall be installed in the storage tray per the fiber optic enclosures manufacturer's instructions.
   7. When installing Armored Fiber Optic cabling, proper telecom bonding techniques to bond the metallic member of the Armored Fiber Optic Cable must be maintained. Armored fiber will be bonded on the CCR/TR/TE end only to the "SBB".

END OF SECTION 27 05 00

SECTION 27 11 00 - COMMUNICATIONS EQUIPMENT ROOM FITTINGS

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes the following structured cabling system components:
   1. Table of Contents
      a. Open Relay/Equipment Racks for CCR/TR's
      b. Category 6 / Category 6A (Wireless) 110 Style Copper Patch Panels
      c. Wire Management Panels
      d. PBB for TEF/CCR
      e. SBB for TR/TE's
   2. Consolidated Computer Room (CCR).
   3. Telecommunications Room (TR).
   4. Telecommunications Enclosure (TE).

B. Related Documents:
   1. Specified in Section 270500 - Common Work Results for Communications.
   2. USPS Structured Cabling System Best Practices, 01 October, 2022.

1.02 REFERENCES

A. Specified in Section 270500 - Common Work Results for Communications.

1.03 SUBMITTALS

A. Specified in Section 270500 - Common Work Results for Communications.

1.04 QUALITY ASSURANCE

A. USPS to approve all Rack Elevations before Issued for Construction (IFC) drawings are distributed.

B. Specified in Section 270500 - Common Work Results for Communications.

PART 2 - PRODUCTS

2.01 OPEN EQUIPMENT / RELAY RACKS WITH VERTICAL WIRE MANAGERS FOR CCR/TR'S

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Chatsworth Products, Inc.
   2. CommScope Uniprise.
   3. Great Lakes Case & Cabinet Co., Inc.
   4. Hoffman.
   5. Ortronics (Legrand) Corporation.
   6. Panduit.
   7. Rittal.
   8. Tripp-Lite by Eaton.
   9. Product options and substitutions. Substitutions: Not permitted.

B. Constructed of aluminum extrusion framework. Dimensions: 84 inch high x 3 inch deep x 19 inch wide. Double sided, 10/24 tapped holes with universal EIA rack unit spacing. Black or Aluminum finish.

1. Each equipment rack shall have two double depth vertical cable managers: Dimensions: No less than: 8 inch x 8 inch x 78 11/16 inch for the front side of the relay rack and no less than 8 inch x 8 inch x 78 11/16 inch for the back side of the relay rack. Black or aluminum finish. Attach to sides of relay racks. Must be able to cover and conceal patch cabling. Each end rack will have outside double depth vertical wire managers attached to each outside end.
2. Each equipment rack shall be connected to the overhead cable tray/wire basket system for added rigidity. Equipment racks shall be properly supported to avoid wobbling.
3. Vertical and horizontal wire managers shall be equipped with opaque covers to completely conceal the patch cords.
4. Provide four post, open equipment racks within the CCR and two post style, open equipment racks within the TRs.

2.02 CATEGORY 6/6A 8-PIN MODULAR (IDC) "110" STYLE PATCH PANELS FOR CCR/TR/TE'S

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. CommScope Uniprise.
   2. Hubbell, Inc.
   3. Ortronics (Legrand).
   4. Panduit.
   5. Product options and substitutions. Substitutions: Not permitted.

B. 48-port/24-port/12-port, Category 6/6A Copper Patch Panels:
   1. Rack mounted 8-pin modular, Category 6/6A, non-keyed.
   2. Complies with ANSI/TIA/EIA-568-C "T568A" pinning configuration.
   3. Install manufacturer supplied strain relief bar assemblies for every 12, 24 and 48 port rear copper terminations. Secure Cat6/6A cable with hook and loop cable straps. Plastic tie wraps are not acceptable.

2.03 WIRE MANAGEMENT PANELS

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Panduit: Preferred.
   2. Chatsworth Products.
   3. CommScope Uniprise.
   4. Leviton.
   5. Ortronics (Legrand).
   6. Product options and substitutions. Substitutions: Not permitted.

B. Cable Management Panels: Rack mounted horizontally and vertically. See the latest "USPS Structured Cabling System Best Practices" for guidelines on rack layouts.
   1. Horizontal management panel for use at top of each CCR equipment rack shall be two separate 2RU panels along the top of each equipment rack.
   2. Horizontal management panels for use at top of TR equipment racks shall be two 2RU panels along the top of each equipment rack.
   3. Each vertical wire management panel will be at least 8 x 8 inches deep on the front side and at least 8 x 8 inches deep on the back side of the equipment rack to form a Full Height Double-depth Vertical Wire Management system. No exceptions.

2.04 PRIMARY BONDING BUSBAR - PBB FOR TEF/CCR (REFER TO TIA-607-D)

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Harger. P/N GBI/14424 TMGB.

2. Chatsworth - P/N 40158-024.
3. Legrand.
4. Product options and substitutions. Substitutions: Not Permitted.

B. Provide one PBB at the Telecom Entrance Facility (TEF), below ceiling acoustic tile, with all bonding leads clearly labeled by machine labeler. All bonding leads shall be 2 hole compression lugs. This PBB shall be bonded to the building grounding electrode system using minimum #1/0/AWG/CU conductor. Size according to number/size of Telecom Bonding Backbone (TBB) leads being attached to the PBB. Minimum size will be 4 inch H x 0.25 inch W x 24 inch L. The PBB shall be mounted as close as possible to the building grounding electrode system busbar to keep the Telecom Bonding Conductor (TBC) as straight and as short as possible.

C. Each 2-lug compression connector shall have antioxidant coating applied to lug and busbar prior to attachment.

D. The maximum value of resistance between any point in the Telecommunications bonding system and the building electrical grounding electrode system shall be 100 Milliohms or .1 ohm to comply with TIA-607-D standard. This resistance value shall be tested and certified, in the presence of the Raleigh IT SME, by an independent third party testing agency, prior to applying power to any telecommunications equipment. Test meter shall be Micro-Ohmmeter Model 6240 manufactured by AEMC Instruments or approved equal.

2.05 SECONDARY BONDING BUSBAR - SBB FOR CCR, TR/TE'S (REFER TO TIA-607-D)

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
1. Harger - P/N GBI/14212 TMGB (TR/TE's), P/N GBI/14424 TMGB (CCR).
2. Chatsworth - P/N CPI 13622.
3. Legrand - P/N OR-GB2X12TGB.
4. Product options and substitutions. Substitutions: Not Permitted.

B. Provide one SBB in the CCR and in every TR/TE, below ceiling acoustic tile (or within each TE), with all bonding leads clearly labeled by machine labeler. All bonding leads shall be 2 hole compression lugs. This SBB will connect to the PBB using minimum #1/0/AWG/CU via Telecom Bonding Backbone (TBB). Size according to number/size of bonding leads being attached to SBB. Minimum size will be 2 inch H x 0.25 inch W x 12 inch L. Provide 4 inch H x 0.25 inch W x 24 inch L secondary bonding busbar within the CCR.

C. Each 2-lug compression connector shall have antioxidant coating applied to lug and busbar prior to attachment.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Consolidated Computer Room (CCR):
1. Do not use the CCR as a TR/TE. If utilized provide an additional, entirely separate wire basket or ladder tray to support all of the Cat6/6A copper for the TR/TE needs. Do not install Cat6/6A copper cabling inside wire basket used for any CCR equipment. Keep the two systems completely separate. A maximum fill ratio of 40% will be adhered to. No exceptions.
2. Furnish, install, and bond, floor mounted, four post, 84 inch high x 3 inch deep x 19 inch wide relay racks shoulder-to-shoulder separated by 8 inch wide, double depth, full height, vertical wire managers perpendicular to wall housing plywood backboards.

a. Mount relay racks in a "side by side" fashion with one double-depth vertical wire management channel between each rack, and one double-depth wire management channel on outside side rail of both end racks.
   1) Place two 2RU horizontal wire managers at the top of each rack.
   2) Supply four (4) 1RU wire managers within each rack for USPS use. These (4) 1RU wire managers are in addition to the 1RU wire managers placed between the patch panels and the 2RU wire managers at the top of each rack.
   3) Supply (500) hook and loop cable straps for USPS use; Hellerman Tyton: P/N GT 50X60C2.
b. The 12 inch wide ladder tray/basket tray shall be mounted to top of relay racks and extend to plywood backboard or other ladder rack/basket tray for each relay rack installed. This ladder rack serves as additional support for relay racks as well as cable routing from relay rack to backboard.
c. Each rack will receive a separate #6 AWG bonding wire home run to the Secondary Bonding Busbar (SBB) in the CCR.
d. Each rack shall be equipped with a factory manufactured power strip equipped with 12 NEMA5-20R receptacles. Preferred rack mounted power strip: Tripp-Lite #RS-1215-20.
e. Each rack shall be provided with an installation kit and isolation pads for securing and isolating the rack to and from the floor.

3. Provide a minimum of two 4 x 8 foot plywood backboards along walls behind and perpendicular to CCR racks and directly below the basket or cable tray. Provide quadraplex power receptacle and quad T/O each mounted 6 feet AFF and center within each sheet of plywood. Additional 4 x 8 foot plywood sheets may be required by USPS.
   a. Plywood: 48 inch x 96 inch x 3/4-inch A/C rated (A = smooth side; C = slight blemishes against wall), fire rated, void-free, smooth side out. Absolutely no knot holes or voids shall be visible on outer face of plywood.
   b. Install plywood with long dimension in vertical orientation with bottom of sheet 8 inches AFF.
   c. Field paint with white or gray enamel fire resistant paint prior to installation of equipment.
   d. Provide an industry approved Secondary Bonding Busbar (SBB) and attach minimum #6 AWG bonding conductors using 2 hole compression type fittings for all bonding needs within the CCR. All bonding cable connections shall be clearly labeled on the SBB indicating where the connection is coming from/going to via machine made labels. All metallic components of CCR shall be bonded to the installed Secondary Bonding Busbar (SBB). Interconnect the SBB to the PBB utilizing minimum ##1/0/AWG/CU bonding conductor.
4. Install 12 inch wide ladder rack/basket tray with 2 inch side bars the entire width of plywood back boards at 7 feet 6 inches to 8 feet AFF (Racks are 84 inches high).
   a. Furnish and install 12 inch wide ladder rack/basket tray with 2 inch side bars at 7 feet 6 inches to 8 feet AFF between plywood backboards and relay racks (racks are 84 inches high). All sections of ladder rack and or basket tray shall be joined with manufacturer approved devices. No sections of ladder rack or basket tray shall be zip tied together. All sections of ladder rack and/or basket tray will be bonded. All wall connections will be made with factory wall mounts. No homemade connectors are permitted.
   b. Provide 2 factory manufactured cable "drop out" fittings at each rack within the "CCR".

5. Provide a minimum of two, four-post, floor mounted, Postal Furnished Equipment (PFE) racks for LAN, WAN electronic components. PFE racks shall be enclosed type with mesh louvered sides and plexiglass hinged door. USPS to specify equipment rack locations. Supply four 1-RU wire managers per rack in addition to the two 2-RU wire manager at the top of each rack and the 1 RU wire managers between the copper patch panels.
6. Install Category 6 48-port patch panels in relay rack(s) that the 4-pair cables serving only the CCR are to be terminated. It is recommended that Telecommunications Outlets within the CCR be installed using the ladder rack/basket tray system. USPS IT will specify final termination points. Power for each MPE cabinet is to be served using a drop cord and 20A, 120 volt connector. The drop cords are to be extended down from outlet boxes secured to the ladder rack/basket tray. Provide one quad receptacle secured to the wire basket/ladder tray behind each USPS PFE rack.
7. All metallic ladder tray, basket tray, equipment racks or enclosures shall be bonded using a #6 AWG stranded bond wire with green insulation using 2 hole compression type fittings approved for basket tray installation. All painted surfaces shall be fully burnished for paint removal to achieve maximum bonding connection. Provide all UL documentation on how the support system should be bonded to form a system.
8. All bonding in CCR shall be made at the Secondary Bonding Busbar (SBB) installed by Contractor. This SBB shall be wall mounted below the acoustic ceiling if one is installed and shall not be installed on the plywood backboards. All bonding wires will be on two lug compression fittings with full machine-made labeling clearly showing where the bond originates.
9. Contractor shall provide enough 10/24 mounting screws for 32 connections per equipment rack in CCR, TR or TE rack for the installation of USPS PFE active electronic components. Example: If 8 new relay racks are installed, provide 256 10/24 pitch screws.
10. The floor surface of the "CCR" shall be covered with static control resilient flooring. A sealed concrete floor is not acceptable. Refer to specification section 096536 for grounding provisions.

B. Telecommunications Room (TR):
1. Provide appropriate number of 4-pair Category 6 UTP cables from each office area and workroom column mounted T/O (telecommunications outlets) to Telecommunications Rooms as indicated on drawings. Terminate 12 each 4-pair Category 6 UTP cables from each Type 1 Consolidation Point to Telecommunications Room as indicated on drawings.
2. Provide one 4-pair Category 6A UTP cable from each Wireless Access Point (WAP) to Telecommunications Enclosures/Rooms as indicated on drawings to 12 or 24 port Category 6A copper patch panels.
3. Provide a minimum 20 foot service loop in a figure eight coil, in the ceiling/wire basket for all copper cables terminated in TE/TR's.
4. Provide and bond, floor mounted, 84 inch high x 3 inch deep x 19 inch wide relay racks shoulder-to-shoulder, separated by double-depth vertical wire managers, perpendicular to wall housing plywood backboards with double-depth vertical wire managers on each outer end of equipment racks.
   a. Racks will be used to house fiber/copper wiring and PFE.
   b. Allow a minimum 16 empty rack units per rack for PFE data equipment.
   c. Provide one factory manufactured cable "drop out" fitting at each rack within the "TR".
   d. Each rack shall be equipped with a rack mounted power strip equipped with 12 NEMA5-20R receptacles. Mount power strip below last 48 port copper patch panel. Preferred: Tripp-Lite #RS-1215-20.
   e. Provide a minimum of 2 racks within each TR.
   f. Each rack shall be provided with an installation kit and isolation pads for securing and isolating the rack to and from the floor.
5. Provide one 2RU rack mounted wire manager at top of rack.

6. Provide one rack mounted, 24 strand fiber optic interconnect, center below 2RU wire manager.
7.
8. Provide needed 48-port Copper Patch Panels separated by 1RU Wire Managers.
9. Provide the following within the Telecommunications Room:
    a. Each rack shall be equipped with separate #6 AWG bond conductor homerun to the Secondary Bonding Busbar (SBB) in that TR.
    b. Provide an industry approved copper Secondary Bonding Busbar (SBB) and attach minimum #1/0 AWG bond conductor from this SBB to the "PBB" in the TEF using the Telecom Bonding Backbone (TBB) and 2 hole compression type fittings. All bonding cable connections shall be clearly labeled on the busbar indicating where the connection is coming from/going to via machine made labels. All metallic components of the "TR" shall be bonded to the installed SBB inside that TR.
10. Provide a minimum of one 3KVA (120V - input/output) uninterruptible rack mountable power supply with 30 minute battery reserve in each TR. Mount on the lowest RU of the right-most open relay rack and ensure power plug is wired as NEMA 5-30P, 3 wire.
11. Contractor shall provide enough 10/24 screws for 32 connections per rack for the installation of USPS PFE active electronic components. Example: If 3 new relay racks are installed, provide 96 10/24 pitch screws.

C. Telecommunications Enclosure (TE) for Column Mounted Applications
1. Constructed of steel or aluminum with Safety Glass or Plexiglas front doors. Cabinet must be rated NEMA-12, designed for front and rear access, have forced fan with filtration for intake and exhaust ventilation, and adjustable vertical mounting rails. The intake fan with filter will be mounted 5 inches from the top center of the rear metal door and a filtered louvered exhaust vent will be mounted 4 inches from the bottom center of the rear door. The fan must be oriented to blow filtered air into the cabinet to create a positive pressure within the cabinet and will be plugged into the power strip (min. 12 outlet NEMA 5-20R with power switch guard) mounted inside of the cabinet; Basis of Design: Tripp-Lite #RS-1215-20. Dimensions: 86 inches high x 36 inches deep x 24 inches wide with 19 inch EIA rack width. The Telecommunications Enclosure and the installation of the enclosure must comply with area seismic zone rating.
    a. Provide appropriate number of conduit risers equipped with 90 degree bends and bushings/collars for incoming backbone and outgoing horizontal cables. Conduit risers shall be minimum 2 inch diameter and sized for 40 percent fill and shall be sealed to maintain positive air pressure within the TE. Provide one spare 3 inch conduit riser sealed with threaded cap for future cabling.
    b. Allocate 16 RU's within the rack for PFE and provide a second separate NEMA 12 cabinet if less than 16 RU's for PFE is available in first TE. Two units can be "ganged/married" together (inner side panels removed and doors adjusted to open from center) to form one TE for workroom floor applications. The doors shall be adjusted to open in opposite directions from the center. Cabinets shall be bolted together in such a manner to maintain the NEMA-12 compliance, and the inner side panels will be removed from each unit creating an open pathway between cabinets. All copper connections will be placed in the left-most cabinet and the fiber along with the 1.5 kVA UPS will be installed in the right-most cabinet.

2. Each NEMA 12 TE Cabinet shall be attached to Work Room Floor Column via double-nutted 5/8-inch. All-Thread/Uni-Strut or a metal platform welded to the column. The bottom of each NEMA-12 cabinet shall be 9 feet AFF. Install a Uni-Strut/All-Thread/Platform designed to support a minimum of 1000 lbs., firmly connected to the column and or the overhead building support structure that meets all codes and seismic requirements. Provide a structural engineer approved platform solution during the Design Review Process and before the Issued For Construction drawings are distributed. There shall not be any installed Bollards blocking or interfering any Verti-Lift access into either the front or rear of the NEMA-12 cabinet TE. The mounted TE shall not sway or swing in any manner.
    a. TE's shall not be mounted on the Workroom floor. TE's shall be column mounted. No exceptions.
    b. The front rails of the TE shall be positioned to avoid the patch cords from contacting with the front plexiglass door.
3. Provide appropriate number of 4-pair Category 6 UTP cables from each office area and workroom T/O (telecommunications outlet) to Telecommunications Enclosure as indicated on drawings. Terminate 12 each 4-pair Category 6 UTP cables from each Type 1 Consolidation Point to Telecommunications Enclosure as indicated on drawings.
4. Provide one rack/cabinet mounted, 1RU, 24 strand fiber optic interconnect, center at top of cabinet. Fiber ports will be laid out ports 1-12, left to right on 1RU only. Duplex ports to be mounted vertically: Ports 1-12; left to right.
5. Provide one 4-pair Category 6A UTP cable from each Wireless Access Point (WAP) to Telecommunication Enclosure as indicated on drawings to Cat6A 24 port Copper Patch Panel, the bottom-most Copper Patch Panel.
6. Provide a minimum 10 foot service loop for all copper cables. Service Loop shall be placed inside TE in between the side TE panel and mounting railing with no obstruction to PFE.
7. All metallic basket tray, equipment racks or enclosures shall be bonded and grounded using a #6 AWG stranded bond wire with green insulation using 2-hole compression type fittings or bond fitting approved for basket tray installation. All painted surfaces shall be fully burnished for paint removal to achieve maximum bonding connection.
8. Provide an industry approved Secondary Bonding Busbar (SBB) and attach minimum #1/0 AWG bond conductor from this SBB to the PBB in the TEF using the Telecom Bonding Backbone (TBB) and 2 hole compression type fittings. All bonding cable connections shall be clearly labeled on the busbar indicating where the connection is coming from/going to via machine made labels. All metallic components of the TE shall be bonded to the installed SBB inside that TE.
9. Provide a minimum 1.5kVA uninterruptible power supply with 30 minute battery reserve rack mounted at the backrail within each TE.
10. Install conduit/EMT through top of cabinet to allow cable entry as needed. Seal openings with an intumescent fire-stop putty when all cables have been installed. Spray foam sealants not permitted.
11. Contractor shall provide enough 10/24 screws or screws/square cage nuts for 32 connections per rack for the installation of USPS PFE active electronic components. Example: If 2 new relay racks are installed, provide 64 10/24 pitch screws.
12. Maximum horizontal cabling distances shall not exceed 230 feet for CP-1 Consolidation Points. This horizontal distance is to include all vertical distances plus required service loops. Total workroom floor coverage is required when designing the TE placements throughout the workroom floor environment.
13. Maximum horizontal cabling distances for telecommunication outlets shall not exceed 295 feet, including vertical distances and required service loops.
14. CCTV or ISIP equipment components of any kind shall not be mounted within a TE.

D. Patch Panels: Install 12-port, 24-port and 48-port, 8-pin module Category 6/6A patch panels at main cross-connect and horizontal cross-connect for termination of cables installed as part of Work of this Section.
   1. Install patch panels inside wall mounted data enclosure for retrofit column mounted TE applications. Install patch panels on floor mounted 19 inch wide by 84 inch high open relay racks at CCR and TR room locations, only.
   2. Provide wire management panel (1RU) on rack or cabinet mounting rails above and below each patch panel for all locations.
   3. Provide 4 additional 1RU wire managers to be used in between PFE.
   4. Provide manufacturers strain relief bars sufficient to maintain UTP bend radius at rear of panels.
   5. Terminate all 4 pairs of each horizontal 4 pair cable to each 8 pin ("T568A") patch panel port.
   6. Consolidation Point One (CP-1) cabling will terminate starting on the first Category 6 copper patch panel followed by the T/O cabling in consecutive order.

END OF SECTION 27 11 00

SECTION 27 13 00 - COMMUNICATIONS BACKBONE CABLING

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes the following structured cabling system components:
   1. Fiber Optic Cabling.
   2. Termination equipment.
   3. Patching equipment.
   4. Fiber Optic Testing.

B. Related Documents:
   1. Specified in Section 270500 - Common Work Results for Communications.

C. Related Sections:
   1. Specified in Section 270500 - Common Work Results for Communications

1.02 REFERENCES

A. Specified in Section 270500 - Common Work Results for Communications.

1.03 SYSTEM DESCRIPTION

A. Specified in Section 270500 - Common Work Results for Communications.

1.04 SUBMITTALS

A. Specified in Section 270500 - Common Work Results for Communications.

1.05 QUALITY ASSURANCE

A. Specified in Section 270500 - Common Work Results for Communications.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Section 016000 - Product Requirements: Transport, handle, store, and protect Products.

B. Deliver in accordance with NEMA WC 26.

PART 2 - PRODUCTS

2.01 CONDUITS, BOXES AND CABLE TRAYS

A. Specified in Section 260533 - Raceway and Boxes for Electrical Systems.

2.02 OS2 ARMORED BACKBONE

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Belden.
   2. Berk-Tek.
   3. CommScope Uniprise.
   4. Corning Cable Systems.
   5. General Cable.
   6. Leviton.
   7. Optical Cable Corp.
   8. Ortronics (Legrand).
   9. Superior Essex
   10. Product options and substitutions. Substitutions: Substitutions: NOT permitted.

B. Conductors: 24 / 48 strands
   1. Terminate fiber strands onto "LC" ports, vertically mount, ports 1 through 12, left to right.
   2. Fiber strands are required to be installed on Fiber Optic Interconnection panels based on the project requirements.
   3. The same port layout orientation must be preserved on the far end strand terminations. All ports must be installed vertically. No horizontal orientation allowed.
   4. All individual Armored Fiber runs are required to be bonded on the CCR end only, connected to the SBB in the CCR and clearly labeled with machine labels.
   5. All backbone fiber strands shall be installed using reverse-pair positioning which allows the use of A-B fiber patch cords. Refer to ANSI-TIA-568.3.D, Annex C.
   6. Provide individually insulated plenum rated strands under common plenum rated sheath.
   7. Complies with individual characteristics established in TIA-568-C including all addendums for fiber optic cable performance specification.
   8. All underground fiber cable shall be indoor/outdoor rated. Loose tube fiber cable, if utilized, shall be equipped with furcation kits.

2.03 FIBER OPTIC RACK MOUNT INTERCONNECT CENTER

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. CommScope Uniprise.
   2. Corning Cable Systems.
   3. Ortronics (Legrand).
   4. Panduit.
   5. Product options and substitutions. Substitutions: Not permitted.

B. Enclosure connector and adapter panels:
   1. LC type connectors.
   2. Rack mount enclosures shall be one RU, two RU or four RU style coupler panels based on the project requirements.
   3. Complies with TIA-568-C specification.

2.04 OS2 FIBER OPTIC PATCH CORDS: 2 STRAND, TIGHT BUFFERED

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Belden.
   2. Berk-Tek.
   3. CommScope Uniprise.
   4. Corning Cable Systems.
   5. General Cable.
   6. Leviton.
   7. Optical Cable Corp.
   8. Ortronics (Legrand).
   9. Superior Essex.
   10. Product options and substitutions. Substitutions: Not permitted.
   11. Fiber patch cord manufacturer shall be the same manufacturer furnishing the backbone fiber. Mixing of manufacturers is not acceptable.

B. Fiber optic duplex patch cords.
   1. USPS to specify connector type and length for patch cords based on the total number of fiber ports being installed. Connectors shall be LC/LC.
   2. Complies with individual characteristics established in TIA-568-C including all addendums for fiber optic patch cable performance specification.
   3. Patch cords shall be factory made and factory tested individually.

4. Provide fiber patch cords for 75 percent of the total fiber ports installed. Example: 50 Duplex fiber ports (100 strands) installed, provide 75 Duplex fiber patch cords. All fiber patch cord colors, lengths and quantities shall be determined by USPS.
5. Fiber optic patch cord connector types, lengths and quantities shall be specified by USPS prior to procurement.
6. Match performance characteristics of installed fiber optic backbone.

2.05 TYPE 1 FIBER OPTIC POINT (FP-1) FOR WORKROOM FLOOR ENCLOSURES (TYPICAL FOR FSS SITES)

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Hoffman Enclosures, Inc.
   2. Rittal Corporation (AE, or EB Series enclosures).
   3. Product options and substitutions. Substitutions: Not permitted.

B. Enclosures:
   1. Metal, no plastic allowed. Minimum 12 inches high x 12 inches wide x 8 inches deep sheet steel NEMA-12 enclosure with hinged, lockable door with rubber gasket, mounted at 14 feet AFF. Alternate size: 14 inches high x 16 inches wide x 6 inches deep.
   2. Door must be oriented so that it opens in a horizontal manner. Enclosure may not be mounted in a manner so that the door opens downward.

C. Connectors and adapter plates:
   1. LC type connectors.
   2. 3 port coupler panel with LC adapters for each FP.
   3. 6 Strands x Number of FPs = size of enclosure needed, and amount of LC adapters for CCR.
   4. Rack mount enclosure is 1.75 inches high. (1 Rack Unit).
   5. Complies with TIA-568-C specification.

PART 3 - EXECUTION

3.01 INSTALLATION - COMPONENTS

A. Type 1 Fiber Optic Point: Furnish and install 6 port fiber optic enclosures (FP-1) at each location identified on Electrical Drawings using NEMA 12 rated enclosures. Mount fiber optic points with uni-strut or equivalent fastening devices to steel ceiling structure (red iron) so that box is vertical and installed between 20 to 23 feet above the finished floor. Door must be oriented so that it opens in a horizontal manner. Enclosure may not be mounted in a manner so that the door opens downward. Contractor must ensure all minimum bend radius specifications for cables can be achieved inside the enclosures. All terminating components shall match the requirements/performance specifications of the cable specified. Install cables from Fiber Optic Points to CCR in wire basket tray, support loops or J-Hooks with a maximum span of 4 feet.

3.02 FIBER OPTIC TESTING

A. OS2 Single-Mode Fiber Optic Cable Testing
   1. Fluke testers are the only allowed fiber tester manufacturer. Tester must be Encircled Flux Compliant.
   2. Test Reference Cords (TRC's) must be used. Test Reference Cord verification must be shown in the final test result submission.
   3. Tier 1, Tier Method B (one jumper) and Tier 2 OTDR testing is required. The Tier 2 OTDR requires bi-directional testing.
   4. The installer shall Set a Reference based on Method B (Single Jumper) which includes both mated connector losses and the loss of the link under test.

5. The installer shall perform Tier 1 Testing with Optical Loss Test Set (OLTS) that includes testing for length.
6. The installer shall perform Tier 2 testing with OTDR to show all splices.
7. The supplier shall perform Bi-directional testing on all installed fiber optic cabling. Supplier test equipment shall perform testing of fiber in accordance with the fiber type being tested, TIA-526-7 for Single mode fiber.
8. The fiber testers and test heads shall have passed calibration within one year of actual test date. Any calibration in excess of one year is not acceptable. Each test set and fiber head must have the recent calibration paper printout from the calibration lab for inspection by USPS, prior to testing. The calibration printout must show actual serial numbers of test sets (main and remote and each fiber tested).
    a. The current calibration for the main and remote fiber units MUST be supplied to USPS prior to any testing.
    b. USPS RITSC representative will determine test labeling format inside the fiber tester prior to actual testing. The Main Unit must be in the CCR.
    Example for fiber strand test: CCR to TE 1-06 14 (for strand 14). All fiber strands will be tested bi-directionally. Any fiber test results that only show testing in one direction will be rejected.
9. Single mode fiber optic cable shall be tested bi-directionally at 1310 nm and 1550 nm.
10. Cable tester test parameter shall be set to correct values for:
    a. Actual manufacturer of fiber being installed. Tester to be specific to the manufacturer's model of fiber cable being tested; tester cannot be a generic fiber type.
    b. Index of Refraction based on manufacturer specifications for cable type being tested.
    c. Quantity of adapters: Test Method B, one Jumper, 2 adapters.
    d. Fiber Type.
    e. Test to Tier 1 as mandated by TIA-568-C.4.
    f. Preferred tester is Fluke Versiv series with Encircled Flux.
11. The Low Voltage Installer shall provide all Fiber tests in one, single file. No multiple files will be accepted.
12. The Supplier shall review test settings with the USPS technical representative. Supplier shall have cable specifications on site for USPS technical review to verify settings are correct on test equipment.
13. Fiber optic cables shall pass all attenuation tests referenced to formulas presented in the listed standards.
14. Perform end-to-end tests of each fiber optic backbone cable as follows (applies to CCR/TR/TE applications only):
    a. Tier 1 Test: Light Source Power meter tests per TIA-568-C specification.
    b. Optical Time Domain Reflectometer (OTDR) tests per TIA-568-C specification including all addendums.
    c. Both the Tier 1 test and the Tier 2 OTDR test results must be uploaded to the "Link Ware Live" cloud based repository for USPS RITSC access.
    d. Performing one test and not the other does not satisfy a complete fiber test. Both tests must be submitted in one file, all at the same time.
    e. Measured effective cable run length.
15. Optical photographs of each fiber end shall be submitted for documentation and warranty purposes.

3.03 CONSTRUCTION

A. Specified in Section 270500 - Common Work Results for Communications.

3.04 FIELD QUALITY CONTROL

A. Specified in 270500 - Common Work Results for Communications.

END OF SECTION 27 13 00

SECTION 27 15 00 - COMMUNICATIONS HORIZONTAL CABLING

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes the following structured cabling system components:
   1. Category 6/6A copper communication cable.
   2. Termination equipment.
   3. Patching equipment.
   4. Category 6/6A copper testing.

B. Related Documents:
   1. Specified in Section 270500 - Common Work Results for Communications.

C. Related Sections:
   1. Specified in Section 270500 - Common Work Results for Communications.

1.02 REFERENCES

A. Specified in Section 270500 - Common Work Results for Communications.

1.03 SYSTEM DESCRIPTION

A. Specified in Section 270500 - Common Work Results for Communications.

1.04 SUBMITTALS

A. Specified in Section 270500 - Common Work Results for Communications.

1.05 QUALITY ASSURANCE

A. Specified in Section 270500 - Common Work Results for Communications.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Section 016000 - Product Requirements: Transport, handle, store, and protect Products.

B. Deliver in accordance with NEMA WC 26.

PART 2 - PRODUCTS

2.01 CATEGORY 6/6A (CATEGORY 6A IS FOR WIRELESS USE ONLY) HORIZONTAL CABLING

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Belden.
   2. Berk-Tek.
   3. CommScope Uniprise.
   4. General Cable.
   5. Leviton.
   6. Ortronics (Legrand).
   7. Panduit.
   8. Hitachi (Drybit).
   9. Product options and substitutions. Substitutions: Not permitted.

B. Conductors: 4 twisted pair, minimum #23 AWG, solid copper.

1. Individually insulated plenum rated conductors under common plenum rated sheath unless entire cable is installed within conduit/EMT or if area where cable is installed is not considered a return air plenum according to any applicable codes.
2. Complies with individual characteristics established in TIA-568-C, and all addendums for Category 6/6A cable performance specification.
3. Nominal Impedance: 100 ohms plus or minus 15 percent.
4. Category 6 cabling shall be "blue" in color and certified as capable of performing to a minimum of 250 MHz.
5. Category 6A cabling shall be "white" in color and certified as capable of performing to a minimum of 500 MHz.

2.02 CATEGORY 6/6A / (CATEGORY 6A WIRELESS USE ONLY), COPPER PATCH CORDS

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Belden.
   2. Berk-Tek.
   3. CommScope Uniprise.
   4. General Cable.
   5. Hitachi (Drybit).
   6. Leviton.
   7. Ortronics (Legrand).
   8. Panduit.
   9. Product options and substitutions. Substitutions: Not permitted.

B. Conductors: Straight through type 4 twisted pair minimum #23 AWG, stranded copper.
   1. Terminated with male 8-pin modular plugs.
   2. Complies with individual characteristics established in TIA-568-C, and all addendums for Category 6/6A cable performance specification.
   3. Nominal Impedance: 100 ohms plus or minus 15 per cent.
   4. Match performance and impedance characteristics of the installed horizontal unshielded twisted pair cable.
   5. Contractor shall provide Category 6/6A copper patch cord for 75 percent of the total copper ports installed. Example: 1000 copper ports installed, provide 750 Category 6/6A copper patch cords. Contractor shall provide manufacturer terminated patch cables. Category 6 patch cord lengths shall be determined by Raleigh IT Service Center SME.
   6. Each patch cord shall have a plastic arch for ease of removal of the connector. Preferred copper patch cord type: Ortronics (Legrand) non-booted.
   7. Patch cords shall be factory made, tested and certified.
   8. All Category 6A patch cords shall be "white" in color and certified as capable of performing to a minimum of 500 MHz. All WAP Category 6A patch cords will be 3 feet on the WAP end.
   9. All category 6 patch cords shall be "blue" in color and certified as capable of performing to a minimum of 250 MHz.

C. Connector:
   1. 8-pin modular, Category 6/6A, non-keyed.
   2. Complies with TIA-568-C "T568A" pinning configuration.
   3. Color: Clear.

2.03 CATEGORY-6 12 PORT MODULAR SURFACE-MOUNTED "110"-STYLE PATCH PANELS CONSOLIDATION POINTS (CP-1). TYPE 1

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. CommScope Uniprise.

2. Ortronics (Legrand).
3. Panduit.
4. Product options and substitutions. Substitutions: Not permitted.

B. Boxes/Panels:
1. Capable of terminating 12 Category 6 cables.
2. Equipped with an 89D surface mounting bracket.
3. Complies with TIA-568-C "T568A" pinning configuration.

C. Connector:
1. 8-pin modular, Category 6, non-keyed.
2. Complies with TIA-568-C "T568A" pinning configuration.
3. Color: Selected by the USPS IT Project Manager.
4. Attached to backboard of CP-1 with 89-D type bracket.

D. Housing
1. Wall or raceway mounted outlet enclosure, CP-1.
2. Able to contain 12 modular 8-pin connectors
3. Installation over single gang junction box, double gang junction box, or raceway knockout as indicated on Drawings.
4. Color: Selected by the USPS IT Project Manager.

2.04 TYPE 1 CONSOLIDATION POINT (CP-1) FOR WORKROOM FLOOR ENCLOSURES

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
1. Hoffman Enclosures, Inc.
2. Rittal Corporation: (AE, or EB Series enclosures).
3. Product options and substitutions. Substitutions: Not permitted.

B. Enclosures:
1. Must be metal, no plastic allowed. Minimum 12 inches high x 12 inches wide x 8 inches deep sheet steel NEMA-12 enclosure with hinged, lockable door with rubber gasket, mounted at 14 feet AFF. Alternate size: 14 inches high x 16 inches wide x 6 inches deep.
2. Door must be oriented so that it opens in a horizontal manner. Enclosure may not be mounted in a manner so that the door opens downward.
3. The total cable length to the TE, TR or CCR shall be permanently labeled on the inside cover.

2.05 OUTLET FACEPLATES/MOUNTING FRAMES

A. Wall mounted, or raceway mounted outlet faceplates or mounting frames, suitable for the following:
1. Mounting required number of 8-pin modular connectors.
2. Use with approved 8-pin modular connectors.
3. Installation over single gang junction box, double gang junction box, or raceway knockout as indicated on Drawings.

B. Color: White with Machine manufactured permanent labeling with Black lettering.

2.06 CONDUITS, BOXES AND CABLE TRAYS

A. Specified in Section 260533 - Raceway and Boxes for Electrical Systems.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Specified in Section 270500 - Common Work Results for Communications.

3.02 INSTALLATION

A. Cables: Provide communications cables as specified, in accordance with Cable Termination Schedules, manufacturer's published instructions, TIA-568-C including all addendums and as indicated on Drawings.
   1. Dress cable to final location, remove sheath to point allowing splaying of conductors, and terminate. Make each termination uniform and precise. Hook and loop cable straps shall be used for bundling and dressing all cabling. No nylon zip ties shall be used for cable bundling or attachment.
   2. Maintain manufacturer's twisting of wire pairs to termination point. Do not attempt to restore, modify, or add to manufacturer's twisting of cable. Do not untwist more than 1/2 inch of the stripped cable.
   3. Label each end with a machine generated, self laminating label.
   4. Mechanical couplers or splices not permitted in copper or fiber cabling.
   5. Cable conductors shall be continuous from originating termination equipment to destination termination equipment.

B. Telecommunications Outlet: Provide appropriate number of female 8-pin modular jack connectors on one face plate at each T/O (telecommunications outlet) as indicated on Drawings.
   1. Install faceplate over duplex outlet box, double duplex outlet box, or raceway knockout, level and in alignment with adjacent faceplates.
   2. Except where entire cable run is in conduit/EMT, provide a minimum 20-foot service loop in the ceiling at the end of the conduit/EMT riser before the cable enters the outlet box.
   3. Coordinate color with Raleigh IT Service Center POC.

3.03 CAT-6/6A COPPER TESTING

A. Section 014000 - Quality Requirements: Field testing and inspection.

B. Testing and Certification Overview
   1. The Contractor shall provide Fluke Copper equipment and materials for the testing of all installed copper transmission media. For Category 6 copper, the supplier shall employ Level III compliant test equipment that stores the test results in internal memory and produces test result reports. For Category 6A, the supplier shall employ Level IV compliant test equipment that stores the test results in internal memory and produces test result reports. The supplier shall provide the USPS, test results in test equipment format (raw electronic). Supplier prepared spread sheets and PDF files are not acceptable.
      a. The USPS technical representative may conduct random tests of copper cable with USPS test equipment as part of the final inspection. Contractor to re-terminate and retest any cable found to be defective.
      b. Provide all equipment and services necessary to secure and provide the USPS a system warranty. Inspect installation of cables and equipment during and at completion of installation.
      c. Test results indicating "Pass*(Star)" or "Fail" shall not be accepted and must be repaired/retested with second set of test results submitted to USPS.
      d. Test results must be uploaded to the "Link Ware Live" cloud based repository for USPS RITSC access.

C. Copper Cable Testing
   1. Test parameters include, but are not limited to:
      a. Wire Map
      b. Length
      c. Propagation Delay
      d. Delay Skew
      e. DC Loop Resistance
      f. Insertion Loss (Attenuation)

g. Return Loss (RL), RL @ Remote
h. NEXT, NEXT @ Remote
i. Attenuation-to-crosstalk Ratio (ACR-N), ACR-N @ Remote
j. ACR-F (ELFEXT), ACR-F @ Remote
k. Power Sum ACR-F (ELFEXT), PS ACR-F @ Remote
l. Power Sum NEXT, PS NEXT @ Remote
m. Power Sum ACR-N, PS ACR-N @ Remote
n. Power Sum Alien Near End Xtalk (PS ANEXT)
o. Power Sum Alien Attenuation Xtalk Ratio Far End (PS AACR-F)
p. Alien Cross-talk

2. Cable test parameters shall be set to the manufacturer's values for NVP and Test Limit (TIA-568-C, Category 6/6A, Permanent Link). If the NVP is not set correctly, test results will be rejected.
3. Perform end-to-end tests of each 4-pair cable as follows:
   a. Pair/conductor for proper pinouts and continuity.
   b. Ground fault.
   c. Proper termination, shorts, and crossed pairs.
   d. Channel attenuation per TIA-568-C, including all addendums.
   e. Channel bi-directional worst case near end cross talk (NEXT) at frequencies up to 250 MHz (category 6) or 500 MHz (category 6A), per TIA-568-C, including all addendums.
   f. Measured effective cable run length.

3.04 INSTALLATION COMPONENTS

A. Specified in Section 270500 - Common Work Results for Communications.

3.05 CONSTRUCTION

A. Specified in Section 270500 - Common Work Results for Communications.

3.06 FIELD QUALITY CONTROL

A. Specified in Section 270500 - Common Work Results for Communications.

END OF SECTION 27 15 00

SECTION 28 13 04 - ENTERPRISE PHYSICAL ACCESS CONTROL SYSTEM (ePACS)

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes specifications for an integrated security management system which shall perform the following general service:
   1. Access control.
   2. Alarm monitoring.
   3. Reporting functions.
   4. Security management functions.
   5. Photo-ID badge issuing.

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other documents including:
   1. ePACS Standard System Configuration - revised March 2017.
   2. Access Control (ePACS) SOP - revised October 2019.

C. Related Sections:
   1. Section 081100 - Metal Doors and Frames.
   2. Section 087100 - Door Hardware.
   3. Section 111415 - Turnstiles.
   4. Section 260500 - Common Work Results for Electrical.
   5. Section 264128 - Surge Protective Devices (SPD).
   6. Section 270500 - Common Work Results for Communications.
   7. Section 281524 - IP Video Intercom and Exterior Gate Control System.
   8. Section 281600 - Intrusion Detection System.
   9. Section 282305 - Integrated Security and Investigative Platform (ISIP) CCTV System.
   10. Section 283100 - Fire Emergency Voice/Alarm Communication System (EVACS).

1.02 SYSTEM DESCRIPTION

A. Enterprise Physical Access Control System:
   1. Access management system (System) shall monitor and control access to areas defined herein.
   2. The system will utilize proximity cards as its primary access device, but will support ISO 14443 Contactless smartcard technology, such as Mifare, and (keypad) technology at each door. It shall also support alarm inputs and control outputs.
   3. System shall consist of computers (servers/workstations), stand-alone microprocessor based controllers, card readers and/or keypads and host software.
   4. The microprocessor based controllers will be capable of controlling 16 card reader inputs and 16 door outputs. It will also be able to monitor a minimum of 92 alarm points, storing a minimum of 5000 events before downloading to the central computer. It will be able to store a minimum of 10,000 cardholders.
   5. System shall be capable of operating in a distributed processing environment with or without host connectivity.
   6. Specific types of devices and their functions shall be addressed in relevant sections.
   7. The system shall support an integrated electronic photo identification (photo-ID) system.
   8. System will utilize an ODBC compliant database, such that it can share or retrieve information from a local database.

9. System shall be able to compare its list with the information from the USPS database and flag discrepancies of listed individuals in either database and have the ability to generate a report listing the discrepancies and records.
10. System will share its database with the electronic photo-ID system to eliminate redundant input of data to the databases for common data fields.
11. The operating system shall be USPS Windows applicable version (ACE standard operating system).
12. The system shall support true multi-user, multi-tasking with a minimum of 3 workstations.
13. The system shall include capability for remote access for off-site support and/or management workstations. Systems that connect to the network must provide remote access via the Postal Service business partner VPN connectivity. Dial-up phone connectivity is not permitted when systems are connected to the network.
14. The system shall utilize standard GUI interface allowing day-to-day operations to be performed using a standard mouse. All graphics shall be dynamic color alarm graphic maps (user definable) created with graphic drawing programs, not vector files. All device names shall be user programmable (minimum of 32 characters, full English).
15. The complete operator instruction manual shall be imbedded in the on-line help and shall be readily accessible using standard "Index," "Help Topics," "Keyword" and "Search" requests.
16. The client shall have the ability to define events for viewing in any one of multiple event viewer screens or any combination of screens. Events shall also be designated for printing to selectable printers.
17. Provide multiple levels of password protected system access with encryption. All passwords will use one-way encryption.
18. Provide operator with configurable reporting of event history and cardholder activity by authorized request only.
19. Provide reports for: Inputs (all or in groups), outputs (all or in groups), alarm messages, instructions, event action, card transaction history, field devices and panel reports, alarm history, alarm suppression.
20. Report generation shall allow for reports to be filtered by time and date as well as by device name, event category and definition and by card holder categories or individual record(s).

B. Description of work:
1. Include all necessary labor, tools, equipment, and ancillary materials required to furnish and install a complete and operational access control and alarm monitoring system.
2. Enterprise Physical Access Control System will manage access to the following building and selected areas using encoded cards and/or coded Keypads.
   a. Employee entrances/exits.
   b. Access to administrative space.
   c. Registry Cage.
   d. Stamp Depository.
   e. Vehicular access (employee and USPS maneuvering area).
3. The extent of Enterprise Physical Access Control System work is defined to include, but not by way of limitation:
   a. ePACS Controllers.
   b. Reader Interface Modules.
   c. Card reading sensors.
   d. Input monitoring modules.
   e. Output relay modules.
   f. Wiring, power supplies, switches, and ancillary equipment.
4. Requirements are indicated elsewhere in these specifications for work including, but not limited to, raceways and electrical boxes and fittings required for installation of control equipment and wiring, not the work of this section.

5. The power supplies and micro-switches controlling the egress electric locks at each of the turnstiles and the RE-4 personnel door located at the employee entry shall be de-energized upon activation of a fire alarm emergency or manual operation of the emergency evacuation pushbutton. Turnstiles shall immediately operate in the "free spin" mode (egress direction only). Inbound entry direction shall remain secure. Coordinate all requirements with Section 111415 - Turnstiles.

1.03 REFERENCES

A. NEC: All electrical wiring work shall comply with the latest edition of the NEC.

B. NEMA**:** Electrical equipment shall comply with applicable portions of NEMA.

C. FCC: All assemblies shall be in compliance with FCC emission standards.
   1. Proximity/Contactless Smartcard Card Reading Sensors: Part 15, Subpart F (field disturbance sensors).
   2. Dial-up modems: Part 68.

D. UL-1012 and CSA: All power supplies shall be in compliance with Underwriters Laboratories standard 1012 and CSA standards for power supplies.
   1. UL-294: The system shall comply with Underwriters Laboratories standard 294 for Enterprise Physical Access Control Systems.

1.04 SUBMITTALS

A. Product Data: Submit for prior approval, Manufacturer's data on Enterprise Physical Access Control System and components, including manufacturer's model numbers, catalog data sheets, power requirements, dimensions, layouts, installation details, single line riser diagram.

B. Shop Drawings: Submit dimensioned drawings of Enterprise Physical Access Control System and accessories including controllers, proximity card reading sensors, keypads, power supplies, switches, and ancillary equipment. Submit separate layout drawings of each terminal cabinet, equipment rack, control panel, interpanel and intrapanel wiring, power supplies, terminal strips, including labeling of all components, point-to-point wiring, and calculations for UPS power. Provide 1/8-inch scale floor plans showing locations of all devices.
   1. Submit dimensioned and scaled elevation drawings for each ePACS terminal cabinet showing the location of the reader interface modules, associated reader power supplies, terminal strips, surge protectors, receptacles and other ePACS components. Elevation drawings shall be submitted and approved prior to ordering the terminal cabinets.

C. Security Riser Diagram: Shall detail the number and location of controllers, reader interface modules, power supplies, indicate all cabling and wiring, host equipment. Riser diagrams shall be submitted to the USPS Project Manager for review and concurrence prior to execution.

D. Operator's Manual: Submit for prior approval, Manufacturer's manual for programming and operating the system and its related components.

E. Submit evidence of training from the manufacturer of the system proposed for installation. Evidence shall include written certificates of training or similar documentation on manufacturer's letterhead demonstrating the installer's qualifications.

1.05 QUALITY ASSURANCE

A. Manufacturer: Manufacturer of products defined in this section must have:
   1. Industry experience: Company must have at least 5 years experience in manufacturing and servicing integrated access control and alarm monitoring systems.

B. Contractor:
   1. Furnish all labor, services, including a Systems Integrator, and materials necessary to furnish and install a complete, functional enterprise physical access control system (ePACS). The System shall comply in all respects with the requirements of the specifications, manufacturer's recommendations, and Underwriters Laboratories Inc. (ULI) listings.
   2. Furnish certification that the entire system has been inspected and tested, is installed entirely in accordance with the applicable codes, standards, manufacturer's recommendations and ULI listings, and is in proper working order.
   3. The USPS requires professional workmanship from an experienced "systems" installer and will reject any faulty workmanship or installation methods not meeting their satisfaction.

C. Systems Integrator:
   1. Company with a minimum of 5 years system design, engineering supervision, and installation experience in the alarm, building automation, or Access Control industry.
   2. The Integrator shall obtain a Sensitive Clearance from the USPS. This clearance will be coordinated by the USPS Project Manager. Use the following email for assistance in obtaining this clearance: pacs-support@usps.gov
      a. An interim clearance will be issued to allow the Integrator to request an ACE login from the USPS Project Manager.
      b. It will take a minimum of two weeks to obtain an interim clearance.
   3. Company that is trained and authorized to install manufacturer products. The ePACS wiring shall be installed by a Systems Integrator trained and authorized to install and wire the manufactured products.
   4. Company that has been successfully installing systems of equal size and complexity for a minimum of 5 years. Submit a minimum of 3 references. System references shall include projects where software and hardware installed is similar to the software and hardware proposed for this project.
   5. The Systems Integrator shall include all necessary labor, tools, equipment, and ancillary materials required to furnish and install a complete and operational access control and alarm monitoring system.
   6. The extent of Enterprise Physical Access Control System work is defined to include, but not be limited to:
      a. Installation of and testing of system including controllers, reader interface modules, proximity/contactless smartcard card readers, keypads, input modules and output modules, software and photo-ID badge issuing system equipment.
      b. Wiring, power supplies, switches, and ancillary equipment.
      c. Programming of system, including creation/translation of database with USPS input, and access levels.
      d. Operator Training for using and programming the system for up to 6 operators and 2 shift supervisors, provide in 2 sessions of 8 hours each. Provide 2 additional 8-hour training sessions 3 months after acceptance. Provide 2 sessions of 4 hours each for separate training for photo-ID badge production operators.
      e. Submitting procedures for installing system on USPS networks and performing cut-over and acceptance testing on the system. Coordinate procedures with USPS Information Technology to ensure no interference with USPS network or systems.
      f. Provide two 8-hour maintenance training sessions.

D. System Checkout:
   1. Burn-in: 1,000 hours at normal operating conditions or equivalency.
   2. On-site testing: Manufacturer trained and authorized Systems Integrator shall functionally test each component in the system after installation to verify proper operation and confirm that the panel wiring and addressing conform to the wiring documentation.

3. Service facility: Systems Integrator shall have service facilities within 4 hours travel time of the installation. Any increase in this time shall be approved by the USPS Project Manager.

1.06 WARRANTY

A. System Components: 12 months from date of acceptance.
   1. Systems Integrator shall provide 24 hour emergency service for all reported system operational failures during such 12-month warranty period. The system must be fully operational within 48 hours. Include all necessary maintenance for the entire integrated system for the 12-month warranty period. On-site service response shall be 4 hours of the initial request for service and shall be provided 24 hours a day,7 days a week inclusive of all holidays.
   2. Service requests shall be reported via phone call to a designated service toll free phone number provided by the Systems Integrator.

PART 2 - PRODUCTS

2.01 USPS-SUPPLIED PRODUCTS

A. The following items are Not In Contract but are supplied by the local USPS facility for incorporation into the Work by the Contractor. Contact USPS via email at pacs-support@usps.gov for assistance.
   1. Proximity access cards/badges.
   2. ACE standard server/workstations and software.
   3. Photo-ID badge printer.
   4. Digital badge camera, backdrop, consumables, and peripherals.

2.02 MANUFACTURERS

A. Enterprise Physical Access Control System Controller:
   1. Contract transfer to Vanderbilt Industries (sole source provider for controller).
      a. Contact the following;
         1) Patrick Shadood, USPS Account Manager, 2 Cranberry Road, Parsippany NJ 07054, office 973-316-3910; mobile 908-432-8806; fax 973-334-4850; PatrickShadood@vanderbiltindustries.com
         2) The Contractor is required to inform the manufacturer that the controller is for a USPS project.
   2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Not Permitted.

B. All other Enterprise Physical Access Control System peripheral components:
   1. Subject to compliance with project requirements, manufacturer offering Products which may be incorporated in the Work, including the following:
      a. Alarm Controls Corporation (800) 645-5538.
      b. Allegion/Schlage (877) 671-7011.
      c. Altronix Corporation (888) 258-7669.
      d. Bosch Security Systems, Inc. (800) 289-0096.
      e. Ditek Corporation (800) 753-2345.
      f. GE Security (800) 428-2733.
      g. George Risk Ind./GRI (800) 523-1227.
      h. HES Innovations (800) 626-7590.
      i. HID Corporation (800) 237-7769.
      j. Hirsch Electronics Corporation/Identiv, Santa Ana, CA (888) 809-8880.
      k. Honeywell Security (800)323-4576.

l. Lenel Systems International (866) 788-5095.
m. Potter Electric Signal Co. (866) 240-1870.
n. Safety Technology International (STI) (800) 888-4784.
o. Software House (800) 507-6268.
p. Vanderbilt Industries: contact G. Patrick Shadood; Office - (973) 316-3910; Mobile - (908) 432-8806.
q. XCeedID Corporation (877) 671-7011.

2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

2.03 MATERIALS AND COMPONENTS

A. Enterprise Physical Access Control System Controller:
1. The Enterprise Physical Access Control System shall include microprocessor based controllers by Vanderbilt Industries #VRCNX-A.
2. The Contractor is required to inform the manufacturer that the controller is for a USPS project.
3. The controller shall be ordered in a NEMA 1, metal enclosure for wall mounting and include integrated battery backup. The controller shall be equipped with keyed door latch.

B. Controllers: provide complete hardware to operate with the following features:
1. The controllers shall support a minimum of 16 card reading sensors and shall be capable of supporting additional input and output modules. Vanderbilt Industries #VRCNX-A.
2. Database: Database shall store all user operating data and handle event reporting for all possible attached devices, and shall contain memory capacity for the following:
   a. Minimum of 10,000 card holder records.
3. Event activity: System shall designate activity as an alarm or non-alarm condition, dependent upon modules installed, and shall report activity for:
   a. Supervised monitor points: 92 minimum.
   b. Outputs: 46 minimum.
4. Relay outputs: System shall initiate relay output commands based on:
   a. Card Access Activity.
   b. Operator Keyboard Inputs.
   c. Pre-programmed Time Periods.
   d. Input activation.
5. System diagnostics:
   a. Automatic system diagnostics and automatic alarming based on detected faults in the controllers, card readers, wiring, and expansion modules. At a minimum, diagnostics shall include faults, card reader errors, input change of state, expansion module faults, host communications, power monitoring and reader communications errors. If a problem is detected, it shall be reported to the host (when communications is restored).
   b. Each time the controller is powered, the panel shall go through an automatic diagnostic cycle. If a problem is detected, it shall be reported to the host. Diagnostics cycle shall include indications for fault, reader error, card swipe, monitor point change of state, host communication, card reader communication, program watchdog and power.
6. Transaction buffer: 5,000 transactions, minimum.
7. Flash memory for real time program updates from the host and/or locally connected computer.
8. Communication: Primary communications shall support TCP/IP protocols for Ethernet using the USPS structured wiring system via an on-board Ethernet port. In addition, the controllers shall have an on-board RS-232 port for local connection and emergency dial-up communications.
9. Tamper Switch: enclosure shall include a SPDT tamper switch wired at the factory.
10. UL-294 rated.

11. Power:
    a. The controllers shall operate on 12 - 24VDC, powered from an external, regulated power supply with battery backup. The controllers shall provide necessary power to all card readers and expansion modules.
    b. Memory Retention: The controllers shall maintain configuration and card holder information for up to 72 hours when operating power is disconnected from the controllers.

C. Reader Interface Module (RIM):
    1. Each card reader sensor shall be interfaced with a dedicated, single reader interface module connected to the controller via RS-485 protocol, Vanderbilt "VRINX". Interface module shall have the following features:
        a. The reader interface module shall support multiple reader technologies including, but not limited to:
            1) Smart Card
            2) Magnetic Stripe (swipe or insertion)
            3) Wiegand (swipe or insertion)
            4) Proximity
            5) Biometric
            6) Barcode
        b. Input/Output configurations:
            1) Inputs: 4 supervised or non-supervised.
            2) Outputs: 2 form "C" SP/DT 1A relays.
        c. RS-485 communication: Up to 4,000 feet.
        d. Power:
            1) Voltage: 14 - 24 VDC predicted upon read head used.
            2) Current: 120 mA @ 24 VDC (without read heads).
        e. Operating temperature: 32 degrees to 120 degrees F.
        f. Operating relative humidity: 10 - 90 percent non-condensing.
        g. The reader interface module shall be mounted within NEMA 1, metal enclosure with keyed door latch.

D. Input/Output Expansion Board:
    1. Input/Output expansion boards shall be utilized to provide additional input and output responses in excess of the available input/output configurations at the reader interface modules. Expansion boards shall also be utilized to supervise multiple door contacts, exit alarms, etc. in lieu of using reader interface modules.
    2. Input/Output expansion board shall be connected to the controller via RS-485 protocol, Vanderbilt #VIONX-8.
    3. The input/output expansion board shall support universal triggers which integrate any input with any or all output responses and shall have the following features:
        a. 16Kb flush memory and 1Kb RAM
        b. Two serial ports (RS232 or RS485)
        c. 8 supervised or unsupervised contact inputs
        d. Contacts can be defined as alarms, door status, egress, or other environmental conditions
        e. 8 Form "C" SP/DT mechanically latching 1 A relay outputs
        f. NEMA 1, metal enclosure with hinged door and keyed door latch
        g. BAA compliant.

E. Card Reading Sensor:
    1. General:
        a. Card Reader (CR) shall read proximity cards and send signal to Controller for processing. The CR shall be compatible with:

1) 125 KHz proximity, such as HID Corp 1000, capable of direct image printing (PVC overlay for direct image printing is acceptable).
2)

b. Reader shall be listed in the FIPS 201 Evaluation Program Approved Product list http://fips201ep.cio.gov/apl.php
c. CR shall comply with the Standards for Facility Accessibility by the Physically Handicapped (USPS Handbook RE-4).
d. CR shall have the means to be electrically isolated to prevent short circuits from disrupting other communications in the data line network.

2. Capacities:
   a. CR shall read digital proximity cards signals to a minimum distance of *2* inches and contactless smartcard to a minimum distance of 1.5 inches (5.08mm) and does not require contact with the sensor.
3. Long Range Proximity Card Readers (LRCR) to be provided at:
   a. High-speed rollup doors.
   b. Automatic impact doors.
   c. Inbound Truck Maneuvering Area Gates (a LRCR on top due to mirror on high vehicles and a LRCR mounted low for cars).
   d. Employee parking area gates.
   e. These LRCR's are exempt from the requirements of FIPS201.
   f. Note that long range card readers require individual power supplies and batteries that must be served with 120 volt power. The power supplies shall be mounted within NEMA 1, metal enclosure equipped with keyed door latch.
4. Specifications: Material shall be Polycarbonate UL94, and shall be UV resistant, sealed, water and weather resistant, and tamperproof.
5. Environmental:
   a. Humidity: 0 percent to 100 percent condensing.
   b. Temperature: -40 degrees to +158 degrees F.
6. Regulatory: Controller shall be designed to meet the following regulatory requirements:
   a. UL294 Listing Standard for Safety.
   b. FCC EMI and EMC Class A.
   c. EN55022 EMI and EMC Class A.
7. Mounting:
   a. CR shall have the capacity to be mounted and operated behind any non-metallic, non-conductive surface, including glass.
   b. CR shall have the capability to be mounted on any metal door frame.
   c. Long range proximity card readers (LRCR):
      1) At high-speed rollup and automatic impact doors mount per manufacturer's recommendations for industrial powered trucks and protect CR from vehicle impacts.
      2) At Vehicle Gates mount per Standard Details.
8. Power:
   a. Source: Via the Wiegand interface cable to the controllers.
   b. The sensor shall emit a low power (less than one microwatt) RF field in up to 6 inches from surface.
9. Wiring: Multiple conductor overall shielded cable (6/C-#18 AWG minimum). Size cable gauge to meet distance requirements from the controllers.
10. Feedback:
    a. Single tri-color LED (green/amber/red) shall provide capability for diagnostic feedback.
    b. Green LED indicates valid card and red LED indicates invalid card.

c. An audio tone shall indicate successful digital proximity/contactless smartcard card read and access granted.

11. Diagnostics: CR and data-line integrity shall be monitored continuously and shall alarm if failure is detected and indicate device and location of fault.
12. Self-protection:
    a. Physical damage, including breaking open sensor housing, shall not allow access to any circuitry which would allow the system to be compromised.
    b. Transmission of any frequency (or set of frequencies) into the sensor at any power level shall not compromise the system.

F. Keypad:
1. General:
    a. The Keypad shall operate in conjunction with CR for an increased level of user authentication.
    b. Where required, the keypad shall be integral to the CR and provided as a single card reader/keypad combination unit.
    c. The system shall have the means to utilize a numeric keypad for entry of a Personal Identification Number (PIN).
2. Capacities:
    a. Keypad only reader shall provide a standard 10 digit numeric entry organized in the standard telephone pad layout.
    b. The user shall be able to enter a 4-digit Personal Identification Number (PIN).

G. Power Supplies with battery backup: Provide separate power supplies for controllers, associated electric locks and reader interface modules not powered by controllers.
1. General:
    a. Uninterruptible Power Supply shall provide continuous power to the controller, card reader, expansion modules, annunciator devices, and electric locks and operate from a 120VAC/60Hz source.
    b. Provide external rechargeable battery(s) to maintain all controller, card reader, expansion module, and electric lock operation for at least 4 hours in event of power failure.
    c. Power supplies and batteries shall be mounted within NEMA 1, metal enclosure equipped with keyed door latch.
2. Capacities: The Power supply shall provide:
    a. 12 Volt DC output to the controller; or 24 volt DC output to the electric locks.
    b. Ampere output current at 12 VDC, 24 VDC - 6 amps continuous.
    c. Power failure output and battery charger output.
3. Environmental:
    a. Humidity: 85 percent at 86 degrees F.
    b. Temperature: 32 degrees to +122 degrees F.
4. Regulatory: UL 294 and CSA.
5. Power: 120VAC/60Hz source.
6. Wiring:
    a. The power supply shall be connected to the controller via wiring of at least 16 AWG.
    b. The power supply shall utilize phoenix type connectors to allow for ease of field wiring and unit replacement or as recommended by the manufacturer.
7. Feedback: A single LED indicates power ON condition.
8. Self-protection: The power supply shall provide the following signals to the Controller:
    a. Power fail.
    b. Battery recharge signal.
9. The electric lock power supplies controlling the exit doors shall be equipped with a fire alarm interface for emergency lock release.

H. Long Range Vehicle and Driver Identification Reader:
   1. Provide long range vehicle and driver I.D. reader based on semi-active RFID technology with the following key features:
      a. Simultaneous vehicle and driver identification.
      b. Read range up to 33 feet.
      c. Object speed up to 125 mph.
      d. Tag authentication based on AES encryption.
      e. Adjustable read range.
      f. Bi-directional communication using two RFID channels.
      g. Robust industrial design.
      h. USB, Wiegand, Magstripe, Barcode Communication interface.
   2. Provide Long Range Vehicle and Driver I.D. Reader complying with the following technical specifications:
      a. Dimensions: 13 x 10.8 x 5.5 inches.
      b. Weight: 8.82 lbs.
      c. Protection class: IP66.
      d. Material: Cover ABS, Housing Die-casting ADC12.
      e. Operating temperature: -22 to +140°F.
      f. Relative humidity: 10% ... 93% relative humidity, non-condensing.
      g. Power supply: 24 VDC, 0.7A; Output 24Vdc, 0.1A.
      h. Power consumption: <20 Watt (on DC).
      i. Read range: Up to 33 feet, message acceptance ratio > 80%.
      j. Object speed: Up to 124 mph at appropriate distance.
      k. Operating frequency: 2.438 - 2.457 GHz, 433.62 & 434.22 MHz (RX-Category 3) Ton <5sec.
      l. Antenna polarization: Circular (LHC) (2450 MHz) integrated antenna; Horizontal (433 MHz); dedicated antenna.
      m. Air interface:
         1) 2.45 GHz: Nedap proprietary encoding standard.
         2) 433 MHz: Encryption based upon diversified AES128; 300kbps/ GFSK 75 kHz.
         3) Duty cycle < 1%; LBT not applicable.
      n. Communication interfaces:
         1) USB, Wiegand, Magstripe (clock & data), Barcode (Code39).
      o. Connectors: PCB screw connectors.
      p. Tamper switch: Magnetic switch, normally closed.
      q. Standards: CE, FCC, IC, ACMA, R-NZ, China_CMIIT, UL294.
      r. Included accessories: 9984364 Wall Mounting Set.
      s. Basis of Design: Nedap Transit Ultimate #9215689.
   3. The Long Range Vehicle and Driver Identification Reader shall be provided with the following accessories:
      a. Weather protection hood - Nedap #9218-327.
      b. Pole mounting kit - Nedap #5626595.
      c. Security key pack - Nedap #9216537.
   4. Long Range Vehicle and Driver Identification Booster Tags shall be provided for communication with the long range RFID reader and easy mounting to the vehicle's windshield. When an authorized photo identification access card is inserted into the booster, it is read, and then boosted to the vehicle and driver identification reader; access is granted and the gate opens automatically.
      a. Basis of Design: Nedap Smartcard Booster Ultimate #9982809.
      b. Quantity required: 50 Boosters Tags.

I. Electric Door Strike
   1. The electric strike shall transmit data to the interface module indicating the bolt is not engaged and the strike mechanism is unlocked. Power line supervision shall incorporate an end-of-line resistor.
      a. Electric strike shall be concealed for use with cylinder locksets.
      b. Electric strike shall be tamper resistant and rated 12 or 24 VDC with internally mounted solenoid.
      c. Electric strike shall accommodate 1/2 inch to 5/8 inch cylindrical latch bolt (5/8 inch with 1/8 inch door gap).
      d. Field selectable "Fail Secure/Fail Safe" (set to "Fail Safe").
      e. Basis of Design: Assa Abloy #HES8000C (complete with faceplate).

J. Magnetic Lock
   1. Provide bracket mounted, magnetic lock with 1200 pounds of holding force suitable for single door leaf installations. Double leaf doors shall be equipped with (2) single magnetic lock assemblies in separate enclosures. The magnetic lock shall have the following features:
      a. U.L. 294, U.L. 10C and U.L. 864 Listed.
      b. Current Draw: 505 mA.@12 VDC.
      c. Magnet Size: 10-1/2 L x 2-7/8 H x 1-1/2 W.
      d. Armature Size: 7-1/4 L x 2-3/8 H x 5/8 W inches.
      e. Clear anodized finish.
      f. Lifetime Warranty.
      g. Magnetic lock to include all necessary mounting plates and brackets.
      h. Built-in ARC suppression.
      i. Instant Release: No hysteresis.
      j. Basis of Design: Alarm Controls #1200S.

K. Door Contact Switch
   1. Furnish and install door contacts at each door location indicated on the drawings. Door contact switches shall be concealed type, recessed in the jamb, opposite the hinged location.
      a. Install per manufacturer's written recommendations and maintain the minimum gap separation.
      b. Door contacts shall be compatible with steel doors and jambs.
      c. Door contacts for new doors shall be concealed type only. Surface mounted door switches shall only be installed on existing doors and frames.
      d. Basis of Design:
         1) Interlogix Magnetic Contacts, #1085TWN with 1K ohm resistor (surface mount).
         2) Interlogix Roller Plunger, #3005-N with 1K ohm resistor (recessed - wood doors).
         3) Interlogix Roller Plunger, #1076CW-N with 1K ohm resistor (recessed-steel doors).
         4) Interlogix Overhead Door Magnet Contacts, #2315A-L with IK ohm resistor (track mounted, overhead door contact - closed loop).

L. Emergency Evacuation Pushbutton
   1. Manual activation of the emergency evacuation pushbutton shall de-energize the power supplies and micro-switches serving the egress electric locks at the turnstiles and RE-4 personnel door. The turnstiles shall immediately operate in the "free spin" mode (egress direction only). Inbound entry direction shall remain secure. Normal exit operation of the turnstiles and RE-4 access gate will be disabled until the manual reset of the pushbutton. Minimum reset time shall be set at 30 seconds, per NFPA 101.
   2. The pushbutton shall have indoor, "blue" polycarbonate housing with protective cover and the following features:

a. Push to activate; turn to reset operation.
b. "Red" LED indicator light.
c. Raised label to read "Emergency Evacuation Pushbutton".
d. 2 form "C" maintained contacts, rated 10 Amps at 125/250 VAC.
e. UL/cUL Listed; ADA compliant.
f. Indoor flush or surface mount.
g. Basis of Design: Safety Technology International, Stopper Station Series SS2429ZA-EN. Substitutions: Permitted.

M. Exit Door Alarm
1. All controlled exit doors requiring emergency egress shall be equipped with an audible and visual alarm station. The horn/strobe exit alarm shall be equipped with a remote key operated "reset" station and shall be 12 VDC powered from the lock power supply and batteries located at the controller.
2. The exit door alarm shall have white polycarbonate housing and blue lens with the following features:
   a. U.L. Listed; CE approved.
   b. Sound output: 101 dBA at 10 ft. (minimum)
   c. Single tone: Piezo Siren-Warble.
   d. Current draw: 748 mA at 12 VDC.
   e. Rating: 12 VDC regulated.
   f. Indoor flush or surface mount.
   g. Flash rate: Same as siren sounding.
   h. Basis of Design: AMSECO #SSX-52SB. Substitutions: Permitted.
   i. Alternate Manufacture: ADI/WBox #OE-SRNSTROBT.
   j. Substitutions: Permitted.
3. Key operated "reset" stations for all door alarms shall be keyed alike and shall be wall mounted top at 60 inches AFF, adjacent to the door.
   a. Basis of Design: Alarm Controls #KA105A.
4. Due to the capacity of the lock power supply, alarm and visual indications shall operate continuously for no more than 45 seconds. The visual/audible alarm shall be field adjusted to operate 30 seconds, if not reset.

N. Door Release Pushbutton
1. Doors equipped with electromagnetic locks requiring free egress shall be released by a pneumatic time delay pushbutton.
2. Pushbutton shall contain 1-1/2 inch diameter, mushroom head and pneumatic time delay with the following features:
   a. U.L. listed components.
   b. Green mushroom pushbutton with single gang, stainless steel plate.
   c. Contacts rated 10A at 35 VDC.
   d. Time range settable 2 seconds to 60 seconds.
   e. One Normally open and one normally closed contact.
   f. Switch time repeatable $\pm$10 percent.
   g. Labeled "Push to Exit".
   h. Basis of Design: Alarm Controls #TS-14.

PART 3 - EXECUTION

3.01 INSTALLATION METHODS

A. Drawings are schematic and diagrammatic. Use judgment and care to install Work to function properly and fit within building construction and finishes. Power and low voltage conductors, conduit, components, not shown or specified, which are required to produce a complete and operative system are required to be furnished and installed. Refer to Section 260500 - Common Work Results for Electrical.

B. Exact location of components is determined from dimensions on the Drawings, manufacturer's shop drawings, or as may be determined at Project Site. Do not scale Drawings for exact location of any item. Verify item mounting heights as required by project conditions, prior to rough-in.

C. Route conduits and wiring associated with equipment and systems above ceilings, in chases, and concealed within building structure.

D. All interior conduits shall be EMT with steel set-screw type couplings. All exposed exterior conduits (and interior conduits mounted below 10 feet - 0 inches AFF within the workroom) shall be heavy wall, rigid galvanized steel type.

E. All concealed and exposed conduit and cabling shall be routed parallel and perpendicular to structural elements.

F. All cables and conduits shall be routed below the roof decking to avoid damage due to future reroofing. Raceways and cables shall not be routed between the decking flutes.

G. Surface mounted raceways or conduit permitted only at locations indicated on Drawings.

H. Proposed equipment or devices, shown mounted on and/or adjacent to equipment, which if installed, would impair proper operation of existing or new equipment, shall be removed and relocated by Contractor as required so equipment will function properly. Notify USPS Project Manager immediately if any such condition exists.

I. Seal and make permanently watertight penetrations by raceways or equipment through ceilings, walls or floors.
   1. Seal penetrations in non-fire rated ceilings, walls or floors material specified in Section 079200 - Joint Sealants.
   2. Seal penetrations in fire rated walls with material specified in Section 078400 - Firestopping.

J. Install equipment and materials to provide required working clearance for servicing, repair and maintenance. Coordinate final location of concealed equipment and devices requiring access with final location of required access panels and doors. Allow required space for removal of parts that require replacement or servicing.

K. Install materials and equipment level and plumb, parallel, and perpendicular to other building systems and components.

L. Coordinate all cutting, patching and site work with the Design Build Entity or General Contractor.

M. Touch-up scratched and marred surfaces to match original finishes; remove all dirt and construction debris.

N. All work areas shall be left in a broom swept condition at the end of each day.

3.02 INSTALLATION - HANGERS AND SUPPORTS

A. Install products in accordance with manufacturer's published instructions. Install all electrical equipment in accordance with Section 260500 - Common Work Results for Electrical.

B. Furnish and install anchors, fasteners, and supports in accordance with NECA SI.
C. Do not fasten supports to pipes, ducts, mechanical equipment, and conduit.
D. Do not use spring steel clips and clamps.
E. Do not use powder-actuated anchors.
F. Obtain permission from structural engineer before drilling or cutting structural members.
G. Fabricate supports from structural steel angle or structural steel channel. Rigidly weld members or use hexagon head bolts to present neat appearance with adequate strength and rigidity. Use spring lock washers under all nuts.
H. Install surface-mounted cabinets with minimum of four anchors.
I. In wet and damp locations use structural steel channel supports to stand cabinets one inch off wall.
J. Use sheet metal channel to bridge studs above and below cabinets recessed in hollow partitions.

3.03 INSTALLATION - TERMINAL CABINETS

A. Terminal cabinets shall be provided to house long range reader power supplies, interface modules, SPDs and other access control system components. Enclosures shall be hinged and lockable with panelboard construction and plywood backboards.

B. ePACS terminal cabinets shall not contain CCTV components. The ePACS shall utilize independent wiring, raceways, and cabinets.
C. Terminal cabinets shall be wall or pedestal mounted with bottom of cabinet at no less than 12 inches A.F.F. or A.F.G. Provide NEMA Type 1 enclosures within interior locations and NEMA Type 4 stainless steel or non-metallic polycarbonate type for exterior locations. Pedestal mounted cabinets shall be supported utilizing 4 inch square concrete posts buried 24 inches below finished grade and set in concrete footing with 6 inches of concrete all around.
D. Terminal cabinets shall be amply sized to accommodate all components without overheating and forced air exhaust fans shall be provided. Cabinets shall be equipped with copper ground busses and those requiring 120 Volt power shall be provided with appropriate number of 20 Amp, 125 Volt receptacles complete with surge protection. Receptacles shall be securely mounted within the cabinet.
   1. Components shall be individually mounted and secured to the backboard. Stacking of components is not acceptable and the use of tie-wraps is prohibited.
   2. Long range, card reader power supplies shall be fed from 120 Volt circuits segregated to serve only the power supplies. Circuits utilized to feed gate controllers, CCTV components or any other non-related loads shall not be utilized to serve the ePACS or intercom equipment.
   3. Multi-output, fused power supplies may be utilized within the terminal cabinets to serve multiple long range car readers, provided each output is fuse protected.

E. Exterior terminal cabinets mounted near or adjacent to vehicular traffic shall be protected using 6 inch dia. x 4 feet high concrete bollards. Exterior terminal cabinets shall be located within the secured area of the facility.

F. Exterior terminal cabinets shall be equipped with a copper ground bus bonded to a driven ground rod using #2/AWG copper grounding electrode conductor.

3.04 EQUIPMENT INSTALLATION AND DOCUMENTATION

A. Installation:
   1. The Enterprise Physical Access Control System shall be installed and wired completely on the USPS structured wiring system as shown on the plans by factory trained and authorized employees of the Systems Integrator.
   2. Systems Integrator shall make all necessary wiring connections to external devices and equipment. Systems integrator shall program anti-pass back modes into the system in accordance with USPS requirements. Use the following e-mail for assistance in obtaining information regarding current USPS requirements: pacs-support@usps.gov.
   3. Systems Integrator employees shall carry proof of manufacturer's certification at all times.
   4. Install systems to conform with the approved submittal data. Where coordination requirements conflict with the system requirements, refer conflicts to the USPS Project Manager.
   5. All Enterprise Physical Access Control System devices shall be securely mounted to the building structure and fastened with tamper resistant screws. Provide USPS with three sets of tamper screw removal tools to be stored locally for service and maintenance.
   6. All wiring connections shall enter enclosures at one location and be neatly dressed.
   7. Device Mounting:
      a. The controllers shall be wall mounted in a secure area.
      b. The power supplies shall be installed in a secure area adjacent to the controllers.
   8. All DC operated locking hardware, relays, and all other inductive loads shall have a diode connected to them to prevent noise and/or any induced currents. All AC operated relays or electric strikes shall have a MOV connected to them to suppress any current induced noise. Diodes and MOVs shall be connected at the strike or relay and shall be of the type recommended by the device manufacturer.
   9. Install PIR request-to-exit sensors such that "corridor pedestrian traffic" will not activate the sensor. Ceiling or wall mount shall be acceptable. Adjust the pattern and sensitivity such that detection is ensured for all egress attempts and such that detection cannot be achieved from the exterior side of the door.

B. Network Communications:
   1. Installer shall coordinate all network communications wiring requirements with the structured cabling system provider to insure transmission pathway through the structured wiring system.
   2. Telecommunications outlets and cabling for equipment as shown on the plans are specified under Section 270500 - Common Work Results for Communications.
   3. Refer to "ePACS Standard System Configuration" and "Access Control (ePACS) SOP" for database configuration and local facility responsibilities.

C. Documentation:
   1. Accurate "as built" drawings shall be furnished before final acceptance is requested, by the Systems Integrator to aid the USPS in programming. These shall indicate the door(s) controlled by each lock output, the monitoring points for the door controlled area, host server, workstation and badge issuing station location, all controllers locations, all electrical circuit and telecommunications outlet designations and any annunciator outputs or special inputs into the system in hard copy and electronic format (AutoCAD-coordinate version requirements with the USPS Project Manager).
   2. The Systems Integrator shall supply 6 copies of operating and maintenance manuals to aid the USPS in the programming of the system.

D. Special Requirements for Cable Routing and Installation:
   1. The majority of the ePACS wiring in this building will be installed above ceilings without conduit. All communications cabling used throughout this project shall comply with the requirements as outlined in the National Electric Code (NEC) article 725. All cabling shall bare CMP and/or appropriate markings for the environment in which they are installed.
   2. Seal openings, existing or created, for cable pass through between floors, through rated fire and smoke walls. Creation of such openings as are necessary for cable passage between locations as shown on the drawings shall be the responsibility of the Contractor's work. Any openings created and left unused shall also be sealed as part of this work.
   3. Cabling routed underground, on the exterior, through inaccessible ceilings or less than 10 feet A.F.F. in the workroom shall be contained in conduit. Provide flush boxes within finished areas and factory boxes in unfinished areas. Provide 3/4 inch conduit risers with 90 degree bend and bushing for all wall mounted devices.
   4. The entire ePACS system shall utilize an independent wiring system not shared with any other building system. The structured cabling system racks, the TE's, the fiber backbone, cable trays, etc. cannot be utilized for any ePACS purpose. Cable trays installed for the ePACS cabling may be utilized to contain the CCTV wiring.

E. Surge Suppression:
   1. Provide individual surge protective devices (power and low voltage) at both ends of all exterior copper ePACS wiring and associated wiring exiting the building. Surge suppression shall be provided for the power and control wiring associated with the barrier arm and sliding gates, exterior card readers, exterior reader interface modules, power supplies, door contacts and magnetic locks. Refer to Section 264128 - Surge Protective Devices (SPD).
   2. Provide high-definition photographs showing the installation of the required surge protection devices at both ends of all exterior power and low voltage conductors. Photographs shall be transmitted to the A/E and USPS Project Manager.

F. Gate and Door Release:
   1. The gate and door release functions required for vehicle and employee entry shall not be controlled or wired as part of the ePACS. Independent wiring from the video intercom CEU is to be provided. The video intercom system shall perform all the gate and door release functions. Loss or interruption of the ePACS shall not affect the operation of the gate or door release functions.

3.05 SERVICE AND SUPPORT

A. Startup:
   1. The Systems Integrator shall coordinate all system database requirements with the USPS and build the system database for the host server and workstations. At a minimum the Systems Integrator shall:
      a. Provide worksheets to the USPS with requested database information a minimum of 4 weeks prior to anticipated system startup.
      b. Load all system device names and system addresses.
      c. Load basic access levels.
      d. Load and test all applications and interfaces.
      e. Load and test sample proximity cards compatible with USPS Standard Card.
   2. After the system has been installed, the documentation delivered to the USPS and network communications is established in compliance with Sections 3.1 & 3.2, A above, the Systems Integrator shall verify correct operation of all system components and demonstrate and test the system for the USPS.

3. Final system acceptance testing shall be conducted by the USPS Project Manager or, at the option of USPS, their authorized representative. Acceptance testing shall demonstrate all aspects of the Enterprise Physical Access Control System as described in the contract documents. The Systems Integrator shall make provisions for testing (any simulations required for testing) and provide a final acceptance test plan a minimum of one week prior to the anticipated testing date.
4. Final acceptance testing shall be conducted on the completed system as described in this specification and configured to the satisfaction of the USPS Project Manager.
5. The Systems Integrator shall guarantee all material and workmanship involving the system for 12 months after startup.

B. Training:
1. After system startup, the Systems Integrator shall instruct USPS personnel in how to program the system and demonstrate a typical operating program for each type of access controlled area.
2. Enterprise Physical Access Control System training sessions shall be arranged with the USPS at least one week prior to the training date. Training manuals shall be delivered for each trainee with one additional copy delivered for archiving on the project site.
3. Training manuals shall consist of an agenda, defined objectives for each lesson, a detailed description of the subject matter of each lesson, and the manufacturer's written operation and system manuals. At a minimum, training agenda shall consist of the following.
   a. An overview of the system components and features.
   b. A detailed description of how the equipment will operate to meet the performance requirements of the Enterprise Physical Access Control System.
   c. A description of the operating system and application software.
   d. Start up and orderly shutdown procedures for the system.
   e. Hands on training on all Enterprise Physical Access Control System software and hardware features.
   f. Basic troubleshooting guide intended to identify the source of system problems.
   g. System configuration and data back-up and restoration procedures.

C. Warranty Support:
1. The Authorized Systems Integrator shall be available during the warranty period to answer programming and application questions to support USPS personnel during this period.
2. The Authorized Systems Integrator shall have the training and capability to provide additional support services including:
   a. Regular testing and inspection of all system components and to submit reports on the results.
   b. Emergency Service for repairs and adjustments to the system and part replacement if necessary.

END OF SECTION 28 13 04

SECTION 28 15 24 - IP VIDEO INTERCOM AND EXTERIOR GATE CONTROL SYSTEM

PART 1 - GENERAL

1.01 SCOPE

A. Provide an IP video intercom system complete with door and gate controls for the exterior motorized gates and employee entrances. Provide multiple master control units and video door stations as shown on the drawings.

1.02 SUMMARY

A. Section Includes:
   1. Video master control station.
   2. High power PoE+ switch.
   3. Video door stations.
   4. Category-6 and fiber cabling.
   5. Fiber distribution patch panels.
   6. Fiber media converters.
   7. UPS unit.
   8. Spare parts.

B. Related Sections:
   1. Section 260500 - Common Work Results for Electrical.
   2. Section 264128 - Surge Protective Devices (SPD).
   3. Section 281304 - Enterprise Physical Access Control System.

1.03 REFERENCES

A. Electronic Industries Association (EIA):
   1. ANSI/TIA/EIA-568 - Commercial Building Telecommunications Cabling Standard.
   2. ISO 9001:2015 - Quality Management Systems - Requirements.

B. National Fire Protection Association (NFPA):
   1. NFPA 70 - National Electrical Code.

1.04 SUBMITTALS

A. Procedures for submittals.
   1. Product Data: Manufacturer's data sheets on each product to be used, including:.
      a. Preparation instructions and recommendations.
      b. Storage and handling requirements and recommendations.
      c. Installation methods.
   2. Shop Drawings: Submit the following:
      a. Wiring Diagrams: Indicate wiring for each item of equipment and interconnections between items of equipment.
      b. Include manufacturer's names, model numbers, ratings, power requirements, equipment layout, device arrangement, complete wiring point-to-point diagrams, and conduit layouts.
   3. Installation and Operation Manuals:
      a. Submit manufacturer's installation and operation manual, including operation instructions and component wiring diagrams.
      b. Provide detailed information required for the Postal Service to properly operate equipment

B. Field Test Reports: Indicate and interpret test results for compliance with performance requirements.

C. Maintenance Data: For equipment to include in maintenance manuals specified in Division 1.

D. Assurance/Control Submittals:
   1. Certificates: Manufacturer's certificate that Products meet or exceed specified requirements.
   2. Qualification Documentation: Submit documentation of experience indicating compliance with specified qualification requirements.

E. Procedures for closeout submittals.
   1. Operating and Maintenance Data: Operating and maintenance instructions, parts lists and wiring diagrams.
   2. Submit written special warranty with forms completed in United States Postal Service name and registered with manufacturer as specified in this Section.

F. Warranty: Submit manufacturer's and installer's 2-year warranty for full parts and labor.

1.05 QUALITY ASSURANCE

A. Qualifications:
   1. Manufacturer: ISO 9001.2015 Certified Company specializing in manufacturing products specified with minimum 5 years documented experience.
   2. Installer: Factory trained and experienced Company specializing in performing the work of this section with minimum 5 years documented experience and an authorized representative of equipment manufacturer for both installation and maintenance of equipment.

B. Regulatory Requirements:
   1. Conform to requirements of NFPA 70 and UL 50.
   2. Products: Listed and classified by Underwriter's Laboratories Incorporated as suitable for the purpose specified and indicated.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Delivery: Deliver materials to site in manufacturer's original, unopened containers and packaging, with labels clearly identifying product name and manufacturer.

B. Storage: Store materials in clean, dry area indoors in accordance with manufacturer's instructions.

C. Handling: Protect materials during handling and installation to prevent damage.

1.07 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Acceptable Manufacturer: Aiphone Corp., which is located at: 6670 185th Ave. NE; Redmond, WA 98052; Toll Free Tel: 800-692-0200; Tel: 425-455-0510; Fax: 425-455-0071; Web: www.aiphone.com.
   1. Regional Sales Manager: Robert Hilt; Tel: 813-365-4403; e-mail: robert.hilt@aiphone.com.

2. Director of Sales - South: Spencer Britenstine; Tel: 614-286-8925; e-mail: spencer.britenstine@aiphone.com.

B. IP Video Intercom System: IX Series Intercom System as manufactured by Aiphone Corporation.

C. Substitutions: Not permitted.

D. On-Site Assistance: Engage a factory-authorized service representative to provide on-site assistance in adjusting sound levels, video images and controls to meet occupancy conditions. Provide up to three on-site assistance visits within one year of Substantial Completion.

2.02 SYSTEM DESCRIPTION

A. IP Network Compatible Video Intercom System: A network-based communication and security system featuring video entry security and internal communication. All master units in the systems shall be able to unlock doors remotely, assist onsite visitors and communicate using a PoE network.
   1. Power Source: Power over Ethernet (802.3af).
   2. Network Interface: 10 BASE-T / 100 BASE-TX Ethernet (RJ-45).
   3. Network Protocols: IPv4, IPv6, TCP, UDP, SIP, HTTP, HTTPS, MJPEG, RTSP, RTP, RTCP, IGMP, MLD, SMTP, DHCP, NTP, DNS.
   4. Bandwidth Usage:
      a. G.711: 64Kbps x 2 per video call.
      b. 64Kbps per monitor.
      c. H.264: 24Kbps ~ 2,048Kbps.
   5. Communication: Hands-free (VOX/PTT) or handset (full-duplex).
   6. Video Display: 7-inch color LCD.
   7. Camera: Type:
      a. 1/3-inch color CMOS. 1.23 Megapixels.
      b. View Area at 0-degree camera angle mounted at 4 feet 11 inches (1500 mm) AFF: 2 feet 3 inches (700 mm) vertical x 3 feet 9 inch (1150 mm) horizontal at 19 inches (500 mm).
   8. Video Stream: ONVIF Profile S.
   9. Door Release: Programmable Form C dry contact, 24V AC/ DC, 1.0A.
   10. Wiring: The audio/video master units and door stations shall be connected to the network node via homerun category-6 wiring and RJ45 jacks.
      a. Maximum distance:
         1) Master station to network node: 330 feet.
         2) Door station to network node: 330 feet.
         3) Utilize fiber optic cable for master and door stations located more than 330 feet from the network node.

2.03 EQUIPMENT

A. Coordinate features to form an integrated system. Match components and interconnections for optimum performance of specified functions.

B. Equipment: Modular type, using solid-state components, fully rated for continuous duty, unless otherwise indicated. Select equipment for normal operation on input power usually supplied at 110 to 130 V, 60 Hz.

C. Waterproof Equipment: Listed and labeled for duty outdoors or in damp locations.

2.04 AUDIO VIDEO MASTER CONTROL STATION(S)

A. Master station(s) shall initiate and answer calls from up to 500 additional master or door substations. Master station(s) shall be wall or desk mounted and shall have the following features:
   1. 7-inch color LCD touchscreen
   2. PoE power source
   3. Camera for 2-way video calls
   4. 8 speed dial buttons
   5. Hands-free VOX/PTT communication and privacy handset (duplex)
   6. All call to other master stations
   7. Selective calling to any station in the system
   8. Privacy prevents other masters from listening in
   9. Door release button activates door strike or magnetic lock (selectable N/O or N/C contacts from door station)
   10. Tone and volume adjustments for audio communication
   11. Brightness and contrast adjustments for video.

B. Basis of Design:
   1. Aiphone #IX-MV7-HW (Master Station) - white finish.

2.05 HIGH POWER POE+ SWITCH

A. The PoE+ switch shall be 24 port gigabit, ethernet, unmanaged, high power PoE+ with 380 Watt PoE budget. Unit shall be expandable and shall have the following features:
   1. Wiring hub for system using category-6 cable
   2. Controls all functionality of "IX" system
   3. Speed: Gigabit
   4. 24 ethernet ports (RJ45)
   5. PoE ports/max. power per port: 30 Watts
   6. Programmable functionality via supplied software
   7. USB to serial adaptor and gender changer, software program and Installation and Operation Manual on CD, and QuikStart Guide.

B. Operation: The PoE+ switch handles the calling and communication signals for the "IX" system, as well as providing selective door release outputs, video outputs, DVR trigger outputs and CO line transfer output. System set-up and a variety of functions are programmed in the PoE+ switch with the use of the supplied software. The serial connection is for uploading the system programming, for raw RS232 data output, or for system monitoring via supplied program.

C. Mounting: The unit shall be mounted in a standard 19 inch EIA rack within the intercom network node cabinet.

D. Basis of Design: Netgear #GS324PP.

2.06 AUDIO/VIDEO DOOR STATIONS

A. The remote audio/video door stations shall include a color camera, microphone, speaker and call button with the following features:
   1. PoE with pass-through port.
   2. Color video camera with audio intercom
   3. 2-way hands-free voice communication with master station
   4. Call button to initiate call to master(s)
   5. White LED illuminator for low light conditions
   6. RJ45 jack for category-6 connection
   7. 2 Form C, dry contacts for door/gate release

B. Operation: When the call button on the door station is pushed, the master station(s) ring and the video screen comes on with the image from the door station's camera. The master station user then initiates communication and the person at the door station speaks hands-free.

C. Basis of Design: Aiphone #IX-DV (surface mounted); Aiphone #IX-DVF (surface or flush wall mounted).
   1. Exterior audio/video door stations to be pedestal or surface wall mounted and containing surge protection devices shall be Aiphone #IX-DVF equipped with stainless steel, surface mount backbox (10 7/16 inch H x 5 15/16 inch W), Aiphone #SBX-IDVF.

2.07 FIBER OPTIC MEDIA CONVERTER MODULES

A. Fiber optic transmission equipment shall be used when category-6 cable lengths (including horizontal and vertical distances) exceed 330 linear feet.

B. Fiber transmitter modules located at field devices shall be low profile totally enclosed type and shall be mounted within individual enclosures or nearby ePACS terminal cabinets containing the long range, card reader power supplies and reader interface modules.
   1. Transmitter requires 60 Watt, 48VDC power supply and 120Volt power feed.
   2. Basis of design: AFI #50SL-PoE+ with #PS-4860 power supply.
   3. Alternate U.S. manufacturers permitted.

C. Fiber receiver modules located at the node location shall be rack mounted.
   1. Basis of design: AFI #RX-50 Series.
   2. Alternate U.S. manufacturers permitted.

D. Fiber optic modules shall conform to the following minimum specifications:
   1. 10/100 MBps RJ45 Ethernet port, SC Fiber Ports.
   2. 62.5/125, OM1, tight-buffered, multimode fiber.
   3. PoE+ device.
   4. Protocol independent.
   5. -31 to 158 deg operating Temperature.
   6. IEEE 802.3, IEEE 803.2u and IEEE 803.2af Complaint.

2.08 CABLING

A. Cabling requirements:
   1. Cable runs from master or call-in stations to the network node that do not exceed 330 feet shall be category-6; utilize plenum or outdoor rated where required.
   2. Cable runs exceeding 330 feet from master or call-in stations to the network node shall be 2 count 62.5/125, OM1, multimode, indoor fiber cable; utilize plenum or outdoor rated where required.
   3. All exterior category-6 and fiber cable runs shall be contained in conduit.
   4. Category-6 cable shall be terminated utilizing 8 pin, male RJ45 jacks at the device and to 8 pin, modular connectors at the patch panels within the network node cabinet to facilitate cable testing prior to installation of the network node. All testing shall be performed only after the cables have been terminated with the male RJ45 jacks.
   5. When fiber optic modules are required, provide fiber optic cable appropriate for the application. Cable shall conform to the following specifications:
      a. "SC" type connectors shall be used on all cable terminations.
      b. Performance characteristics (including optical attenuation) shall be such that the fiber optic modules deliver signals end-to-end with sufficient bandwidth and quality to meet the specified application.
      c. At no time shall fiber optic cable have loose ends terminated and left loose. All fiber optic cable shall be looped, stored, connected and permanently mounted in appropriate LIU cabinets/devices prior to testing. Fiber distribution, patch boxes (interconnect centers) shall be provided for termination of the fiber cabling.

1) Fiber patch box shall be (6) port complete with adapter plate, appropriate number of "SC" connectors, splice tray, protection sleeves and enclosure.
   a) Basis of Design: RLH Industries #PWSN-A-1 (Slimline patch panel: 4 inch W x 5 1/2 inch H x 2 inch D.
   b) Alternate U.S. manufacturers permitted.
2) Fiber patch boxes shall be equipped with covers, so the fiber strands are not exposed.

6. Fiber cabling not routed within conduit shall be encased within appropriately sized inner ducts; utilize plenum rated where required.
7. The Contractor shall terminate and test the fiber optic cable and connectors.

2.09 CATEGORY-6, 12 OR 24 PORT MODULAR SURFACE-MOUNTED "110"-STYLE PATCH PANELS

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. CommScope Uniprise
   2. Ortronics (Legrand)
   3. Panduit
   4. Product options and substitutions. Alternate U.S. manufacturers permitted.

B. Boxes/Panels:
   1. Capable of terminating 12 or 2) category-6 cables.
   2. Equipped with an 89D surface mounting bracket.
   3. Complies with TIA-568-C "T568A" pinning configuration.

C. Connector:
   1. Rack mounted, 8-pin modular, category-6, non-keyed.
   2. Complies with TIA-568-C "T568A" pinning configuration.

2.10 CATEGORY 6 CABLING

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the work include the following:
   1. Belden.
   2. Berk-Tek.
   3. CommScope Uniprise.
   4. General Cable.
   5. Leviton.
   6. Ortronics (Legrand).
   7. Panduit.
   8. Product options and substitutions. Alternate U.S. manufacturers permitted.

B. Conductors: 4 twisted pair, minimum 23 AWG, solid copper.
   1. Individually insulated plenum rated conductors under common plenum rated sheath unless entire area where cable is installed is not considered a return air plenum according to any applicable codes. Provide outdoor, "OSP" rated cable and patch cords for exterior applications.
   2. Complies with individual characteristics established in TIA-568-C, and all addendums for category-6 cable performance specification.
   3. Nominal Impedance: 100 ohms plus or minus 15 percent.
   4. Certified and capable of performing to a minimum of 250 MHz.
   5. Maintain manufacturer's twisting of wire pairs to termination point. Do not attempt to restore, modify, or add to manufacturer's twisting of cable. Do not untwist more than ½ inch of the stripped cable.
   6. Provide footage markings and "yellow" colored insulation.

C. Basis of Design:
   1. Indoor riser rated (yellow): General Cable #7133802 or approved equal.
   2. Indoor plenum rated (yellow): General Cable #7131802 or approved equal.
   3. Outdoor "OSP" rated (black): General Cable #7136100 or approved equal.

2.11 CATEGORY 6, COPPER PATCH CORDS

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Belden.
   2. Berk-Tek.
   3. CommScope Uniprise.
   4. General Cable.
   5. Hitachi (Drybit).
   6. Leviton.
   7. Ortronics (Legrand).
   8. Panduit.
   9. Product options and substitutions. Alternate U.S. manufacturers permitted.

B. Conductors: Straight through type 4 twisted pair minimum 23 AWG, stranded copper.
   1. Terminated with male 8-pin modular plugs.
   2. Complies with individual characteristics established in TIA-568-C, and all addendums for category-6 cable performance specification.
   3. Nominal Impedance: 100 ohms plus or minus 15 per cent. Certified and capable of performing to a minimum of 250 MHz.
   4. Match performance and impedance characteristics of the installed horizontal unshielded twisted pair cable. Utilize plenum or outdoor, "OSP" rated where required.
   5. Each patch cord shall have a plastic arch for ease of removal of the connector (rubber boots are not acceptable). Preferred Copper Patch type: Ortronics (Legrand) #OR-MC615-*XX*.
   6. Patch cords shall be factory made, tested and individually factory wrapped within non-clear plastic bags.
   7. All category-6 patch cords shall be yellow in color.

C. Connector:
   1. 8-pin modular, category-6, non-keyed.
   2. Complies with TIA-568-C "T568A" pinning configuration.
   3. Color: Clear.

2.12 OM1 FIBER CABLING

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Belden.
   2. Berk-Tek.
   3. CommScope Uniprise.
   4. Corning Cable Systems.
   5. General Cable.
   6. Leviton.
   7. Optical Cable Corp.
   8. Ortronics (Legrand).
   9. Superior Essex.
   10. Product options and substitutions: Alternate U.S. manufacturers permitted.

B. Conductors: 2 / 6 strand.
   1. Provide multi-strand, 62.5/125 micron, tight-buffered, multimode, OM1 fiber cabling rated as follows:

a. 1 Gb/s ≤ 150m @ 850 nm.
b. 1 Gb/s ≤ 1000m @ 1300 nm.

2. The fiber cabling shall meet the following specifications:
   a. EIA/TIA-492AAAA-A-1997, "Detail Specification for 62.5 micron Core Diameter/125 micron Cladding Diameter Class 1a Graded-Index Multimode Optical Fibers."
   b. IEC 60793-2-10, "Product specifications - Sectional specification for category A1 multimode fibers", Type A1b 62.5/125 micron graded index fiber.
3. Provide individually insulated plenum rated strands under common plenum rated sheath unless entire area where cable is installed is not considered a return air plenum according to any applicable codes.
4. Fiber cabling shall comply with individual characteristics established in TIA-568-C including all addendums for fiber optic cable performance specification.
5. All exterior or underground fiber cable shall be indoor/outdoor rated.
6. Provide footage markings to easily identify the cable lengths.

C. Basis of Design:
1. Indoor riser rated: General Cable #CG0021PNR or approved equal.
2. Indoor plenum rated: General Cable #CG0021PNU or approved equal.
3. Indoor/Outdoor, plenum rated: General Cable #CG0021ANU.BK or approved equal.

2.13 OM1 FIBER OPTIC PATCH CORDS: 2 STRAND, TIGHT BUFFERED

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
1. Belden.
2. Berk-Tek.
3. CommScope Uniprise.
4. Corning Cable Systems.
5. General Cable.
6. Leviton.
7. Optical Cable Corp.
8. Ortronics (Legrand).
9. Superior Essex.
10. Product options and substitutions. Alternate U.S. manufacturers permitted.

B. Fiber optic duplex patch cords.
1. Fiber connectors shall be SC/SC type.
2. Complies with individual characteristics established in TIA-568-C including all addendums for fiber optic patch cable performance specification. Utilize plenum or outdoor, "OSP" rated where required.
3. Patch cords shall be factory made and factory tested individually, and factory wrapped within non-clear plastic bags.

2.14 WIRE MANAGEMENT PANELS

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
1. Panduit.
2. Chatsworth Products.
3. CommScope Uniprise.
4. Leviton.
5. Ortronics (Legrand).
6. Product options and substitutions: Alternate U.S. manufacturers permitted.

B. Cable management panels shall be "2RU" high, horizontally rack mounted and equipped with opaque covers to conceal the patch cords.

2.15 ACCESSORIES

A. Lightning/Surge Protection: Products shall utilize internal or external (power and low voltage) surge protection such that a normally occurring power surge shall not void any manufacturer's warranty.
   1. Provide individual surge protective devices (power and low voltage) at both ends of all exterior copper intercom wiring exiting the building. Surge suppression shall be provided for the power and control wiring associated with the barrier arm and sliding gates, exterior call stations and power supplies. Refer to specification section 264128.
      a. Rack mounted surge protectors shall be provided within the network node cabinet to protect the category-6 cabling serving the exterior, call-in stations. Modular surge protectors shall also be provided at the call-in stations end of the category-6 cable.
      b. The Contractor shall provide high-definition photographs showing the installation of the required surge protection devices at both ends of all exterior power and low voltage conductors. Photographs shall be transmitted to the A/E and USPS Project Manager.

B. Intercom network node equipment rack shall utilize a standalone, UPS sized for a minimum of 18 minutes of battery run-time. The UPS shall be provided with a dedicated 20 Amp, 120VAC power and NEMA L5-20R receptacle.
   1. The UPS shall be line-interactive, rack mounted and rated 1000VA/900Watt with an 18 minute battery reserve at 450 Watts; Tripp-Lite #SMART1000RMXL2U.

C. Wall mounted rack(s): Provide and install wall mounted, open equipment rack(s) to provide sufficient mounting space for the required video intercom equipment.
   1. Racks shall be all metal construction conforming to EIA standards with 19 inch equipment mounting opening and 1-3/4 inch vertical spacing of equipment. Rack rails shall be punched with captive nuts, 10-32 screws and nylon washers.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.

B. Verification of Conditions: Verify that field measurements, surfaces, substrates, and conditions are as required, and ready to receive Work.

C. Report in writing to the USPS Project Manager prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.

D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the USPS.

E. Provide required power outlets, low voltage power supplies, interconnecting cables, hardware, and equipment for a complete and operable system.

F. Master and call-in station locations are to be reviewed and approved by the Facility and Postal Inspector, through the USPS Project Manager, prior to installation of conduit and cabling.

3.02 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Verify that electrical connections are made correctly.

C. When not installed in cable trays, cable (category-6, fiber optic, and low voltage power) shall be supported with wide base cable hangers rated for proper support of category-6, fiber optic, and innerduct cables (compliant with UL and NEC requirements for structured cabling).
   1. Cable hangers shall be installed every 3 to 6 feet and shall be rated to support the weight of the cable multiplied by a factor of three.

D. Install exposed conduits and cables parallel and perpendicular to surfaces or exposed structural members, and follow surface contours.

E. Wiring within Enclosures: Bundle, lace, and train conductors to terminal points with no excess. Use lacing bars in cabinets.

F. Cables are to be secured to the rack frames at sufficient intervals to ensure that the weight of the cable will not contribute to fatigue or early failure of that cable or the device and connector to which it is attached.

G. Sufficient excess cable shall be provided in "service loop locations" to ensure that the cable may be re-connected without requiring the addition of extension pieces.

H. All permanent cabling shall be mechanically numbered to provide system documentation. Apply wire and cable marking tape to designate wires and cables to identify media in coordination with system wiring diagrams.

I. All wiring to include category-6 and Fiber Optic cables shall utilize hook and loop fasteners to eliminate the risk of over-tightening cable bundles and affecting the strength or rated performance of the cable. The use of tie wraps is not acceptable.

J. Double-sided foam tape shall not be used to secure any equipment, terminal blocks, or accessory devices. All device mounting shall be of a permanent nature.

K. All excess length AC cords are to be tie-wrapped out of the way. Where possible, they shall be routed in a separate bundle a minimum of 6 inches away from any signal or control cable.

L. Care shall be exercised at all times to protect Postal Service property. For example, ladders shall not be placed against wallpapered or finished surfaces, equipment or furnishings; desks or countertops shall not be used in lieu of ladders.

M. Arrange all components to be mounted in the network node rack(s)to provide a neat appearance and accessibility for servicing equipment.

N. Provide required power outlets, interconnecting cables, hardware and equipment for a complete and operable system.
   1. Power, 120VAC: As required by codes and standards for the facility.
   2. Where conduit is used, a minimum of 40% excess capacity shall be provided for future use.

O. Control-Circuit Wiring: Install number and size of conductors as recommended by system manufacturer for control functions indicated.

P. Weatherproof Equipment: Install units that are mounted outdoors, in damp locations, or where exposed to weather consistent with requirements of weatherproof rating. Provide surge protection where required.

3.03 FIELD TESTING CATEGORY-6 COPPER AND FIBER OPTIC CABLE

A. Section 014000 - Quality Requirements: Field testing and inspection.

B. Field Testing Procedures:
   1. Provide all equipment and services necessary to test the cabling.
   2. Test and calibrate instruments before testing.
   3. Re-terminate and retest any cable found to be defective.

4. Perform cable testing and submit report prior to installation of any cameras or node cabinets.

C. Category-6 Copper Cable Testing:
1. Use Level III Compliant test equipment.
2. Test parameters shall include:
   a. Wire map.
   b. Insertion loss (attenuation).
   c. DC loop resistance.
   d. Return loss at camera.
   e. NEXT, NEXT at camera.
3. Perform end-to-end tests of each 4-pair cable as follows:
   a. Pair/conductor for proper pinouts and continuity.
   b. Ground fault.
   c. Proper termination, shorts, and crossed pairs.
   d. Channel attenuation per TIA-568-C, including all addendums.
   e. Channel bi-directional worst case near end cross talk (NEXT) at frequencies up to 250 MHz, per TIA-568-C, including all addendums.
   f. Measured effective cable run length.

D. Fiber Optic Testing:
1. Use 62.5/125 micron, OM1, multimode fiber optic cable testing.
2. Perform testing of fiber in accordance with the fiber type being tested, TIA-526-14-A, Method B for Multimode Fiber (One Jumper/Two Adapters).
3. Multimode fiber optic cable shall be tested bi-directionally at wavelengths of 850nm and 1300nm.
4. The fiber testers and test heads shall have passed calibration within one year of actual test date.
5. Tests include:
   a. Tier 1 Testing with Optical Loss Test Set (OLTS) that includes testing for length.
   b. Tier 2 Testing with OTDR to show all splices.

3.04 FIELD QUALITY CONTROL

A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect field-assembled components and equipment installations, including connections. Report results in writing.

B. Testing:
1. The Contractor shall perform tests and provide test equipment, tools, and personnel required to conduct system tests and inspections.
2. The Contractor shall conduct system acceptance test upon completion of installation using pre-approved procedures. Test shall consist of system, subsystem, and device level acceptance tests, including software.
3. The Contractor shall ensure that test procedures confirm each specification statement and manufacturer requirement has been met or exceeded. An actual demonstration of each system function and a simulation of each system failure shall be provided.
4. An acceptance test period of thirty days shall begin at the start of the acceptance test. Any system failure during the acceptance test period will suspend the acceptance test. The thirty-day test period will restart when the required repairs have been made and certified.
5. Perform all tests in the presence of the USPS Project Manager. The USPS reserves the right to accept any portion or activate any phase prior to acceptance of entire system.
6. Schedule tests with at least seven days advance notice of test performance.

3.05 ADJUST AND CLEAN

A. Adjust equipment for proper operation.

3.06 PROTECTION

A. Protect finishes until substantial completion.

3.07 DEMONSTRATION

A. Engage a factory-authorized service representative to train maintenance personnel to adjust, operate, and maintain equipment as specified below:
   1. Train maintenance personnel on programming equipment for starting up and shutting down, troubleshooting, servicing, and maintaining equipment.
   2. Review data in maintenance manuals.
   3. Schedule training with Postal Service at least seven days in advance.

END OF SECTION 28 15 24

SECTION 28 23 05 – INTEGRATED SECURITY AND INVESTIGATIVE PLATFORM (ISIP) CCTV SYSTEM

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Provide and install a complete IP Video System including, but not limited to:
      a. IP Video Surveillance Cameras, housings, mounts, power supplies, cabling, and related equipment.
      b. Video management software.
      c. Video monitoring and recording equipment.
      d. Equipment enclosures and remote node cabinets.
      e. Network equipment including routers and switches.

B. Related Documents: The Contract Documents, as defined in Section 01 10 00 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other Documents including:
   1. System Installation Manuals (provided by the manufacturer) shall be left on-site during the final acceptance. Manuals will not be provided prior to installation completion.

C. Related Sections:
   1. Section 26 05 00 - Common Work Results for Electrical.
   2. Section 26 05 33 - Raceway and Boxes for Electrical Systems.

1.02 REFERENCES

A. The references listed below form a part of this specification:
   1. NFPA 70 - National Electrical Code.
   2. ANSI / TIA / EIA 568-C Commercial Building Telecommunications Cabling Standard.
   3. ANSI / TIA / EIA 569-B Commercial Building Standard for Telecommunications Pathways.

1.03 SYSTEM DESCRIPTION

A. All equipment including the cameras, camera housings, camera mounts, NVR's, monitors, network switches, headend racks, node cabinets, CCTV terminal cabinets, fiber equipment enclosures, etc.

B. Power, conduit, cable tray, cable, and cable pulling and NEMA 1/ NEMA 4 Enclosures to be used as part of the installation.

1.04 DEFINITIONS

A. "Non Blue Sky": The following camera locations are considered to be non blue sky applications:
   1. Interior cameras.
   2. Exterior building mounted cameras (covered or uncovered).
   3. Exterior cameras covered by an overhang or canopy or similar protection.

B. "Blue Sky": Exterior cameras mounted remote from the building exterior wall are to be considered "blue sky" applications.

1.05 SUBMITTALS

A. Section 01 33 00 - Submittal Procedures:

1. Product Data: Manufacturer's specification sheets for each component shall be required for all products provided.
2. Due to USPS security requirements, submittals will be limited to one electronic copy of the block diagram and one copy of the shop drawings.
3. Final As-Built Drawings and Installation and Operation Manuals will be supplied directly to USPS at uspiscctv@usps.gov and stored within the headend rack per USPS Project Manager.

B. Shop Drawings:
1. Provide a Standard Drawing Package that shall be utilized for the installation of the CCTV system. This package shall include:
   a. Block Diagram: System block diagrams noting major system components and interrelationships of each component.
   b. Headend Equipment Racks and Remote Node Cabinets: Elevation drawings showing the arrangement of all components within the remote node cabinets and headend equipment rack(s).
   c. AutoCAD electronic copies of the final camera placement drawings, camera schedules, headend and remote node locations and all monitors.
   d. Dimensioned and scaled elevation drawings for each CCTV terminal cabinet and fiber equipment enclosures showing the location of fiber media converters, fiber patch boxes, power supplies, receptacles, ethernet extenders, surge protectors and other CCTV components. Elevation drawings shall be submitted and approved prior to ordering the terminal cabinets and equipment enclosures.
   e. Cable routing plans shall be provided prior to installation of the cabling.

C. Field Testing Reports for Category 6 Copper and Fiber Optic Cable
1. Test reports: All testing must be performed using industry standard, compliant test equipment that stores the test results in internal memory and produces test result reports. Test results shall be provided in test equipment format (RAW Electronic). Vendor generated spreadsheets are not acceptable. Paper test results are not acceptable.
2. Submit test reports prior to installation of any cameras, node cabinets or the headend equipment.

D. Sequence and Scheduling Plan: Provide installation scheduling plan for review and approval.

E. Section 01 77 04 - Closeout Procedures and Training:
1. Product Quick Reference cards for the operation of all key system components.
2. Project Record Documents: Provide field-accurate drawings that reflect actual locations of cameras, indicating cable routings, cable identifiers, layout, location and numbering of system devices to reflect as-built conditions.
3. Provide a final materials list of equipment installed. Materials list shall include model number, serial number, and date installed.
4. Project Completion Certification: Document signed by a USPIS Physical Security Specialist indicating that the project is fully complete with all punch-listed items resolved.
5. Operating Instruction:
   a. Provide on-site instruction, upon request, to review the operation of the system and detail any common troubleshooting or maintenance that is required to ensure normal operation. Authorized USPS (USPIS & USPS OIG) Representatives will receive this training.
   b. Provide one complete set of equipment operating and installation manuals that will be stored in the headend rack per USPS Project Manager.

1.06 QUALITY ASSURANCE

A. Installation Contractor:

1. Company with a minimum of five years system design, engineering supervision, and installation experience in the CCTV industry.
2. Company that is trained and authorized to install "Axis", "March Networks" and "Hanwha Techwin" products. The CCTV wiring shall be installed by a CCTV systems installer trained, authorized and certified as a Network Camera Technician to install and wire the Axis, March Networks, and Hanwha Techwin products.
3. Company that has been successfully installing CCTV systems of equal size and complexity for a minimum of five years. Submit a minimum of three references. System references shall include projects where software and hardware installed is similar to the software and hardware proposed for this project.
4. Furnish all labor, services, and materials necessary to furnish and install a complete, functional CCTV system. The System shall comply in all respects with the requirements of the specifications, manufacturer's recommendations and Underwriters Laboratories Inc. (ULI) listings.
5. Furnish certification that the entire CCTV system has been inspected and tested, is installed entirely in accordance with the applicable codes, standards, manufacturer's recommendations and UL listings, and is in proper working order.
6. The USPS requires professional workmanship from an experienced "CCTV systems" contractor and will reject any faulty workmanship or installation methods not meeting their satisfaction.

1.07 DELIVERY, STORAGE AND HANDLING

A. Section 01 60 00 - Product Requirements: Transport, handle, store, and protect products.

B. Keep devices and equipment in manufacturer's packaging in a secured, air conditioned location until system is ready for installation.

C. Comply with manufacturer's requirements. Coordinate storage location with the Postal Service.

D. The equipment delivered must be insured at the Contractor's expense through acceptance.

1.08 PROJECT CONDITIONS

A. Verify customer location has 10X network available for installation of system via USPS site project manager or USPS IT.

1.09 WARRANTY/SERVICE/TECHNICAL SUPPORT PLAN

A. Warranty:
   1. Include manufacturer warranty for one year after facility acceptance and project completion certification for materials and labor.
      a. Service plan shall include all parts and labor, the cost of utilizing a lift truck (if required) and shall include return shipping. Failed equipment shall be repaired or replaced at no charge to the Postal Service during the warranty period.
      b. USPS shall not be required to process any paperwork in order to be entitled to service plan coverage. It is the General Contractor's sole responsibility to monitor and comply with warranty eligibility.
      c. Any communication issues identified by the USPS and mutually agreed upon as 'level one" bugs (impacting operation with no work-around) shall be rectified within two weeks of their being reported.
      d. Any communication issues identified by the USPS and mutually agreed upon as 'level two" bugs (impacting operation but with a work-around) shall be rectified within 90 days of their being reported.
      e. Turnaround time for all repairs (warranty and out-of-warranty) shall not exceed 72 hours.

B. Technical Support:
   1. The General Contractor shall provide toll-free, 24/7 telephone number and/or email address for technical support at no charge throughout the warranty period.
   2.
   3. Data Recovery: Provide a service to assist the USPS in recovering data from digital recording system hard drives and removable storage media in the event of a failure.
      a. Turnaround time for data recovery shall be less than seven days from receipt of hard drives at "March Networks" data recovery center.

PART 2 - PRODUCTS

2.01 MANUFACTURER

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Axis Communications
   2. Black Box
   3. Cisco Systems, Inc.
   4. Costar Video Systems
   5. Dell Technologies
   6. Digital Watchdog
   7. Ditech Surge Protection
   8. Great Lakes Case and Cabinet Co., Inc.
   9. Hanwha Techwin, Co. LTD
   10. ICC
   11. Lantronix
   12. March Networks Corporation
   13. Middle Atlantic Products
   14. Oncam Global Limited
   15. Optical Cable Corporation (OCC)
   16. Orion Images Corporation
   17. Ortronics (Legrand) Corporation
   18. Panduit Corporation
   19. Transitions Networks
   20. Tripp-Lite by Eaton

B. Section 01 60 00 - Product Requirements:
   1. Product options and substitutions are not permitted without a written and USPS/USPIS approved deviation.
   2. All equipment to be supplied under this specification shall be new and the current model of the manufacturer.
   3. Systems and components shall have been thoroughly tested and proven in actual use.

2.02 NETWORK VIDEO RECORDERS AND STORAGE

A. Based on the Construction Documents, purchase all equipment from the authorized manufacturer.

B. Headend and Network Video Recorder requirements:
   1. NVR/Storage Requirements: NVR storage, processor, and RAM requirements will be based off a mathematical formula from the information obtained during the site survey process. Once the number, type and classification of cameras are approved by all parties, it will calculate the required NVR(s) fit for the site.
      a. Storage for 30 Days continuous video with 30% expansion capability and motion identified assuming a frame rate of no less than 15 fps.

b. Dual Network Interface Cards on board and 4 additional GB NIC ports via PCIe card per USPS requirements. The USPS requires 1 connection on the USPS network per NVR. Thus, each NVR will have (at minimum) 2 USPS 10X Network connections.
c. UPS Power Supplies for NVR and Storage.
d. Input Power: 120VAC, 60Hz.
e. Software: "March Networks Command Client" Video Management System.
f. All items rack mounted.

2. Network Video Recorders (NVRs) shall be high capacity, high definition IP recording with RAID and have the following minimum features:
   a. Video
      1) Channel Capacity: 30 fps at up to 1080 primary channel; 15 fps at D1 secondary channel.
      2) Video Compression: H.264 & H.265, up to 4K resolution.
      3) Max IP Camera Inputs: 32, 48 or 64.
      4) IP Camera Performance: Standalone 600 Mbps.
      5) Video Output: HDMI.
   b. Network
      1) Network 1: GigE port for connection to corporate network.
      2) Network 2: GigE port for connection to camera network.
      3) Network 3: GigE port for connection to redundant network, metadata capture.
      4) Security: TLS 1.2 with strong AES encryption, HTTPS encryption, strong hashing algorithms SHA256, continuous security vulnerability assessment.
      5) Interface: 10/100/1000Base-T Ethernet (3 x RJ-45).
   c. Systems/Network Management
      1) Operating system: Linux.
      2) Bandwidth Throttle.
      3) Bandwidth Scheduling.
      4) Adaptive Bandwidth.
      5) Remote Connection: TCP/IP (IPv4).
      6) Concurrent Remote Connections: No set limit.
      7) Internal Battery.
   d. Analytics to include motion detection and face detection.
   e. Storage
      1) Internal Hard Drives: up to (8) 3.5 inch HDDs.
      2) Individual Drive Capacity: Up to 20 TB.
      3) Total Storage Capacity: Up to 160 TB.
      4) Drive Type: Serial ATA (SATA).
      5) Mirroring.
   f. Hardware
      1) SOC: Nvidia Jetson Platform.
      2) CPU: 4 ARM core 64 bit processors.
      3) GPU: Up to 256 GPU cores.
   g. Alarm Inputs/Outputs
      1) Alarm Inputs: 8 current loop inputs, supporting open and closed detection.
      2) Relay Inputs: 4 relay switch outputs (60 VDC, 1A); programmable event driven.
   h. Physical
      1) Dimensions (HxWxD): 3.5x19.0x21.0 in.
      2) Mounting: 2U high, 19 inch rack mount.
   i. Environmental
      1) Operating Temperature: 0 degrees C to 40 degrees C (32 degrees F to 104 degrees F).
      2) Heat Dissipation: Maximum 188 BTU per hour (system plus 4 hard drives).

3) Humidity: 5 percent to 95 percent RH (non-condensing).
4) Storage Conditions: -40 degrees C to 70 degrees C (-40 degrees F to 158 degrees F).

j. Electrical
1) Internal Power Supply: Dual redundant hot-swap,115 VAC, 2.5A, 60 Hz.
2) Power Consumption: 85W typical, 150W Max.

k. Warranty: 3 years - parts/labor/shipping.

l. Basis of Design:
1) March Networks #X32 (32 Channel).
2) March Networks #X48 (48 Channel).
3) March Networks #X64 (64 Channel).
4) Product substitutions not permitted.

2.03 IP VIDEO SWITCH

A. CISCO Network Switch (IP Video):
1. CISCO Systems (CISCOPRO) Switch.
2. Five (5) year service warranty.
3. Basis of Design: CISCO Catalyst 9200L, (24) port PoE+, 4x1G Network Advantage Switch #C9200L-24P-4G-A with the following options:
a. C9200L Network Advantage, (24) port license; #C9200L-NW-A-24.
b. North American AC type "A" power cable; #CAB-TA-NA (2 required per switch).
c. C9200L CISCO DNA Advantage, (24) port, (5) year term license, (60) month service duration; #C9200L-DNA-A-24-5Y.
d. Prime infrastructure lifecycle and assurance term-smart license; #PI-LFAS-T.
e. Prime infrastructure development license for lifecycle and assurance term - 5Y service duration, (60) months; #PI-LFAS-AP-T-5Y.
f. CISCO Catalyst 9200L stack module; #C9200L-Stack-Kit (required at headend rack only).
g. Catalyst 9200 stack module; #C9200-Stack (required at headend rack only).
h. 50CM Type 4 stacking cable; #Stack-T4-50CM (required at headend rack only).
i. Network plug-n-play connect for zero touch device deployment; #NETWORK-PNP-LIC (required at node cabinets only).
4. CISCO Switches shall be furnished, installed and wired by the General Contractor. Programing of the switch 'may be' performed by USPIS-CCTV or a certified March Networks installer.

2.04 VIDEO DECODERS

A. Video Decoders shall be (64) channel and support up to (2) remote monitors with full screen camera views; (4) to (16) camera views per monitor.

B. Camera displays approved only by OIG and USPIS.
1. Video Output - HDMI.
2. Video Decoding - H.265, H.264 and MPEG-4 Unicast and Multicast.
3. Security - Password protected user access HTTPS encryption.

C. Decoder shall be wall mounted behind the CCTV monitor(s) utilizing factory wall brackets. The decoder shall be supplied with 120 Volt obtained from the monitor's UPS unit.

D. Basis of Design: Hanwha Techwin #SPD-152.

E. Alternate manufacturer: Vicon #VECA-SF00N0-00.

2.05 VIDEO CAMERAS

A. IP Cameras for video surveillance and monitoring of specific areas as shown on the drawings and confirmed with Postal Inspection Service and/or OIG through the USPS Project Manager.

B. Fixed, indoor/outdoor, 5 MP (minimum), dome type camera shall be a network camera with WDR, light finder, remote focus and zoom and shall incorporate Power over Ethernet. The camera shall meet or exceed the following requirements:
   1. Be equipped with a 10BaseT/100BaseTX Ethernet interface.
   2. Include a vandal resistant, indoor/outdoor casing with smoked transparent cover where required.
   3. Equipped with pixel counter.
   4. Image sensor: 1/2.7 inch Progressive scan RGB CMOS.
   5. Lens:
      a. Varifocal, 3 to 8mm, F1.3
      b. Horizontal field of view: 104 degrees - 40 degrees
      c. Vertical field of view: 74 degrees - 29 degrees
      d. Minimum focus distance: 1 m (3.28 ft.)
      e. IR corrected, remote zoom and focus, P-Iris control.
   6. Day and night: Automatically removable infrared-cut filter.
   7. Minimum illumination:
      a. Color: 0.13 LUX @ 50 IRE, F1.3.
      b. B/W: 0.0 LUX @ 50 IRI, F1.3.
   8. Shutter speed: 1/33500 to 1/5 second; 60 Hz.
   9. Pan/Tilt/Zoom: Digital PTZ, preset positions, guard tour.
   10. Angle Adjustment: Pan ±180 degrees, tilt -10 to +80 degrees, rotation ±190 degrees.
   11. Resolution: 2592x1944 (5 MP minimum).
   12. Video compression:
      a. H.264 (MPEG-4 Part 10/AVC) Baseline, Main and High Profiles.
      b. H.265 (MPEG-H Part 2/HEVC), Main Profile.
      c. Motion JPEG.
   13. Frame Rate: 30 fps in all resolutions; 60 Hz
   14. Video streaming: Multiple individually configurable streams in H264 and H265 and Motion JPEG.
   15. Support both unicast and multicast MPEG-4.
   16. Support Power over Ethernet according to IEEE802.3af.
   17. Support both IPv4 and IPv6.
   18. Provide multiple user passwords, support for HTTPS and SSL/TLS and incorporate IEEE802.1X authentication.
   19. Be equipped with 1 alarm input and 1 alarm output.
   20. Include embedded event functionality, which may be triggered by alarm input or by video motion or audio detection.
   21. Be supported by an open and published API.
   22. Casing: Indoor/Outdoor; IP52, NEMA 4X and IK10 impact resistant, polycarbonate dome with encapsulated electronics (1.8 lbs).
   23. Processor and Memory: ARTPEC-8, 2048 MB RAM, 8192 MB Flash.
   24. Connectors: RJ45 10 BASE - T/100BASE-TX PoE terminal block for (1) alarm input and (1) alarm output.
   25. Operating Conditions: Indoor/Outdoor; -40 to 122 degrees F; 10 to 100 percent RH.
   26. Accessories: Mounting plate, smoked transparent cover. Provide ceiling, pendant or wall bracket mounting and connector kits.
   27. Basis of Design:
      a. Indoor: Axis #P3267-LV.
      b. Outdoor: Axis #P3267-LVE.
   28. Alternate manufacturers:
      a. Hanwha Techwin #XND-C8083RV (indoor)
      b. Hanwha Techwin #XNV-C8083R (outdoor)

c. March Networks #ME6 - indoor/outdoor IR dome

C. Indoor/Outdoor PTZ camera shall be a network dome camera and shall incorporate 25x (minimum) optical zoom, day/night functionality, and simultaneous Motion JPEG and MPEG-4 video streams. Camera Shall meet or exceed the following requirements:
1. Be equipped with a 10BaseT/100BaseTX Ethernet interface.
2. Include a vandal resistant, indoor/outdoor casing with smoked transparent cover.
3. Feature a progressive scan CMOS sensor with Wide Dynamic Range (WDR), electronic image stabilizer and day/night functionality.
4. Be equipped with 25x optical zoom (minimum).
5. Image Sensor: 1/2.8" Progressive scan RGB CMOS.
6. Lens: Varifocal, F1.4 to F4.0, 4.3 to 137.6 mm, angle of view: Horizontal - 58.3 to 2.4 degrees, vertical - 35.5 to 1.4 degrees.
7. Day and night: Automatically removable infrared cut-out filter.
8. Minimum Illumination:
   a. Color: 0.1 LUX @ 30 IRE F1.4.
   b. B/W: 0.01 LUX @ 30 IRE F1.4.
9. Shutter Time: 1/66,500s to 2s.
10. PTZ:
    a. E-Flip, 256 preset positions.
    b. 32x optical zoom and 12x digital zoom, total 384x zoom.
    c. Pan: 360 degrees endless, 0.1 to 350 degrees/s.
    d. Tilt: 180 degrees, 0.1 to 350 degrees/s.
11. Video Compression: H264 (MPEG - 4 part 10/AVC) baseline, main and high profiles motion J-PEG.
12. Resolution: 1920 x 1080 (2 MP minimum).
13. Frame Rate: Up to 60 fps in all resolutions.
14. Video streaming: Multiple individually configurable streams in H264 and H265 and Motion JPEG.
15. Support multiple, motion JPEG4, H264, and H265.
16. Support Power over Ethernet according to IEEE802.3af.
17. Support both IPv4 and IPv6.
18. Provide multiple user passwords, support for HTTPS and SSL/TLS and incorporate IEEE802.1X authentication.
19. Be equipped with full memory card for alarm triggers.
20. Include embedded event functionality, which may be triggered by alarm input or by video motion or audio detection.
21. Be supported by an open and published API.
22. Casing: Indoor/Outdoor; IP66, IK10 and NEMA 4X impact - resistant aluminum.
23. Processor and Memory: 1024 MB RAM, 512 MB Flash.
24. Connectors: RJ45 10 BASE - T/100BASE-TX PoE push-pull connector for (2) alarm input and (2) alarm output.
25. Operating Conditions: -22 to 122 degrees F; 10 to 100 percent RH.
26. Security: Password protection, IP address filtering, HTTPS encryption, IEEE 802.1x network access control.
27. Power: 24 to 28 VDC max 18W; power over Ethernet IEEE 802.3at.
28. Accessories: Mounting plate, smoke transparent cover. Provide ceiling, pendant, recessed housing, or wall bracket mounting and connector kits.
29. Basis of Design: Indoor/Outdoor, Axis #P5655-E.
30. Alternate manufacturers:
    a. Hanwha Techwin #XNP-9250
    b. March Networks #SE2

D. Indoor, 360 degree panoramic (fisheye) view, fixed dome camera shall be 12 mega pixel , network type with WDR, light finder, remote focus and zoom and shall incorporate Power over Ethernet (PoE). The camera shall meet or exceed the following requirements:
   1. Be equipped with a 10BaseT/100BaseTX Ethernet interface.
   2. Include a vandal proof resistant casing with smoked transparent cover.
   3. Equipped with pixel counter.
   4. Image Sensor: 12 MP (4000 x 3000), 1/2.3 inch progressive scan RGB CMOS.
   5. Lens:
      a. 1.2 mm, F2.2
      b. Horizontal field of view: 182 degrees
      c. Vertical field of view: 182 degrees
      d. Fixed iris, fixed focus, IR corrected.
   6. Day and night: Automatically removable infrared-cut filter.
   7. Minimum Illumination:
      a. Color: 0.19 LUX @ 50 IRE, F2.2
      b. BW: 0.04 LUX @ 50 IRE, F2.2
   8. Camera angle adjustment: Digital roll: ±180 degrees.
   9. Shutter speed: 1/8,100s to 1/2s
   10. Video compression: H264 (MPEG-4 part 10/AVC), baseline, main and high profiles, H265 (MPEG-H Part2).
   11. Resolutions:
      a. Overview: 2992x2992 to 160x160 (1:1)
      b. Panorama: 3840x2160 to 192x72 (8:3, 16:9 or 32:9)
      c. Double panorama: 3584x2688 to 384x288 (4:3 or 16:9)
      d. Quad view: 3584x2688 to 384x288 (4:3 or 16:9)
      e. View area 1-4: 2048x1536 to 256x144 (4:3 or 16:9)
      f. Corner left/right: 3200x1600 to 192x72 (2:1 or 8:3)
      g. Double corner: 2880x2880 to 384x288 (1:1 or 4:3)
      h. Corridor: 2560x1920 to 256x144 (4:3 or 16:9)
   12. Frame rate:
      a. 360° overview, only up to max. resolution without WDR: 30 fps @ 60 Hz
      b. 360° overview and 4 dewarped views up to max. resolution with WDR: up to 20 fps @ 60 Hz
   13. Video Streaming: Multiple, individually configurable streams in H264, H265, and Motion JPEG.
   14. Multi-view streaming: 360° overview, dewarped panorama, corridor, corner left/right and quad views.
   15. Pan/Tilt/Zoom: Digital PTZ of view areas, digital PT of panorama, corner, corridor and quad views, preset positions, guard tour.
   16. Support Power over Ethernet according to IEEE802.3at.
   17. Support both IPv4 and IPv6.
   18. Provide multiple user passwords, support for HTTPS and SSL/TLS and incorporate IEEE802.1X authentication.
   19. Be equipped with 1 alarm input and 1 alarm output.
   20. Include embedded event functionality, which may be triggered by alarm input or by video motion or audio detection.
   21. Be supported by an open and published API.
   22. Casing: IK08 impact resistant aluminum and plastic casing with polycarbonate hard-coated dome.
   23. Processor and Memory: ARTPEC-8, 2048 MB RAM, 8192 MB Flash.
   24. Connectors: RJ45 10 BASE - T/100BASE-TX PoE terminal block for (1) alarm input and (1) alarm output.

25. Power: 24 to 34 VDC, max 5.1 Watts, PoE (IEEE802.3af) class 2.
26. Operating Conditions: 32 to 104 degrees F, Humidity 10 to 85 percent RH (condensing).
27. Accessories: Mounting plate, vandal casing. Provide ceiling pendant and recessed housing where indicated.
28. Basis of Design: Axis #M4328-P.
29. Alternate manufacturers:
    a. Hanwha Techwin #QNF-9010

E. Outdoor, 180 degree panoramic view, fixed dome camera shall be 13 mega pixel , network type with forensic WDR, light finder, remote focus and zoom and shall incorporate Power over Ethernet (PoE). The camera shall meet or exceed the following requirements:
1. Be equipped with a 10BaseT/100BaseTX Ethernet interface.
2. Include a vandal proof resistant casing with smoked transparent cover, fan, and heater.
3. Equipped with pixel counter.
4. Image Sensor: 3 x 5 MP, 1/2.8 inch progressive scan RGB CMOS.
5. Lens:
    a. Fixed 3.2 mm, F2.0
    b. Horizontal field of view: 180 degrees
    c. Vertical field of view: 90 degrees
6. Day and night: Automatically removable infrared-cut filter.
7. Minimum Illumination:
    a. Color: 0.16 LUX, F2.0
    b. BW: 0.05 LUX, F2.0
8. Camera angle adjustment:
    a. Pan: ±180 degrees
    b. Tilt: 0 degrees to 50 degrees.
    c. Roll: ±5 degrees.
9. Shutter speed: 1/91000s to 1/25s with WDR at 1/50000 to 1/25s.
10. Video compression: H264 (MPEG-4 part 10/AVC), baseline, main and high profiles, H265 (MPEG-H Part 2/HEVC) main profile. Motion JPEG.
11. Resolutions: 5120 x 2560 (13.1 MP) to 256x128.
12. Frame rate:13.1 MP @ 30 fps (60 Hz) WDR.
13. Video Streaming: 13.1 MP: 1 configurable stream in H.264, H.265, and Motion JPEG in full frame rate.
14. Support Power over Ethernet according to IEEE802.3at.
15. Support both IPv4 and IPv6.
16. Provide multiple user passwords, support for HTTPS and SSL/TLS and incorporate IEEE802.1X authentication.
17. Be equipped with 1 alarm input and 1 alarm output.
18. Include embedded event functionality, which may be triggered by alarm input or by video motion or audio detection.
19. Be supported by an open and published API.
20. Casing: Outdoor, IP66/IP67 and NEMA 4X rated, IK10-rated impact resistant casing with polycarbonate hard-coated dome, aluminum base and dehumidifying membrane.
21. Processor and Memory: ARTPEC-7, 2048 MB RAM, 512 MB Flash.
22. Connectors: RJ45 1000 BASE - T with terminal block for two configurable supervised inputs/digital outputs (12 VDC output, max load 50 mA), 3.5mm analog/digital mic/line in.
23. Power: Camera with built-in fan and heater, 24 to 34 VDC, max 12.95 Watts, PoE (IEEE802.3af) class 3.
24. Operating Conditions: -40 to 122 degrees F, Humidity 10 to 100 percent RH (condensing).
25. Accessories: Outdoor, weather shield, cable shield, 16 foot network cable with pre-mounted gasket. Provide pole attachment, pendant, and recessed housing where indicated.

26. Basis of Design: Axis #P3818-PVE.
27. Alternate manufacturers: Hanwha Techwin #XNF-9010RV.

F. Outdoor, 360 degree panoramic (fisheye) view, fixed dome camera shall be 12 mega pixel, network type with WDR, light finder, remote focus and zoom and shall incorporate Power over Ethernet (PoE). The camera shall meet or exceed the following requirements:
1. Be equipped with a 10BaseT/100BaseTX Ethernet interface.
2. Include a vandal proof resistant casing with smoked transparent cover, fan, and heater.
3. Equipped with pixel counter.
4. Image Sensor: 12 MP (4000 x 3000), 1/2.3 inch progressive scan RGB CMOS.
5. Lens:
   a. 1.2 mm, F2.2
   b. Horizontal field of view: 182 degrees
   c. Vertical field of view: 182 degrees
   d. Fixed iris, IR corrected, fixed focus.
6. Day and night: Automatically removable infrared-cut filter.
7. Minimum Illumination:
   a. Color: 0.19 LUX @ 50 IRE, F2.2
   b. BW: 0.04 LUX @ 50 IRE, F2.2
8. Camera angle adjustment: Digital roll: ±180 degrees.
9. Shutter speed: 1/8100s to 0.5s
10. Video compression: H264 (MPEG-4 part 10/AVC), baseline, main and high profiles, H265 (MPEG-H Part2).
11. Resolutions:
    a. Overview: 2992x2992 to 160x160
    b. Panorama: 3840x2160 to 192x72
    c. Double panorama: 3584x2688 to 512x288
    d. Quad view: 3584x2688 to 384x288
    e. View area 1-4: 2048x1536 to 256x144
    f. Corner left/right: 3200x1200 to 192x72
    g. Double corner: 2880x2880 to 384x288
    h. Corridor: 2560x1920 to 256x144
12. Frame rate:
    a. 360° overview only, up to 2992x2992 without WDR: 30 fps @ 60 Hz
    b. 360° overview and dewarped views with WDR: up to 20 fps @ 60 Hz
13. Video Streaming: Multiple, individually configurable streams in H264, H265, and Motion JPEG.
14. Pan/Tilt/Zoom: Digital PTZ of view areas, digital PT of panorama, corner, corridor and quad views, preset positions, guard tour.
15. Support Power over Ethernet according to IEEE802.3at.
16. Support both IPv4 and IPv6.
17. Provide multiple user passwords, support for HTTPS and SSL/TLS and incorporate IEEE802.1X authentication.
18. Be equipped with 1 alarm input and 1 alarm output.
19. Include embedded event functionality, which may be triggered by alarm input or by video motion or audio detection.
20. Be supported by an open and published API.
21. Casing: Outdoor, IP66 and NEMA 4X, impact resistant aluminum and plastic casing with polycarbonate hard-coated dome.
22. Processor and Memory: ARTPEC-8, 2048 MB RAM, 8192 MB Flash.
23. Connectors: RJ45 10 BASE - T/100BASE-TX PoE terminal block for (1) alarm input and (1) alarm output.

24. Power: Camera with built-in fan and heater, 24 to 34 VDC, max 12.95 Watts, PoE (IEEE802.3af) class 3.
25. Operating Conditions: -40 to 122 degrees F, Humidity 10 to 100 percent RH (condensing).
26. Accessories: Outdoor, weather shield, cable shield, 16 foot network cable with pre-mounted gasket. Provide pole attachment, pendant, and recessed housing where indicated.
27. Basis of Design: Axis #M4318-PLVE.
28. Alternate manufacturers: Hanwha Techwin #XNF-9010RV.

G. Exterior fixed camera shall be 4 MP (minimum), outdoor, network type with WDR, light finder, remote focus and zoom and shall incorporate Power over Ethernet (PoE). The camera shall meet or exceed the following requirements:
1. Be equipped with a 10BaseT/100BaseTX Ethernet interface.
2. Include a vandal proof resistant casing with fan and heater.
3. Equipped with pixel counter.
4. Image Sensor: 1/2.7inch progressive scan RGB CMOS.
5. Lens:
   a. Varifocal 2.8-8 mm, F1.2
   b. Horizontal field of view: 90 - 38 degrees
   c. Vertical field of view: 67 - 28 degrees
   d. IR corrected, C5-mount lens, P-Iris.
6. Day and Night: Automatic IR filter removal in low light conditions.
7. Minimum Illumination:
   a. Color (HDTV): 0.13 LUX @ 50 IRE, F1.2.
   b. BW (HDTV): 0.03 LUX @ 50 IRE, F1.2.
8. Shutter speed:
   a. WDR: 1/33500s to 1/5s.
   b. No WDR: 1/50,000s to 1/5s
9. Video compression:
   a. H.264 (MPEG-4 Part 10/AVC) Baseline, Main and High Profiles.
   b. H.265 (MPEG-H Part 2/HEVC), Main Profile.
   c. Motion JPEG.
10. Resolutions:
    a. 2592 x 1944 (5 MP)
    b. 268 x 1512 (4 MP).
11. Frame rate:
    a. Capture mode 5 MP: 30 fps
    b. Capture mode 4 MP: 30 fps.
12. Video Streaming: Multiple, individually configurable streams in Motion JPEG, H.264, and H.265.
13. Support Power over Ethernet according to IEEE802.3af.
14. Support both IPv4 and IPv6.
15. Provide multiple user passwords, support for HTTPS and SSL/TLS and incorporate IEEE802.1X authentication.
16. Be equipped with 1 alarm input and 1 alarm output.
17. Include embedded event functionality, which may be triggered by alarm input or by video motion or audio detection.
18. Be supported by an open and published API.
19. Casing: Outdoor; IP66, IP67 and NEMA 4X, IK10 impact resistant polymer enclosure and aluminum base with integrated humidifying membrane.
20. Processor and Memory: ARTPEC-7, 1 GB RAM, 512 MB Flash.
21. Connectors: RJ45 10 BASE - T/100BASE-TX PoE terminal block for (1) alarm input and (1) alarm output.

22. Power: Camera with built in fan and heater, 12 to 28VDC, max 25.5 Watts, PoE (IEEE802.3af) class 4.
23. Operating Conditions (extreme cold weather): -40 to 140 degree F, Humidity 10 to 100 percent RH (non-condensating).
24. Accessories: Outdoor, weather shield, cable shield, 16 -foot network cable with pre-mounted gasket. Provide wall mount pole attachment.
25. Basis of Design: Axis #P1377-LE.

H. Exterior, long range, bullet type fixed camera shall be 4 MP (minimum), outdoor, network type with WDR, light finder, remote focus and zoom and shall incorporate Power over Ethernet (PoE). The camera shall meet or exceed the following requirements:
1. Be equipped with a 10BaseT/100BaseTX Ethernet interface.
2. Include a vandal proof resistant casing with fan and heater.
3. Equipped with pixel counter.
4. Image Sensor: 1/1.8" progressive scan RGB CMOS.
5. Lens:
   a. 4.3 - 137 mm, F1.4 - 4.0
   b. Horizontal field of view: 60 - 2.3 degrees
   c. Vertical field of view: 39 - 1.3 degrees
   d. Autofocus, auto-iris, automatic day/night
   e. Optimized IR with 80 meter (262 ft) range
6. Day and Night: Automatic IR filter removal in low light conditions.
7. Angle of view: 60 to 2.3 degree horizontal; 39 to 1.3 degree vertical.
8. Minimum Illumination:
   a. Color (HDTV): 0.18 LUX @ 50 IRE, F1.4.
   b. BW (HDTV): 0.04 LUX @ 50 IRE, F1.4.
9. Shutter speed: 1/100,000s to 2s.
10. Video compression:
    a. H.264 (MPEG-4 Part 10/AVC) Baseline, Main and High Profiles.
    b. Motion JPEG.
11. Resolutions:
    a. 2560x1400 to 160x120
    b. Maximum pixel desnit with 32x optical zoom:
       1) 25 m (82 ft): 2551 px/m
       2) 30 m (164 ft.): 1275 px/m
       3) 250 m (820 ft.): 255 px/m
12. Frame rate:
    a. WDR: 30 fps, 60 Hz.
    b. No WDR: 60 fps, 60 Hz.
13. Pan/Tilt/Zoom: 32x optical zoom.
14. Video Streaming: Multiple, individually configurable streams in Motion JPEG and H.264, VBR/ABR/MBR H.264.
15. Support Power over Ethernet according to IEEE802.3af.
16. Support both IPv4 and IPv6.
17. Provide multiple user passwords, support for HTTPS and SSL/TLS and incorporate IEEE802.1X authentication.
18. Be equipped with 1 alarm input and 1 alarm output.
19. Include embedded event functionality, which may be triggered by alarm input or by video motion or audio detection.
20. Be supported by an open and published API.
21. Casing: Outdoor; IP66, IP67 and NEMA 4X, IK10 impact resistant aluminum with integrated dehumidifying membrane. IK08 impact-resistant glass front window and weathershield.

22. Processor and Memory: 1024 MB RAM, 512 MB Flash.
23. Connectors: RJ45 10 BASE - T/100BASE-TX PoE terminal block for (1) alarm input and (1) alarm output.
24. Power: Camera with built in fan and heater, 20 to 28VDC, max 12.95 Watts, PoE (IEEE802.3af) class 3.
25. Operating Conditions: -40 to 140 degree F, Humidity 10 to 100 percent RH (non-condensating).
26. Accessories: Outdoor, weather shield, cable shield, 16 -foot network cable with pre-mounted gasket. Provide NEMA 4, watertight factory, cast aluminum backbox.
27. Basis of Design: Axis #Q1786-LE.

I. Products shall utilize internal or external surge protection such that a normally occurring power surge shall not void any manufacturer's warranty.

J. Product model numbers indicated with the cameras are for convenience only. Errors or obsolescence shall not relieve the furnishing of cameras, which meet the technical description given in specifications noted or required by function designated. Cameras of equal or better specifications shall be provided for those cameras found to be discontinued by the manufacturer.

2.06 VIDEO MONITORS

A. Provide 27 inch LCD flat-panel color monitor(s) with the following minimum capabilities.
   1. Product Requirements:
      a. Video Interface Connections: HDMI - 1 in, display port in
         1) Switching between inputs shall be performed using a front panel control.
         2) Resolution shall be equal to the native resolution of the installed Network Video Recorder, if applicable.
      b. Input Power: 120VAC, 60Hz (a power adaptor may be used to provide this voltage).
      c. Mounting: Each monitor shall be wall or desktop mounted. VESA mounting holes shall be provided and a series of optional VESA compliant mounts shall be made available at extra cost.
      d. Operating Temperature: Range shall be equal to or greater than 50 to 104 degrees Fahrenheit.
      e. Humidity: Withstand a minimum of 10% to 80% humidity.
      f. Resolution: 1920 x 1080 FHD.
      g. Pixel Pitch: 0.3114x0.3114 mm.
      h. Brightness: 250 nits (typ.) 200 nits (min).
      i. Contrast Ratio: 1000:1.
      j. Panel Type: IPS.
      k. Panel Aspect Ratio: 16:9.
      l. Warranty: 3 years - parts/labor.
      m. Adjustments: Must support on-screen display for setup and adjustment of monitor parameters.
      n. Colors: Must support a minimum of 16.7 million colors.
      o. Basis of Design: Samsung #F27T450FQN.
      p. Alternate Models: Dell #P2722H

B. Provide 50 inch LCD flat-panel color monitor(s) with the following minimum capabilities.
   1. Product Requirements:
      a. Video Interface Connections: HDMI - 1 in, display port in
         1) Switching between inputs shall be performed using a front panel control.
         2) Resolution shall be equal to the native resolution of the installed Network Video Recorder, if applicable.
      b. Input Power: 120VAC, 60Hz (a power adaptor may be used to provide this voltage).

c. Mounting: Each monitor shall be wall or desktop mounted. VESA mounting holes shall be provided and a series of optional VESA compliant mounts shall be made available at extra cost.
d. Operating Temperature: Range shall be equal to or greater than 0 to 40 degrees Celsius.
e. Humidity: Withstand a minimum of 10% to 80% humidity.
f. Resolution: UHD (3840x2160).
g. Pixel Pitch: 0.285375x0.285375 mm.
h. Brightness: 300 cd/m^2.
i. Contrast Ratio: 4000:1.
j. Panel Type: LED BLU.
k. Warranty: 3 years - parts/labor.
l. Adjustments: Must support on-screen display for setup and adjustment of monitor parameters.
m. Colors: Must support a minimum of 16.7 million colors.
n. Basis of Design: Samsung #QE50T.
o. Alternate Models:
   1) Dell #C5522QT.

C. Provide a wall mounted UPS unit at each monitor station location.
   1. The UPS shall be line-interactive, rated 1000VA/900W with (18) minute battery reserve at 450 Watts; Tripp-Lite #SMART1000RMXL2U and #2POSRMKITWM wall bracket.

2.07 CAMERA POWER SUPPLIES

A. Interior Fixed Cameras: Camera shall be powered by PoE from network switch. Maximum total cable length (including horizontal and vertical distances) from switch to camera is 300 feet. Provide fiber cabling for cable runs exceeding 300 feet.
   1. Network switch shall be equipped with UPS power supply.

B. Interior and Exterior PTZ Cameras (non "Blue Sky"): Camera shall be powered by PoE from network switch. Maximum total cable length (including horizontal and vertical distances) from switch to camera is 300 feet. Provide fiber cabling for cable runs exceeding 300 feet.
   1. Network switch shall be equipped with UPS power supply.

C. Exterior Building Wall Mounted Fixed Cameras (non "Blue Sky"): Camera and enclosure shall be powered by PoE from network switch. Maximum total cable length (including horizontal and vertical distances) from switch to camera is 300 feet. Provide fiber cabling for cable runs exceeding 300 feet.
   1. Network switch shall be equipped with UPS power supply.
   2. Camera enclosures shall be equipped with integral heaters and defoggers.
   3. All exterior building wall mounted cameras are to be considered as "non blue sky" type.

D. Exterior Cameras ("Blue Sky"): Exterior cameras mounted remote from the building exterior wall are considered "blue sky" type.
   1. Wall mounted environmental enclosure power supplies shall be located in a suitably protected area near the camera. Provide individually fused power supplies.
   2. Pole mounted environmental enclosure power supplies shall be located within a NEMA 4 enclosure at the pole. Provide individually fused power supplies.
   3. Camera enclosures shall be equipped with integral heaters and defoggers.
   4. Equip environmental enclosures for exterior cameras with individual 120 VAC / 12 VDC power supplies when required.

E. Cameras requiring cable runs in excess of 300 feet and all exterior cameras not building wall mounted and exposed to the elements ("blue sky" type) shall utilize fiber optic transmission equipment and shall be powered by individually fused power supplies.

F. Provide a means for disconnecting camera power supplies from main power at the power supply enclosure, either through a detachable power cord, master fuse or circuit breaker located in the power supply cabinet, or other UL approved switching device.
   1. Provide dedicated 120VAC lockable panelboards as required, located in the Criminal Investigative Office (CIO) to serve all Investigative CCTV cameras. Comply with National Electric Code clearance requirements.
   2. Camera power supplies shall be fed from 120 Volt circuits segregated to serve only the power supplies. Circuits utilized to feed gate controllers, ePACS components or any other non-related loads shall not be utilized to serve the CCTV equipment.

G. Power supplies shall be rated to support 200 percent of the actual (nominal) power loading and shall be as recommended by the camera manufacturer, equipped with ESD protection for data and video feeds.

H. Enclosures housing camera power supplies, media converters, fiber patch box and 120 Volt receptacles shall contain interior back planes for mounting of all components. NEMA type 4, stainless steel or non-metallic polycarbonate, hinged and lockable enclosures shall be provided for exterior applications serving 3 or less pole mounted cameras. CCTV terminal cabinets shall be used to house the components serving more than 3 cameras. Refer to paragraph 2.16.

2.08 VIDEO CAMERA HOUSINGS AND MOUNTS

A. Provide arm brackets, recessed housings, surface mounts, ceiling mounts, pendant kits and surface conduit back boxes as required for all camera types with the following minimum capabilities:
   1. Interior Cameras:
      a. All cameras shall be in a housing that is coordinated with adjacent finishes with the appropriate mounting hardware. Selection of housings and mounts, including incremental changes to paint colors, dome materials, and cosmetic finishes shall be approved by the USPS or their authorized agent.
      b. All housings shall be sufficiently dust and moisture resistant to withstand normal environ-mental conditions in their chosen installation location.
      c. Hardware shall be provided to ensure tamper-resistant mounting in public access areas after normal business hours without modification to the integrity of the housing.
      d. Pendant mounts shall be suitable for use as wall, ceiling and column mounts. Pendant mounts shall attach to the appropriate camera housing using standard threaded, rigid aluminum (type IMC) pipes. Pipes are to be a minimum of 1-1/2 inch in diameter. Where indicated, furnish and install 1-1/2 inch pipe to pendant kit at each pendant hung camera (length as required).
      e. All pendant mounts shall incorporate installer provided safety chain or cable of sufficient endurance to support 2 times the weight of the camera and mounting hardware. Safety chain or cable shall be securely attached to the building structure at one end and to the bottom of the pendant stem at the other end.
      f. Camera mounts must be bolted or clamped, not welded, to allow simple relocation. If mounting height requires stems of 4 ft. or more, then additional bracing will be required to prevent camera movement.
   2. Exterior Cameras:
      a. Environmental: Thermostatically controlled heaters and blowers with defrosting capabilities.
      b. Moisture: Rainproof seals and gaskets.
      c. Wind Resistance: Rated for 80mph sustained winds, minimum.
      d. Ambient Temperature Rating: -22 to 131 degrees F.
      e. Areas with more demanding environmental conditions will be granted a deviation from this specification.

f. Exterior building mounted cameras shall be provided with surge protection at the camera and at the node or headend location.
g. All exterior housings, mounts and components including arm brackets, pole mounting kits, cabling, connectors, seals, etc. shall be rated NEMA 4 watertight. Provide factory termination kits and seals.

2.09 FIBER OPTIC MEDIA CONVERTER MODULES

A. Provide fiber optic transmitters, receivers and associated power supplies as recommended by the camera manufacturer.

B. Fiber optic transmission equipment shall be used when one or more of the following conditions are met:
1. Camera cable lengths (including horizontal and vertical distances) exceed 300 linear feet.
2. The camera is located outdoors, is not building wall mounted and is exposed to the elements ("blue sky" type).
   a. Building wall mounted cameras and any cameras protected by canopies or other architectural elements that shield them from direct view of the overhead sky are excluded from this requirement.
3. Cabling from the remote node cabinets to the CCTV headend.
4. The cable path is within 20 feet of a TIME or MIMS aerial.

C. Fiber transmitters and power supply modules located at field devices shall be low profile "miniaturized" type and shall be mounted in the NEMA 1 (indoor) or NEMA 4 (outdoor) enclosures containing the PoE Injectors for both fixed and 360 degree panoramic (fisheye) cameras.
1. AC power is required for the transmitter power supply and PoE injector.
2. The power cords shall be 24 inches long to avoid large cable bundles within the enclosures.

D. Fiber receiver modules located at the node or head-end locations shall be rack mounted.
1. If more than one fiber optic rack is used, modules shall be distributed as evenly as possible among the racks to reduce the load on the rack power supply and minimize the impact of a failed rack.

E. Fiber optic modules shall conform to the following minimum specifications:
1. 10/100 MBps RJ45 Ethernet port, LC Fiber Port.
2. 50/125, OM3, tight-buffered, multimode fiber.
3. PoE (PD) device or locally powered.
4. Protocol independent.
5. -31 to 158 deg operating Temperature.
6. IEEE 802.3, IEEE 803.2u and IEEE 803.2af Complaint.

F. Patch Cables
1. Provide fiber optic patch cables for patched connections within the node cabinets and NEMA 1/NEMA 4 fiber enclosures.

G. Alternate manufacturers: As listed in paragraph 2.1A.

2.10 REMOTE NODE CABINET

A. Remote Node Cabinet shall be 25 inch W x 42 inch H x 12 inch D, lockable and shall house an IP video system network switch, category 6 and fiber patch panels, fiber media converters, surge protectors, midspan injectors, plug-in strips, UPS and camera power supplies and accessories as indicated on the drawings.
1. Basis of Design: Great Lakes #DC07844A with forced fan ventilation.
2. Alternate Manufacturer: Middle Atlantic #VWM-4-5-42K-PW.

3. Supply 1000VA/900W, line-interactive, rack mounted UPS with 18 minute battery reserve rated at 450 Watts. Tripp-Lite #SMART1000RMXL2U.
4. Provide (2) eight position, 15 Amp, plug-in power strips and feed from a dedicated 20 Amp, 120 Volt circuit for each remote node cabinet.

B. Remote node cabinets shall be mounted high enough to deter unauthorized tampering, but low enough to avoid the use of motorized lifts for future repair or warranty work.
1. Node cabinets mounted within ePACS secured rooms shall typically be wall mounted top at 6 feet AFF.
2. Node cabinets mounted within the workroom and platform areas must be wall or column mounted bottom at no less than 9 feet AFF and no more than 14 feet AFF.

C. Interior cable runs from remote node cabinets to the CCTV headend rack shall be (6) count, OM3, interlocked armor, tight buffered, multi-mode, plenum rated indoor fiber cable (Superior Essex #L4006N401). Note that armor jacketed fiber cabling need not be contained within innerduct or conduit.

D. Each remote node cabinet containing surge protection devices shall be bonded to the overhead building steel or to the nearest telecommunication ground bar (TMG). Provide copper ground bus (Square "D" #PK27GTACU with compression lugs or equal) within these node cabinets and bond to building steel or "TGB" utilizing #6/THWN/CU/GRND conductor.

2.11 MIDSPAN INJECTORS / ETHERNET CABLE EXTENDERS

A. Provide Ethernet Midspan Injectors as required and recommended by the camera manufacturer.

B. When adding or replacing cameras to an existing CCTV system, camera total cable lengths (including horizontal and vertical distances) in excess of 300 feet and no more than 800 feet may be equipped with Ethernet cable extenders.

C. All camera cable runs exceeding 300 feet shall be fiber optic.

D. Modules located at the cameras shall be located within a properly sized junction box mounted near the camera. Field device modules require local 120 Volt power.
1. The power cords shall be 24 inches long to avoid large cable bundles within the enclosures.

E. Midspan injectors located at the head-end and node cabinets are standalone modules rack mounted within the equipment rack.

F. Basis of Design: Axis 30W Midspan.
G. Alternate manufacturers:
1. Hanwha Techwin
2. March Networks.

2.12 CABLING

A. Cabling requirements:
1. Interior cable runs from cameras to node cabinets or to the CCTV headend that do not exceed 300 feet shall be category 6 (purple in color); plenum rated.
2. Interior cable runs exceeding 300 feet from cameras to node cabinets or to the CCTV headend shall be 2 count OM3, interlocked armored, tight buffered, multimode, plenum rated indoor fiber cable.
3. Exterior cable runs routed to remotely located "blue sky" cameras shall be 2 count, OM3, non-armor jacketed, tight buffered, multi-mode, indoor/outdoor, plenum rated fiber cable. Multiple strand fiber cabling (6 or 12 strand) may be utilized to serve more than one camera provided (2) strands are dedicated for each camera.

4. Interior cable runs from remote node cabinets to the CCTV headend rack(s) shall be 6 counts, OM3, interlocked armored, tight buffered, multi-mode, plenum rated indoor rated fiber cable. Note that armor jacketed fiber cabling need not be contained within innerduct or conduit.
5. All exterior category 6 and fiber cable runs shall be contained in conduit or an approved raceway.

B. Camera Ethernet Data Cabling:
1. 4-Pair Category 6, plenum rated unshielded twisted pair cable (purple in color).
2. Provide and install the RJ45 male jack with coupler jack module attached at the camera end and RJ45 male jack at the patch panel end of each cable as indicated in paragraph 3.2 G.
   a. Basis of Design: Belden #AX104210 (coupler jack).
   b. Acceptable Alternate Manufacturer: ICC #IC107CP6BK (coupler jack).
3. Category 6 cable shall be terminated utilizing male RJ45 jacks on both ends of the cable to facilitate cable testing prior to installation of the node cabinets or headend. All testing shall be performed only after the cables have been terminated with the male RJ45 jacks.
4. Complies with individual characteristics established in ANSI/TIA/EIA-568-B terminated to T568A and all addendums for Category 6 cable performance specification.
5. Cabling and wire ways shall be installed in accordance with section 26 05 33.
6. Final category 6 cabling routed from the pole mounted NEMA 4 enclosures and the terminal cabinets, serving the "blue sky" cameras and the patch cords utilized within the exterior enclosures and cabinets, shall be "outside plant (OSP)" rated, gel-filled, direct burial type.

C. Power cable shall be appropriately sized to ensure that any signal loss as a function of cable length does not prohibit the delivery of sufficient voltage and current from the power supply to the powered device. A separate power cable may be required by the design engineer as shown on the drawings.

D. Cable shall have footage markings to Identify CCTV system cable lengths.

E. Fiber Optic - When fiber optic modules are required, provide fiber optic cable appropriate for the application. Cable shall conform to the following specifications:
1. OM3, interlocked armored, tight buffered, multimode, plenum rated indoor fiber.
2. OM3, non-armor jacketed, tight buffered, multimode, indoor/outdoor, plenum rated fiber.
3. "LC" type connectors shall be used on all cable terminations, including junction boxes and break-out trays.
4. Performance characteristics (including optical attenuation) shall be such that the fiber optic modules specified in Section 2.9 function to deliver signals end-to-end with sufficient bandwidth and quality to meet the specified application.
5. Physical characteristics such that the cable has sufficient strength and endurance to withstand installation and environmental conditions without adversely affecting optical performance.
6. At no time shall fiber optic cable have loose ends terminated and left loose. All fiber optic cable shall be looped, stored, connected and permanently mounted in appropriate LIU cabinets/devices prior to testing. All fiber cabling shall be terminated utilizing factory manufactured fan-out kits.
7. Fiber cabling not armor jacketed and routed within interior spaces shall be encased within appropriately sized plenum rated, innerducts.
8. Terminate and test the fiber optic cable and connectors.
9. Fiber distribution, patch boxes (interconnect centers) shall be provided within the NEMA 4 enclosures at the pole mounted, blue-sky type cameras for termination of the fiber cabling.

a. Provide (6) port fiber patch box complete with adapter plate, "LC" connectors, splice tray, protection sleeves, fan-out kit and enclosure.
    1) Basis of Design: RLH Industries #PWSN-A-56-63-1 (Slimline patch panel: 4 inch W x 5 1/2 inch H x 2 inch D.
    2) Alternate U.S. manufacturers permitted.

10. Fiber cabling shall be well managed and protected and never directly connected to any camera device or component.
11. Fiber patch boxes (interconnect centers) shall be equipped with covers, so the fiber strands are not exposed.
12. Cap all unused fiber ends to maintain cleanliness and avoid physical damage; all fiber strands are to be terminated.
13. Clean out any metal shavings or other debris from the enclosure.
14. Do not run fiber strands through any brackets or around other cables.
15. Label each fiber cable with to/from destination. Attach a panel label to the inside of the patch panel cover.
16. Allow access to optical fiber cabling for testing.
17. Protect connections against accidental contact with foreign objects that may disturb optical continuity.
18. Properly handle optical fiber cables and patch cords and promote their orderly management.
19. A fiber optic service loop of sheathed fiber no less than 20 feet at each end shall be installed at each termination point. All service loops shall be installed so that the minimum bend radius (10 times the outside diameter of the fiber) is maintained and shall be installed outside of the fiber optic termination housing. Once the fiber reaches the entrance point of the fiber optic patch panel, there shall be no less than 3 feet of unsheathed fiber installed neatly in the fiber optic storage tray prior to terminations being installed. Unsheathed fiber shall be installed in the storage tray per the fiber optic enclosures manufacturer's instructions.

Cable Type	Signal	Use
(2) Count, OM3, interlocked armored, tight buffered, multimode, indoor plenum rated fiber cable (Superior Essex #L4002N301 or Approved Equal)	Camera Data	Interior camera cable runs exceeding 300 feet. See Sections 2.9 and 2.14
(6) Count, OM3, interlocked armored, tight buffered, multimode, indoor plenum rated fiber cable (Superior Essex #L4006N401 or Approved Equal)	Data	Interior cable runs - node cabinets to headend. See Sections 2.9 and 2.14
(2) Count, OM3, non-armor jacketed, tight buffered, multimode, indoor/outdoor, plenum rated fiber optic (Superior Essex #W4002NG01 or Approved Equal)	Camera Data	Exterior cable runs to "blue sky" cameras. See Sections 2.9 and 2.14
Category 6 plenum rated cable with footage markings (purple) (Superior Essex #77-240-7B or Approved Equal)	Camera Data	Interior camera cable runs less than 300 feet. See Section 2.13
Category 6 "OSP" rated cable with footage markings (black) (Superior Essex #04-001-68 or approved equal)	Camera Data	Exterior final cable runs (50 feet maximum) to cameras. See Section 2.13.

2.13 CATEGORY 6 CABLING

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the work include the following:
   1. Belden
   2. Berk-Tek
   3. CommScope Uniprise
   4. General Cable
   5. Leviton
   6. Ortronics (Legrand)
   7. Panduit
   8. Superior Essex
   9. Product options and substitutions. Substitutions: Permitted if approved by the A/E and Manufacturer.

B. Conductors: 4 twisted pair, minimum 23 AWG, solid copper.
   1. Individually insulated plenum rated conductors under common plenum rated sheath. Provide outdoor, "OSP" rated cable and patch cords for exterior applications.
   2. Complies with individual characteristics established in TIA-568-C, and all addendums for Category 6 cable performance specification.
   3. Nominal Impedance: 100 ohms plus or minus 15 percent.
   4. Certified and capable of performing to a minimum of 250 MHz.
   5. Maintain manufacturer's twisting of wire pairs to termination point. Do not attempt to restore, modify, or add to manufacturer's twisting of cable. Do not untwist more than ½ inch of the stripped cable.

2.14 OM3 FIBER CABLING

A. Manufacturers: Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:

1. Belden
2. Berk-Tek
3. CommScope Uniprise
4. Corning Cable Systems
5. General Cable
6. Leviton
7. Optical Cable Corp.
8. Ortronics (Legrand)
9. Superior Essex
10. Product options and substitutions. Substitutions: Permitted if approved by the A/E and Manufacturer.

B. Conductors: 2 / 6 strand
1. Provide multi-strand, 50/125 micron, laser optimized, tight buffered, multimode, OM3 fiber cabling rated as follows:
a. 1 Gb/s ≤ 1000m @ 850 nm.
b. 1 Gb/s ≤ 600m @ 1300 nm.
2. The fiber cabling shall meet the following specifications:
a. TIA/EIA-455-78, ANSI/TIA-455-176-A-2003, "Detail Specification for 50 micron Core Diameter/125 micron Cladding Diameter Class 1a Graded-Index Multimode Optical Fibers."
3. Terminate fiber strands onto "LC" ports.
4. Provide individually insulated plenum rated strands under common plenum rated sheath.
5. Fiber cabling shall comply with individual characteristics established in TIA-568-C including all addendums for fiber optic cable performance specification.
6. Interior fiber cabling shall be indoor rated and interlocked armor jacketed.
7. All exterior and underground fiber cable shall be indoor/outdoor, plenum rated. Provide indoor/outdoor, "OSP" rated, fiber patch cords for exterior applications.

2.15 SECONDARY BONDING BUSBAR - SBB BEHIND HEADEND

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
1. Harger - P/N GBI/14212 G
2. Chatsworth
3. Legrand
4. Product options and substitutions permitted if approved by the A/E.

B. Provide and install one SBB behind the CCTV headend rack(s), below ceiling acoustic tile, with all bonding leads clearly labeled by machine labeler. All bonding leads shall be 2 hole compression lugs. This SBB will connect to the nearest telecommunication or electrical system grounding electrode ground bar using minimum #1/0/AWG/CU bonding conductor. Minimum size will be 2 inch H x 0.25 inch W x 12 inch L.

C. Each headend rack shall be bonded to the SBB using a #6/AWG/CU stranded bond wire.
D. Each (2) lug compression connector shall have antioxidant coating applied to lug and busbar prior to attachment.

2.16 ACCESSORIES

A. Lightning/Surge Protection: Products shall utilize internal or external (power and low voltage) surge protection such that a normally occurring power surge shall not void any manufacturer's warranty.

1. Rack mounted surge protectors shall be provided within the headend and remote node cabinets to protect the category 6 cabling serving the exterior, building wall mounted cameras. Modular surge protectors shall also be provided at the camera end of the category 6 cable.

B. The headend equipment racks shall house the IP video system network switches, NVR's, category 6 and fiber patch panels, fiber media converters, surge protectors, midspan injectors, laptop computer, plug-in power strips, power supplies, necessary accessories and utilize a standalone UPS. Provide dedicated 30 Amp, 120VAC power and NEMA L5-30R twist-lock receptacle.
   1. The UPS shall be line-interactive, rack mounted and rated 3kVA/2.88kW with a 10 minute battery reserve at 1440 Watts; Tripp-Lite #SMART3000RMXL2U.
   2. Provide (2) vertical mounted, sixteen position, 20 Amp, plug-in power strips within each rack.
   3. Provide tri-plex Telecommunications Outlet (T/O) bottom mounted at 24 inch AFF behind the headend rack.

C. Upright Racks: Provide and install appropriate number of 25 inch W x 72 inch H x 36 inch D, enclosed, upright equipment racks to provide sufficient mounting space for the required equipment.
   1. Racks shall be all metal construction conforming to EIA standards with 19 inch equipment mounting opening and 1-3/4" vertical spacing of equipment. Rack rails shall be punched with captive nuts, 10-32 screws and nylon washers.
   2. Basis of Design:
      a. Great Lakes #GL7202436 - Diebold4 with forced fan ventilation and rolling casters.
      b. Product substitutions not permitted.

2.17 CCTV TERMINAL CABINETS

A. CCTV terminal cabinets are typically used to house the fiber media converters and injectors serving more than (3) exterior pole mounted cameras. CCTV terminal cabinets shall be utilized to serve pole mounted cameras located no more than 50 feet (horizontally) from the cabinet. Underground runs of category 6/OSP/Wet cables serving pole mounted cameras in excess of 50 feet from the terminal cabinets are not permitted. Cameras located more than 50 feet from the terminal cabinets must utilize fiber cabling and pole mounted, NEMA 4 enclosures to avoid the increased risk of lightning induced transients.

B. CCTV terminal cabinets shall not contain ePACS components. The CCTV system shall utilize independent wiring, raceways, and cabinets.

C. Terminal cabinets shall be provided to house media converters, mid-spans, power supplies, SPD's and other CCTV system components. Enclosures shall be hinged and lockable with panelboard construction and plywood backboards.

D. Terminal cabinets shall be wall or pedestal mounted with bottom of cabinet at no less than 12 inches A.F.F. or A.F.G. Provide NEMA type 1 enclosures within interior locations and NEMA type 4 (watertight) stainless steel or non-metallic polycarbonate type for exterior locations. Pedestal mounted cabinets shall be supported utilizing 4 inch square concrete posts buried 24 inches below finished grade and set in concrete footing with 6 inches of concrete all around.

E. Terminal cabinets shall be amply sized to accommodate all components without overheating and forced air exhaust fans shall be provided. Cabinets requiring 120 Volt power shall be provided with appropriate number of 20 Amp, 125 Volt receptacles complete with surge protection. Receptacles shall be securely mounted within the cabinet.
   1. Components shall be individually mounted and secured to the backboard. Stacking of components is not acceptable and the use of tie-wraps is prohibited.

F. Exterior terminal cabinets mounted near or adjacent to vehicular traffic shall be protected using 6 inch dia. x 4 -foot high concrete bollards. Exterior terminal cabinets shall be located within the secured area of the facility.

G. Exterior terminal cabinets shall be equipped with a copper ground bus bonded to a driven ground rod using #2/AWG copper grounding electrode conductor.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 01 73 00 - Execution: Verification of existing conditions before starting work.

B. Verification of Conditions: Verify that field measurements, surfaces, substrates, and conditions are as required, and ready to receive work.
   1. Verify that power and video outlets are in correct locations.
   2. Verify that building structure for attachment of equipment mounting devices is in place.

C. Report in writing to the USPS Project Manager any prevailing conditions that will adversely affect satisfactory execution of work of this section. Do not proceed with work until unsatisfactory conditions have been corrected.

D. By beginning work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the Postal Service.

E. Provide required power outlets, low voltage power supplies, interconnecting cables, hardware and equipment for a complete and operable system.

F. Camera locations are to be reviewed and approved by a Postal Inspector and/or OIG, through the USPS Project Manager, prior to installation of conduit and cabling.

3.02 INSTALLATION

A. Install all equipment in accordance with Manufacturer's published instructions. Installation must be done by a Manufacturer Certified Installer to assure proper installation and accountability. This includes, but is not limited to the following:
   1. All hardware used to secure equipment to racking shall include nylon or other non-metallic washer or grommet between the screw head and equipment panel to prevent any damage to the equipment.
      a. Rack mount screws shall be self-centering, Philips-head configuration unless specialized tamper-resistant hardware has been specified.
      b. Screws shall be tightened in such a manner as to allow their removal with common hand tools.
   2. Any equipment placed on shelving mounted on an incline of greater than 2 degrees shall be secured to the rack or shelving in such a manner as to prevent movement of the equipment in the direction of the incline. Such fastening shall be done in a manner as to preserve the integrity of the equipment case and chassis, and shall in no way jeopardize warranty coverage of the device.
   3. All equipment cabling shall be dressed in such a manner as to ensure a neat and clean appearance.
   4. Cable breakouts shall be at 90-degree angles from the harness or chase, and all chases shall be parallel to or at 90-degree angles from the rack frame.
   5. Cables are to be secured to the rack frames at sufficient intervals to ensure that the weight of the cable will not contribute to fatigue or early failure of that cable or the device and connector to which it is attached.

6. Sufficient excess cable shall be provided in "service loop locations" to ensure that the cable may be re-connected without requiring the addition of extension pieces. Provide 20 foot, coiled service loops at the camera and headend/node termination points.
7. All permanent cabling shall be mechanically numbered in a manner consistent with the written system documentation.
8. All wiring to include category 6 and fiber optic cables shall utilize hook and loop fasteners to eliminate the risk of over-tightening cable bundles and affecting the strength or rated performance of the cable. The use of tie wraps is not acceptable.
9. Where wiring is routed through sheet metal or over frame members, the metal edges shall be covered with flexible grommeting or edge dressing (such as automobile door edge trim).
10. Double-sided foam tape shall not be used to secure any equipment, terminal blocks, or accessory devices. All device mounting shall be of a permanent nature.
11. All excess length AC cords are to be tie-wrapped out of the way. Where possible, they shall be routed in a separate bundle a minimum of 6 inches away from any signal or control cable.
12. Exposed wires run to wall mounted cameras shall be fed through tubing or the body of the mount to present a professional appearance.
    a. Any accessible cables that can be reached by an individual standing on the floor, a stool, or a small stepladder shall be encased in protective tubing or armored sheathing to prevent tampering or cutting with common hand tools.
13. Care shall be exercised at all times to protect Postal Service property. For example, ladders shall not be placed against wallpapered or finished surfaces, equipment, or furnishings; desks or countertops shall not be used in lieu of ladders.
14. Each camera shall be labeled by a numbering system requiring no more than three digits. Use the camera numbering system as shown on the Drawings to provide a consistent, matching, and accurate as-built documentation.
    a. Each pendant or arm mounted, PTZ camera shall be labeled on three sides with 3 inch high numbers.
    b. Each ceiling, wall or pendant mounted, 360 degree panoramic camera shall be labeled on three sides with 1 inch high numbers.
    c. Each ceiling, wall or pendant mounted, fixed camera shall be labeled on two sides with 1-1/2 inch high numbers.
    d. The outer shroud of each wall or pole arm mounted, fixed or 180 degree panoramic camera shall be labeled on two sides with 2 inch high numbers.
    e. The flange of each recessed ceiling mounted PTZ, fixed or 360 degree panoramic camera shall be labeled on two sides with 1 inch high numbers.
    f. Labeling shall be stenciled or laminated vinyl in a contrasting color to the camera housing.
    g. Labeling shall not be placed on lower dome or any area that would obstruct camera viewing.
15. Ensure that pendant mounted cameras are hung from stable, vibration free mounting platforms, using guy-wires or other support mechanisms to ensure stability where required. Mount cameras below any suspended lighting to avoid glare or reflection on camera dome and/or lens.
16. Perform complete programming of the system, in coordination with the Postal Inspector and USPS Project Manager or designated representative. Programming shall include, but not be limited to, elimination of duplicate or redundant titling information, synchronization of system clocks, camera sequences, dome presets, salvos and tours. Programming of any system passwords or limiting of accessibility prior to commissioning and training is prohibited.
17. Provide the "as-built" redlined drawings with job condition changes required to provide accurate close-out documentation.

B. Power requirements shall be determined by actual equipment used.

C. Ensure that:
   1. All applicable statutes, ordinances, regulations, license requirements and codes are fully complied with.
   2. All required permits are obtained.
   3. All required inspections are conducted.
   4. All necessary certificates are issued, obtained, and delivered to the Postal Service.
   5. All equipment installations and mounting are in strict accordance with requirements for applicable seismic classification.

D. Arrange all components to be mounted in the console(s)/rack(s) in accordance with the Manufacturer's recommendations and/or Postal Service provided System Elevation drawings. Design shall provide a neat appearance and accessibility for servicing equipment.

E. Provide required power outlets, interconnecting cables, hardware and equipment for a complete and operable system.
   1. Power, 120VAC: As required by codes and standards for the facility.
   2. Where conduit is used, a minimum of 40% excess capacity shall be provided for future use.
   3. All interior conduits shall be EMT with steel set-screw type couplings. All exposed exterior conduits (and interior conduits mounted below 10 feet - 0 inches AFF within the workroom) shall be heavy wall, rigid galvanized steel type.
   4. All concealed and exposed conduit and cabling shall be routed parallel and perpendicular to structural elements. Conduit penetrations through fire rated partitions shall be properly fire stopped.
   5. All cables and conduits shall be routed below the roof decking to avoid damage due to future reroofing. Raceways and cables shall not be routed between the decking flutes.

F. Install cameras as shown on the drawings and in accordance with the USPS specifications.
   1. Provide 84-inch minimum headroom below cameras and their mountings. Where necessary modify mounting type to maintain clearance.

G. All category 6 cable connections must be made to 8 pin coupler jacks at the device and to 8 pin, feed-thru coupler jack patch panels at the head end or node cabinets per T568A standard.
   1. The category 6 CCTV cabling shall be equipped with an RJ45/CAT-6 male jack with coupler jack attached at the camera end ready for final patch cord connection to the camera. The node or headend connection of the cable shall be a CAT-6/RJ45 male jack.
   2. The remote node cabinets and headend rack(s) shall be equipped with feed-thru, coupler jack patch panels to accept the RJ45 male jacks terminated on the camera cabling.

H. When not installed in cable trays, cable (category 6, fiber optic, and low voltage power) shall be supported with wide base cable hangers rated for proper support of category 6, fiber optic, and inner-duct cables (compliant with UL and NEC requirements for structured cabling).
   1. Cable hangers shall be installed every 3 to 6 feet and shall be rated to support the weight of the cable multiplied by a factor of three (3).
   2. Cable tray for camera wiring shall not include any low voltage AC wiring.
   3. Interior fiber optic cabling not equipped with armor jacketing shall be contained within plenum rated innerduct.

I. The entire CCTV system shall utilize an independent wiring system not shared with any other building system. The structured cabling system racks, the TE's, the fiber backbone, cable trays, etc. cannot be utilized for any CCTV system purpose. Cable trays installed for the CCTV cabling may be utilized to contain the ePACS wiring.

3.03 DOCUMENTATION

A. Provide high definition photographs showing the interior components of all equipment enclosures, terminal cabinets, remote node cabinets and the headend rack(s). Photographs shall show wiring and placement of the midspan injectors, fiber media converters, surge protectors, fiber patch boxes, power supplies, power strips and receptacles. Photographs shall be transmitted to the A/E and USPS Project Manager.

3.04 FIELD TESTING CATEGORY 6 COPPER AND FIBER OPTIC CABLE

A. Section 01 40 00 - Quality Requirements: Field testing and inspection.

B. Field Testing Procedures:
   1. Provide all equipment and services necessary to test the cabling.
   2. Test and calibrate instruments before testing.
   3. Re-terminate and retest any cable found to be defective.
   4. Perform cable testing and submit report prior to installation of any cameras or node cabinets.

C. Category 6 Copper Cable Testing:
   1. Use Level III Compliant test equipment.
   2. Test parameters shall include:
      a. Wire map.
      b. Insertion loss (attenuation).
      c. DC loop resistance.
      d. Return loss at camera.
      e. NEXT, NEXT at camera.
   3. Perform end-to-end tests of each 4-pair cable as follows:
      a. Pair/conductor for proper pinouts and continuity.
      b. Ground fault.
      c. Proper termination, shorts, and crossed pairs.
      d. Channel attenuation per TIA-568-C, including all addendums.
      e. Channel bi-directional worst case near end cross talk (NEXT) at frequencies up to 250 MHz, per TIA-568-C, including all addendums.
      f. Measured effective cable run length.

D. Fiber Optic Testing:
   1. Use 50/125 micron, OM3, multimode fiber optic cable testing.
   2. Perform testing of fiber in accordance with the fiber type being tested, TIA-526-14-A, Method B for Multimode Fiber (One Jumper/Two Adapters).
   3. Multimode fiber optic cable shall be tested bi-directionally at wavelengths of 850nm and 1300nm.
   4. The fiber testers and test heads shall have passed calibration within one year of actual test date.
   5. Tests include:
      a. Tier 1 Testing with Optical Loss Test Set (OLTS) that includes testing for length.
      b. Tier 2 Testing with OTDR to show all splices.

3.05 CONSTRUCTION COORDINATION

A. Interface the CCTV installation with Other Work that may be required.

3.06 FIELD QUALITY CONTROL

A. Section 01 40 00 - Quality Requirements: Inspection and testing procedures.

B. Inspection:

1. Inspect equipment installation, interconnection with system devices, mounting locations, and mounting methods.
2. Verify that units and controls are properly installed, connected, and labeled and that interconnecting wires and terminals are identified.

C. Testing:
1. Perform tests and provide test equipment, tools, and personnel required to conduct system tests and inspections. These tests shall include video quality and PTZ operation (where applicable) for all cameras.
2.
3. Conduct system acceptance test upon completion of installation using pre-approved procedures. Test shall consist of system, subsystem, and device level acceptance tests, including software.
4. Utilize USPS accepted CCTV Checklist for system testing.
5. Ensure that test procedures confirm each specification statement and manufacturer requirement has been met or exceeded. An actual demonstration of each system function and a simulation of each system failure shall be provided.
6. An acceptance test period of thirty days shall begin at the start of the acceptance test. Any system failure during the acceptance test period will suspend the acceptance test. The thirty-day test period will restart when the required repairs have been made and certified.
7. Perform all tests in the presence of the USPIS/OIG Personnel. The USPIS/OIG reserves the right to accept any portion or activate any phase prior to acceptance of entire system.

3.07 ADJUSTMENTS

A. Adjust manual lens irises to meet lighting conditions.

B. Adjust field of view for each camera per USPIS/OIG direction.

END OF SECTION 28 23 05

SECTION 31 23 00 - EXCAVATION AND FILL

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Excavating and backfilling for structures, utilities, and pavement.
   2. Pipe bedding.
   3. Compacting fill materials.
   4. Borings and casings under roads.

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other Documents.

C. Related Sections:
   1. Section 312000 - Earth Moving: Cutting, filling, and grading for proposed site improvements.
   2. Section 312317 - Rock Excavation: Removal of rock during excavation.
   3. Section 313200 - Soil Stabilization: Lime, cement, fly ash, and geotextile subgrade stabilizers.

1.02 REFERENCES

A. American Society for Testing and Materials (ASTM):
   1. ASTM D698 - Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort.
   2. ASTM D1557 - Test Method for Laboratory Compaction Characteristics of Soil Using Modified Effort.

B. American Association of State Highway and Transportation Officials (AASHTO):
   1. AASHTO T 180 - Moisture-Density relations of Soils Using a 10 Pound Rammer and an 18 Inch Drop.

C. American Water Works Association (AWWA):
   1. AWWA C 200 - Steel Water Pipe, 6 Inch and Larger.
   2. AWWA C 206 - Field Welding of Steel Water Pipe.

D. National Fire Protection Association (NFPA):
   1. NFPA 70 - National Electric code.

1.03 DEFINITIONS

A. Building Area Subgrade Pad: Portion of site directly beneath and within a line 10 feet beyond building and appurtenances including limits of any future building expansion areas indicated on Drawings.

1.04 SUBMITTALS

A. Section 013300 - Submittal Procedures: Procedures for submittals.
   1. Shop Drawings:
      a. Submit drawings or details indicating proposed alternate earthwork procedures or proposed procedures not indicated in Contract Documents.
      b. Shop Drawings or details pertaining to Site Utilities are not required unless required by regulatory authorities or unless use of materials, methods, equipment, or procedures are contrary to Drawings or these specifications are proposed. Do not perform work until required shop drawings have been approved by Contracting Officer.

2. Assurance/Control Submittals:
   a. Material Source: Submit name of imported materials suppliers. Provide materials from same source throughout the work. Change of source requires Contracting Officer approval.
   b. Test Reports: Submit the following reports directly to Contracting Officer from Testing Laboratory, with copy to Contractor:
      1) Test reports on borrow material.
      2) Verification of each footing subgrade.
      3) Field density test reports.
      4) Optimum moisture-maximum density curve for each type of soil encountered.
      5) Report of actual unconfined compressive strength and bearing tests/results for each strata tested. Give "three-dimensional" description of each test location.
   c. Certificates: Gradation and certification of aggregate material for Testing Laboratory review.
   d. Qualification Documentation: Submit earthwork company documentation of experience indicating compliance with specified qualification requirements.

B. Section 017704 - Closeout Procedures and Training: Procedures for closeout submittals.
   1. Project Record Documents: Accurately record the following.
      a. Spot elevations for building area subgrade pad.
      b. Location of existing utilities remaining, re-routed utilities, new utilities by horizontal dimensions, elevations or inverts, and slope gradients.

1.05 QUALITY ASSURANCE

A. Qualifications: Earthwork company specializing in performing the Work of this Section with minimum 5 years documented experience.
B. Regulatory Requirements: Perform earthwork in accordance with applicable requirements of governing authorities having jurisdiction.

1.06 PROJECT CONDITIONS OR SITE CONDITIONS

A. Existing Conditions: Requirements specified in Section 312000.
B. Existing Utilities: Requirements specified in Section 312000.

PART 2 - PRODUCTS

2.01 MATERIALS

A. Stockpiled on-site fill and backfill material specified in Section 312000, tested by Testing Laboratory and approved by Contracting Officer.
B. Imported off-site fill and backfill material specified in Section 312000, tested by Testing Laboratory and approved by Contracting Officer.
C. Pipe Bedding Material: Processed sand and gravel free from clay lumps, organic, or other deleterious material complying with the following gradation requirements:

SIEVE SIZE	PERCENT PASSING
1 Inch	100
3/4 Inch	90 to 100
3/8 Inch	20 to 55
No. 4	0 to 10
No. 8	0 to 5

D. Steel Casing Pipe: AWWA C 200, minimum grade B; size and wall thickness as indicated on Drawings.

E. Stabilization Fabrics and Geogrids:
   1. Mirafi 500X or 600X.
   2. Amoco Style #2002 Woven.
   3. Reemay Typar 3401 and 3601.
   4. Trevira S1114 and S1120.
   5. Tensar 1100 and 1200.

F. Filter/Drainage Fabrics:
   1. Mirafi 140 N.
   2. Amoco Style #4546.
   3. Reemay Typar 3341.
   4. Carthage Mills, Carthage 6%.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.
B. Verification of Conditions: Verify that field measurements, surfaces, and conditions are as required, and ready to receive Work.
C. Report in writing to Contracting Officer prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.
D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to United States Postal Service.

3.02 PREPARATION

A. Identify required lines, elevations, levels, contours, grades, and datum necessary to perform earthwork operations as indicated on Drawings.
B. Verify that survey benchmark and intended elevations for the Work are as indicated on Drawings.
C. Locate, identify, and protect existing utilities to remain and previously installed utilities that may be damaged by construction operations.
   1. Notify Contracting Officer, municipality, and utility company immediately of utilities, not indicated on Drawings, encountered.
   2. Maintain existing utilities, active utilities, and drainage systems in operating condition.
   3. Comply with utility company requirements and directions of Construction Manager to keep utilities in operation.
   4. Repair damage to utilities as directed by Contracting Officer.

D. Protect plant life, lawns, fences, existing structures, sidewalks, paving and curbs from earthwork operations, excavating equipment, and vehicular traffic.
E. Protect bench marks, property corners, and other survey monuments from damage or displacement. Where markers are required to be removed, provide removal and reinstallation by licensed land surveyor licensed in State where project is located.
F. Overexcavate areas of building subgrade found consisting of unsuitable materials as determined by Testing Laboratory and Contracting Officer. Prepare, fill with suitable material, and compact as specified. Stabilize areas as specified in Section 313200.

3.03 EXCAVATION

A. Excavation for filling and grading specified in Section 312000.

B. Rock excavation specified in Section 312317.
C. Excavation for Structures:
   1. Excavate subbase for building foundations, slabs-on-grade and site structures to width and depth indicated on Drawings.
      a. Cut excavation banks vertically.
      b. Remove rocks, loose soil, and debris from bottom of excavation.
      c. Overexcavate wet or unsuitable soil from bottom of excavation.
      d. Provide stable base for concrete reinforcing installation and concrete placement.
      e. Hand trim to indicated lines and grades just prior to concrete reinforcing installation.
   2. Provide protection for workers within trench areas in accordance with local, state, and national Occupational Safety and Health requirements and regulations.
      a. Trenches minimum 4 feet in depth.
   3. During excavation, stockpile materials suitable for backfilling away from excavation to prevent overloading, slides, or cave-ins.
   4. Remove material encountered in excavating operations that is unsuitable for backfilling, subgrade or foundation purposes as determined by Testing Laboratory and Contracting Officer. Dispose of materials off-site in an approved manner in accordance with requirements of authorities having jurisdiction.
   5. Prevent surface water from flowing into excavations by temporary grading or other approved methods.
      a. Do not allow water to accumulate in excavations.
      b. Remove accumulated water in excavations.
      c. Provide and maintain pumps, well points, sumps, suction and discharge lines, and other dewatering system components required to remove water from excavations.

D. Excavation for Utilities:
   1. Excavate trench width and depth required for laying pipe, conduit, or cable. Cut trench banks vertical. Remove stones from bottom of trench as required to avoid point-bearing. Over excavate wet or unstable soil, if encountered, from trench bottom as required to provide suitable base for continuous and uniform bedding.
   2. During excavation, stockpile materials suitable for backfilling away from trench bank to prevent overloading, slides, or cave-ins.
   3. Remove material encountered in trenching operations that is unsuitable for backfilling, subgrade or foundation purposes as determined by Testing Laboratory and Contracting Officer. Dispose of materials off-site in an approved manner in accordance with requirements of authorities having jurisdiction.
   4. Prevent surface water from flowing into trenches or other excavations by temporary grading or other approved methods.
      a. Do not allow water to accumulate in excavations.
      b. Remove accumulated water in excavations.
      c. Provide and maintain pumps, well points, sumps, suction and discharge lines and other dewatering system components required to remove water from excavations.
   5. Open cut excavation using trenching machine or backhoe. Do not use dirt clods for backfill created by use of machines other than ladder or wheel-type trenching machines.
   6. Grade trench bottom to provide uniform bearing and support for each section of pipe on bedding material along entire trench length, except where necessary to excavate for bell holes, proper sealing of pipe joints, or other required connections. Dig bell holes and depressions for joints after trench bottom has been graded. Do not excavate trench deeper, longer, or wider than required to make proper joint connection.
   7. Excavate trench width below the top of pipe minimum 300 mm wide and maximum 460 mm wider than outside surface of pipe or conduit installed to elevations and grades indicated on Drawings. Excavate trench width for other pipe, conduit, or cable to least practical width allowing for proper compaction of trench backfill.

8. Excavate trench depth measured from finished grade or paved surface to the following requirements or applicable codes and ordinances:
    a. Water Mains: 30 inches to top of pipe barrel or 6 inches below frost line established by local building official, whichever is deeper.
    b. Sanitary Sewer: Elevations, and grades indicated on Drawings.
    c. Storm Sewer: Depths, elevations, and grades indicated on Drawings.
    d. Electrical Conduits: 24 inches minimum to top of conduit or as required by NFPA 70, or local utility company requirements, whichever is deeper.
    e. TV Conduits: 18 inches minimum to top of conduit or as required by local utility company, whichever is deeper.
    f. Telephone Conduits: 18 inches minimum to top of conduit, or as required by local utility company, whichever is deeper.
    g. Gas Mains and Service: 30 inches minimum to top of pipe, or as required by local utility company, whichever is deeper.
9. Provide shoring, sheeting, and bracing, as required, in trenches and other excavations where protection of construction personnel is required. Sheeting may be removed after sufficient backfilling to protect against damaging or injurious caving.

E. Excavation for Pavement:
1. Excavate roadway and pavement areas to line and grade indicated on Drawings.
2. Stockpile excavated material suitable for backfilling on-site.
3. Remove excavated materials not required or not suitable for backfill from site.
4. Overexcavate areas of pavement subgrade found to contain unsuitable material. Prepare, fill with suitable material, and compact as specified. Stabilize areas as specified in Section 313200.

3.04 PIPE BEDDING

A. Excavate trenches, for pipe or conduit installed to elevations indicated on Drawings, 4 inches below bottom of pipe and to width as specified. Place 4 inches of bedding material, compact in bottom of trench, and shape to conform to lower portion of pipe barrel. After pipe installation, backfill and compact to top of trench.
B. Place geotextile fabric as indicated on Drawings.

3.05 BACKFILLING AND SUBGRADE PREPARATION

A. Backfilling:
1. Verify that imported off-site fill and stockpiled on-site fill is tested and approved.
2. Verify that foundation perimeter drainage installation is inspected and approved.
3. Verify that foundation or below grade structure walls are braced to support surcharge forces imposed by backfilling operations.
4. Verify that backfill areas are free of debris, snow, ice, or water, and that ground surfaces are not frozen.

B. Prepare building area subgrade pad in accordance with foundation subsurface preparation information indicated on Drawings and specified herein. Do not use rock larger than 6 inches for building subgrade fill.
C. Areas Exposed by Excavation or Stripping:
1. Scarify areas exposed by excavation or stripping on which building subgrade preparations are to be performed to minimum 8 inch depth.
2. Compact to minimum 95 percent optimum density in accordance with ASTM D1557 (Modified Proctor) at minimum moisture content 1 percent below and maximum 3 percent above optimum moisture content.

3. Proofroll to detect any areas of insufficient compaction by making minimum of 2 complete passes with fully-loaded tandem-axle dump truck, or Contracting Officer approved equivalent, in each of two perpendicular directions under supervision and direction of Contracting Officer.
4. Excavate and recompact areas failing to meet specified requirements.

D. Fill Material Placement:
1. Place in 8 inch maximum lifts compacted minimum 95 percent optimum density in accordance with ASTM D1557 (Modified Proctor) at minimum moisture content of 1 percent below and maximum moisture content 3 percent above optimum moisture content.
2. Maximum allowable values for plasticity index (PI) and liquid limit (LL) of suitable fill materials to be used as fill in the specified areas, unless indicated otherwise on Drawings:

	LOCATION	PI	LL
a.			
b.	Building area, below upper 4 feet	30	40
c.	of proposed subgrade elevation		
d.	Building area, upper 4 feet	20	30
e.	of proposed subgrade elevation		
f.	Paving area, below upper 4 feet	30	40
g.	of proposed subgrade elevation		
h.	Paving area, upper 4 feet	20	30
i.	of proposed subgrade elevation		

E. Provide material imported from off-site with CBR (California Bearing Ratio) or LBR (Limerock Bearing Ratio) value equal to or above pavement design subgrade CBR or LBR value indicated on Drawings.

3.06 MAINTENANCE OF SUBGRADE

A. Verify finished subgrades for elevations indicated on Drawings and specified conditions for construction above subgrade.
B. Protect subgrade from excessive wheel loading during construction, including concrete trucks and dump trucks.
C. Remove areas of finished subgrade found to have insufficient compaction density. Replace in a manner that will comply with compaction requirements as directed by Contracting Officer. Provide hard, uniform, smooth, stable surface, true to grade and cross-section after completion of compaction.

3.07 BORINGS AND CASINGS UNDER ROADS

A. Install street, road, or highway crossings for utility mains by jacking and boring method in accordance with requirements of governing authorities having jurisdiction.
B. Locate approach pits and trenches within right- of-way of street, road, highway, or railroad distance from paving permitting traffic to pass without interference. Tamp backfill for approach pits and trenches within right- of-way in layers not greater than 6 inches thick for entire length and depth of trench or pit. Compact backfill to 95 percent of maximum density obtained at optimum moisture as determined by AASHTO T 180, Method A (Modified Proctor). Mechanical tampers may be used after cover of 6 inches has been obtained over top of pipe barrel.
C. Use commercial type boring rig providing hole bored to proper alignment and grade within 2 inches of same diameter as largest outside joint diameter of pipe installed. Install pipe in hole immediately after bore has been made, and in no instance shall hole be left open while unattended.

D. Clean and prime interior and exterior of casing pipe; apply two coats of asphalt in accordance with requirements of governing authorities having jurisdiction.
E. Butt weld steel casing. Weld using full penetration single butt-welds in accordance with AWWA C 206.
F. Install casing and utility pipe with end seals, vent pipe, and other special equipment in accordance with requirements of governing authorities having jurisdiction.
G. Paving Damage Caused by Contractor Construction Operations:
    1. Repair paving where cracks occur on either side of line where pipe was installed by removing damaged paving between cracks, sawcutting paving in straight line at a point sufficiently beyond location of cracks for repair, and placing new paving to match existing in areas where paving removed.
    2. Make repairs to the satisfaction of paving owner.
    3. Make repairs at no additional cost to United States Postal Service within one year from Date of Substantial Completion.

3.08 FIELD QUALITY CONTROL

A. Section 014000 - Quality Requirements: Field testing and inspection.
B. Excavation: Notify Testing Laboratory and Contracting Officer for visual inspection of bearing surfaces, 48 hours prior to backfilling and other subsequent Work.
C. Site Tests:
    1. Specified in Section 312000.
    2. Tests for Building Area Subgrade Pad:
        a. Cut Areas: Minimum one compaction test for every 2500 square feet.
        b. Fill Areas: Minimum one compaction test for every 2500 square feet for each 8 inch lift measured loose.
    3. Tests for areas outside building area subgrade pad specified in Section 312000.
D. If tests indicate the Work does not meet specified requirements, remove Work, replace, compact and retest at no additional cost to United States Postal Service.

3.09 PROTECTION

A. Protect building subgrade pad and building related earthwork from damage by construction operations and erosion.
B. Prohibit vehicles from entering building subgrade pad area. Vehicles not permitted.
C. Scarify surface, reshape, and compact areas damaged by construction operations or weather erosion.

END OF SECTION 31 23 00

SECTION 32 12 16 - ASPHALT PAVING

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Bituminous concrete paving.
   2. Surface course.
   3. Binder course.
   4. Paving base course.

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other Documents.

C. Related Sections:
   1. Section 312000 - Earth Moving: Earthwork for Pavement.
   2. Section 321313 - Concrete Paving: Concrete paving, curbs and sidewalks.
   3. Section 321723 - Pavement Markings: Painted pavement markings.

1.02 REFERENCES

A. Asphalt Institute (AI):
   1. AI MS-2 - Mix Design Methods for Asphalt Concrete and Other Hot Mix Types.
   2. AI MS-3 - Asphalt Plant Manual.
   3. AI MS-8 - Asphalt Paving Manual.
   4. AI MS-19 - Basic Asphalt Emulsion Manual.

B. American Society for Testing and Materials (ASTM):
   1. ASTM D 242 - Specification for Mineral Fiber for Bituminous Paving Mixtures.
   2. ASTM D 698 - Test Methods for Moisture-Density Relations of Soils and Soil-Aggregate Mixtures, Using 5.5 Pound Rammer and 12 inch Drop.
   3. ASTM D 1188 - Method for Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Paraffin-Coated Specimens.
   4. ASTM D 1557 - Test Methods for Moisture-Density Relations of Soils and Soil-Aggregate Mixtures Using 10 Pound Rammer and 18 inch Drop.
   5. ASTM D 1560 - Test Method for Resistance to Deformation and Cohesion of Bituminous Mixtures by Means of Hveem Apparatus.
   6. ASTM D 2397 - Specification for Cationic Emulsified Asphalt.
   7. ASTM D 2399 - Practice for Selection of Cutback Asphalt.
   8. ASTM D 2726 - Test Method for Bulk Specific Gravity and Density of Nonabsorbative Compacted Bituminous Mixtures.
   9. ASTM D 3381 - Specification for Viscosity-Graded Asphalt Cement for Use in Pavement Construction.
   10. ASTM D 3549 - Test Method for Thickness or Height of Compacted Bituminous Paving Mixture Specimens.
   11. ASTM D 4318 - Test Method for Liquid Limit, Plastic Limit, and Plasticity Index of Soils.

C. American Association of State Highway and Transportation Officials (AASHTO):
   1. AASHTO T 88 - Particle Size Analysis of Soils.

1.03 SYSTEM DESCRIPTION

A. Design Requirements: Provide asphalt-aggregate mixture as recommended by local or state paving authorities to suit project conditions. Use locally available materials and gradations

which meet standard state highway specifications and exhibit satisfactory records of previous installations.

1.04 SUBMITTALS

A. Section 013300 - Submittal Procedures: Procedures for submittals.
   1. Assurance/Control Submittals:
      a. Design Data:
         1) Submit design mix following format indicated Asphalt Institute Manual MS-2, Marshall Stability Method; including type/name of mix, gradation analysis, grade of asphalt cement used, Marshall Stability (pounds), flow, effective asphalt content (percent), and direct references to applicable state highway department specification sections for each material.
         2) Provide design mixture listed in current edition of applicable state highway department specifications.
         3) Use mix designs prepared within 3 years maximum.
         4) Provide documentation of state highway limitations, if any, on use of recycled content materials.
      b. Certificates: Submit materials certificate to Testing Laboratory signed by material supplier and Contractor, certifying that materials comply with, or exceed, the requirements specified herein.
      c. Qualification Documentation: Paving installer documentation of experience indicating compliance with specified qualification requirements.

1.05 QUALITY ASSURANCE

A. Perform Work in accordance with AI MS-8
B. Installer Qualifications: Company specializing in performing the Work of this Section with minimum 5 years documented experience.
C. Regulatory Requirements:
   1. Conform to applicable requirements for paving work on public property.
   2. Maintain access for vehicular and pedestrian traffic as required for other construction activities. Use temporary striping, flagmen, barricades, warning signs, and warning lights as required.

1.06 PROJECT CONDITIONS OR SITE CONDITIONS

A. Jobsite Requirements:
   1. Apply prime and tack coats when ambient temperature is above 40 degrees F, and when temperature has been above 35 degrees F for 12 hours immediately prior to application. Do not apply when base is wet, contains excess moisture, or during rain.
   2. Construct bituminous concrete paving when atmospheric temperature is above 40 degrees F.

1.07 ENVIRONMENTAL REQUIREMENTS

A. Resource Management:
   1. Recycled Content: Provide aggregate fabricated from a minimum of 30% recycled rubble or concrete. Provide asphalt cement fabricated from recycled content asphalt.

PART 2 - PRODUCTS

2.01 MATERIALS

A. Base Course: As indicated on Drawings, complying with applicable state highway specifications regarding source, quality, gradation, liquid limit, plasticity index and mix proportioning.
   1. Unless otherwise specified in applicable state highway specifications, provide base course aggregate fabricated from minimum 30 percent recycled rubble or concrete.

B. Asphalt Cement: Fabricated from minimum 15 percent recycled asphalt and complying with ASTM D 3381; Table 2 AC-10, AC-20, or AC-30, viscosity grade, depending on local mean annual air temperature as indicated below:

TEMPERATURE CONDITION	ASPHALT GRADES
Cold, mean annual air temperature at 45 degrees F or lower	AC-10 85/100 pen.
Warm, mean annual air temperature between 45 degrees F and 75 degrees F	AC-20 60/70 pen.
Hot, mean annual air temperature at 75 degrees F or higher	AC-30

C. Prime Coat: A medium curing cut-back asphalt or an asphalt penetrating prime coat consisting of either ASTM D 2397 or ASTM D 2399, MC- 30 or SS-1h.

D. Tack Coat: Emulsified asphalt; ASTM D 2397 or ASTM D 2399, SS-1h, CSS-1, or CSS-1h, diluted with one part water to one part emulsified asphalt.

E. Mineral Filler: Rock or slag dust, hydraulic cement, or other inert material complying with AASHTO M-17/ASTM D 242, if recommended by applicable state highway department standards.

F. Asphalt-Aggregate Mixture: Unless otherwise indicated on Drawings, the Design Mix shall have a minimum stability based on a 50-blow Marshall complying with ASTM D 1559 of 1000 pounds with a flow between 8 and 16. The Design Mix shall be within sieve analysis and bitumen ranges below:

G. Asphalt-Aggregate Mixture: Unless otherwise indicated on Drawings, the Design Mix shall have a minimum stability based on CALTRANS AR4000. The Design Mix shall be within sieve analysis and bitumen ranges below:

SIEVE ANALYSIS OF MIX

Square Sieve	Total Percent Passing	Percent Tolerance
1/2 inch	80 - 100	5
3/8 inch	65 - 93	4
No. 8	0 - 55	4
No. 50	2 - 27	2
No. 200	0 - 10	2

Percent Bitumen by Weight of Total Mix: 5.0 - 8.5.
Percent Air Voids: 3-6.
Percent Aggregate Voids Filled with Asphalt Cement: 70 - 82.
Allowable Variance of Percent Bitumen by Weight of Total Mix: 0.4.

2.02 EQUIPMENT

A. Maintain equipment in satisfactory operating condition and correct breakdowns in a manner that will not delay or be detrimental to progress of paving operations.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.
B. Verification of Conditions: Verify that field measurements, surfaces, substrates and conditions are as required, and ready to for earthwork operations to begin.
   1. Verify gradients and elevations of base are correct, and base is dry.
C. Report in writing to Contracting Officer prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.
D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the United States Postal Service.

3.02 BASE COURSE PLACEMENT

A. Perform base course construction in a manner that will drain surface properly at all times and at the same time prevent runoff from adjacent areas from draining onto base course construction.
B. Compact base material to not less than 98 percent of optimum density as determined by ASTM D 698 or 95 percent of optimum density, as determined by ASTM D 1557, unless otherwise indicated on the Drawings.
C. Granular Base: Construct to thickness indicated on Drawings. Apply in lifts or layers not exceeding 8 inches, measured loose.
D. Sand/Shell Base: Construct to thickness indicated on Drawings. Apply in lifts or layers not exceeding 4 inches, measured loose.
E. Asphalt Institute Type IV Mix for Full Depth Asphalt Base: Construct to thickness indicated on Drawings in lifts or layers not exceeding 3 inches, measured loose.
F. Asphalt Institute Type VI, VII, or VIII Mixes for Hot-Mix Sand Asphalt Bases: Construct to thickness indicated on Drawings. Apply in lifts or layers not exceeding 3 inches, measured loose.
G. Soil Cement Stabilized Base: Construct to thickness and strength as indicated on Drawings and in accordance with applicable state highway specifications. If not indicated on the Drawings, the minimum compressive strength shall be 500 pounds per square inch, tested at 28 days.

3.03 APPLICATIONS

A. Prime Coat:
   1. Apply bituminous prime coat to all base material surfaces where bituminous concrete paving will be constructed.
   2. Apply bituminous prime coat in accordance with applicable state highway specifications.
   3. Apply at minimum rate of 0.25 gallon per square yard over compacted base material. Apply to penetrate and seal, but not flood surface.
   4. Make necessary precautions to protect adjacent areas from overspray.
   5. Cure and dry as long as necessary to attain penetration of compacted base and evaporation of volatile substances.
B. Tack Coat:
   1. Apply to contact surfaces of previously constructed bituminous concrete base courses or portland cement concrete and surfaces abutting or projecting into bituminous concrete or into bituminous concrete pavement.

2. Apply tack coat to bituminous concrete base course or sand asphalt base course. Apply emulsified asphalt tack coat between each lift or layer of full depth bituminous concrete and sand asphalt bases and on surface of all such bases where bituminous concrete paving will be constructed.
3. Apply emulsified asphalt tack coat in accordance with applicable state highway specifications.
4. Apply at minimum rate of 0.05 gallon per square yard of surface.
5. Allow to dry until at proper condition to receive paving.

3.04 BITUMINOUS CONCRETE PLACEMENT

A. Place bituminous concrete mixture on completed compacted subgrade surface, spread, and strike off. Spread mixture at following minimum temperatures:
   1. When ambient temperature is between 40 degrees F and 50 degrees F, mixture temperature equal to 285 degrees F.
   2. When ambient temperature is between 50 degrees F and 60 degrees F, mixture temperature equal to 280 degrees F.
   3. When ambient temperature is higher than 60 degrees F, mixture temperature equal to 275 degrees F.

B. Whenever possible, all pavement shall be spread by a finishing machine; however, inaccessible or irregular areas may be placed by hand methods. The hot mixture shall be spread uniformly to the required depth with hot shovels and rakes. After spreading, the hot mixture shall be carefully smoothed to remove all segregated course aggregate and rake marks. Rakes and lutes used for hand spreading shall be of the type designed for use on asphalt mixtures. Loads shall not be dumped faster that they can be properly spread. Workers shall not stand on the loose mixture while spreading.

C. Paving Machine Placement: Apply successive lifts of bituminous concrete in transverse directions with the surface course placed in the direction of surface-water flow. Place in typical strips not less than 10 feet wide.

D. Joints: Make joints between old and new pavements, or between successive days and work in a manner that will provide a continuous bond between adjoining work. Construction joints shall have same texture, density, and smoothness as other sections of bituminous concrete course. Clean contact surfaces of all joints and apply tack coat.

3.05 ROLLING AND COMPACTION

A. The mixture, after being spread, shall be thoroughly compacted by rolling as soon as it will bear the weight of the rollers without undue displacement. The number, weight, and types of rollers and sequences of rolling operations shall be such that the required density and surface are consistently attained while the mixture is in a workable condition.

B. Compact mixture with hot hand tampers or vibrating plate compactors in areas inaccessible to rollers.

C. Breakdown Rolling: Accomplish breakdown or initial rolling immediately following rolling of joints and outside edge. Check surface after breakdown rolling, and repair displaced areas by loosening and filling with hot material.

D. Second Rolling: Follow breakdown rolling as soon as possible, while mixture is hot. Continue second rolling until mixture has been thoroughly compacted.

E. Finish Rolling: Perform finish rolling while mixture is still warm enough for removal of roller marks. Continue rolling until roller marks are eliminated and course has attained maximum density.

F. Patching: Remove and replace paving areas mixed with foreign materials and defective areas. Cut out such areas and fill with fresh, hot bituminous concrete. Compact by rolling to maximum surface density and smoothness.
G. Protection: After final rolling, do not permit vehicular traffic on pavement until it has cooled and hardened. Erect barricades to protect paving from traffic until mixture has cooled enough not to become marked.

3.06 CONSTRUCTION

A. Site Tolerances:
   1. Paving Surface Smoothness: Maximum allowable 10 foot straightedge tolerance for smoothness.
      a. Base Course Surface: 1/4 inch.
      b. Wearing Surface Course: 3/16 inch.

3.07 FIELD QUALITY CONTROL

A. Section 014000 - Quality Requirements: Field inspection and testing procedures
B. Site Tests:
   1. Paving Base Course: Perform testing of in-place base courses for compliance with requirements for thickness, compaction, density, and tolerance.
      a. Moisture/Density Test: ASTM D 698 or ASTM D 1557.
      b. Mechanical Analysis Test: AASHTO T-88.
      c. Plasticity Index Test: ASTM D 4318.
      d. Base Material Thickness Test: Minimum one test for every 20,000 square feet.
      e. Base Material Compaction Test: Minimum one test for every 20,000 square feet.
      f. Field Density Tests: Perform testing of in-place base courses for compliance with requirements for density using one of the following methods:
         1) Sand-cone Method: ASTM D 1556.
         2) Balloon Method: ASTM D 2167.
         3) Nuclear Method: ASTM D 2922, Method B (Direct Transmission).
      g. Test each source of base material for compliance with applicable state highway specifications.
   2. Asphalt Concrete Paving: Perform testing of in-place asphalt concrete paving courses for compliance with requirements for thickness, compaction, and surface smoothness.
      a. Thickness: ASTM D 3549; Thickness shall not be less than thickness specified on Drawings.
      b. Surface Smoothness: Testing shall be performed on the finished surface of each asphalt paving course using 10 foot straightedge applied parallel with, and at right angles to centerline of paved areas. Smoothness shall not be less than tolerances specified herein.
   3. Compaction: Field density test for in place materials shall be performed by examination of field cores in accordance with one of the following standards:
      a. Bulk Specific Gravity of Paraffin-Coated Specimens: ASTM D 1188, minimum one core per 20,000 square feet.
         1) Standard Duty Areas: Minimum 3 cores.
         2) Heavy Duty Areas: Minimum 3 cores.
      b. Bulk Specific Gravity Using Saturated Surface-Dry Specimens: ASTM D 2726, minimum one core per 20,000 square feet.
         1) Standard Duty Areas: Minimum 3 cores.
         2) Heavy Duty Areas: Minimum 3 cores.

END OF SECTION 32 12 16

SECTION 32 13 13 - CONCRETE PAVING

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Concrete Pavement
   2. Concrete walks and terraces.
   3. Concrete curbs, and curb and gutters.

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other documents.

C. Related Sections:
   1. Section 312000 - Earth Moving: Earthwork for pavement.
   2. Section 321216 - Asphalt Paving.
   3. Section 033000 - Cast-In-Place Concrete: Concrete requirements for pavement.

1.02 REFERENCES

A. American Concrete Institute (ACI):
   1. ACI 301 - Specifications for Structural Concrete.
   2. ACI 308 - Standard Practice for Curing Concrete.

B. American society for Testing and Materials (ASTM):
   1. ASTM A 185 - Specification for Steel Welded Wire, Fabric, Plain, for Concrete Reinforcement.
   2. ASTM A 615 - Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement.
   3. ASTM C 494 - Standard Specification for Chemical Admixtures for Concrete.
   4. ASTM C 618 - Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Concrete.
   5. ASTM D 1751 - Specification for Preformed Expansion Joint Fillers for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types).

1.03 SUBMITTALS

A. Section 013300 - Submittal Procedures: Procedures for submittals.
   1. Product Data: Submit product data for the following:
      a. Joint filler.
      b. Joint sealant.
      c. Concrete admixtures.
      d. Concrete curing compounds.
   2. Assurance/Control Submittals:
      a. Concrete Mix Design: Submit three copies of each proposed mix design for each class of concrete in accordance with ACI 301, Sections 3.9 "Proportioning on the basis of previous field experience or trial mixture", or 3.10 "Proportioning based on empirical data". Submit separate mix design for concrete to be placed by pumping, in addition to the mix design for concrete to be placed directly from the truck chute.
      b. Include the following information in concrete mix design:
         1) Proportions of cement, fine and coarse aggregate, and water.
         2) Water-cement ratio, 28-day compressive design strength, slump, and air content.
         3) Type of cement and aggregate.
         4) Aggregate gradation.

5) Type and dosage of admixtures.
6) Special requirements for pumping.
7) Range of ambient temperature and humidity for which design is valid.
8) Special characteristics of mix which require precautions in mixing, placing, or finishing techniques to achieve finished product specified.

1.04 QUALITY ASSURANCE

A. Perform work in accordance with ACI 301.
B. Conform to ACI 305R when mixing and placing concrete during hot weather.
C. Conform to ACI 306R when mixing and placing concrete during cold weather.
D. Regulatory Requirements:
   1. Conform to applicable requirements for paving work on public property.
   2. Contractor shall maintain access for vehicular and pedestrian traffic as required for other construction activities. Utilize temporary striping, flagmen, barricades, warning signs, and warning lights as required.

PART 2 - PRODUCTS

2.01 FORM AND REINFORCING MATERIAL

A. Forms: Steel, wood, or other suitable material of size and strength to resist movement during concrete placement and to retain horizontal and vertical alignment until removal. Use straight forms, free of distortion and defects. Use flexible spring steel forms or laminated boards to form radius bends as required.
   1. APA Exterior Plyform BB with a medium density, smooth, hard, fused resin fiber overlay, or metal forms.
   2. Form Oil: Coat forms with nonstaining type coating that will not discolor or deface surface of concrete. Subject to compliance with requirements, manufacturers offering specified items which may be incorporated in the work include the following.
      a. "Eucoslip" - Euclid Chemical Co., Cleveland, OH (800) 321-7628.
      b. "Form Coating" - Nox-Crete Chemicals, Omaha, NE (800) 669-2738.
      c. Substitutions: Under provisions of Section 016000.
B. Curb, Curb and Gutter Forms: Use flexible spring-steel forms or laminated boards to form radius bends. Tolerance: Not to deviate more than 1/4 inch in 10 feet in grade and alignment.
C. Reinforcing:
   1. Welded Wire Mesh: Welded plain cold-drawn steel wire fabric, ASTM A 185. Furnish in flat sheets, not rolls, unless otherwise acceptable to Owner.
   2. Reinforcing Bars: Deformed steel bars, ASTM A 615, Grade 60.
   3. Fiber reinforced concrete mixtures having the same strength or exceeding as specified for concrete mixes, as verified by Manufacturer's testing laboratory procedures, shall be considered as an alternate for welded wire mesh in exterior flat work, curbs and sidewalks.
D. Reinforcing Accessories:
   1. Reinforcing Accessories: Subject to compliance with requirements, manufacturers offering specified items which may be incorporated in the work include the following.
      a. Dayton Superior Corp., Miamisburg, OH (800) 745-3700.
      b. Heckmann Building Products, Inc., Chicago, IL (800) 621-4140.
      c. Hohmann & Barnard, Inc., Hauppauge, NY (800) 645-0616.
      d. Richmond Screw Anchor Co., Inc., Ft. Worth, TX (817) 284-4981.
   2. Conform to Concrete Reinforcing Steel Institute Manual of Standard Practice. Include spacers and chairs with plastic tipped legs, ties and other devices necessary for properly assembling, placing, spacing and supporting forms and reinforcement in place.

3. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

2.02 CONCRETE MATERIALS

A. Comply with requirements of applicable Section 033000 for concrete materials, admixtures, bonding materials, curing materials, surface sealers and others as required.

B. Cement:
   1. Portland Cement: ASTM C150 Type 1.
   2. High-early Strength Portland Cement: ASTM C150, Type III.

C. Aggregates: ASTM C33.
   1. Fine aggregate shall be natural sand, or sand prepared from stone or gravel. Grains shall; be clean, hard, durable, uncoated and free from silt, loam and clay.
   2. Coarse Aggregates: Crushed stone, gravel, or other approved inert materials of similar characteristics, or combinations thereof, having hard, strong, durable pieces free from adherent coatings. Maximum size of pieces shall be 3/4" to #4 except for footings, which may be 1-1/2". The maximum size of aggregate may also be not larger than one fifth of the narrowest dimension between forms, nor larger than three fourths of the minimum clear spacing between reinforcing bars.

D. Water: Clean and free from injurious amounts of oil, acids, salts, organic or other deleterious matter.

E. Air Entrainment: ASTM C260.
   1. Use air-entrained concrete for exterior exposed concrete including walls, walks, paving, etc. where minimum daily temperatures are expected below 38 degrees F during pouring or subsequent 38 day curing period.
   2. Proportion air-entraining concrete to attain minimum 28-day compressive strength specified.
   3. Total Air Entrainment in Concrete: Not less than four percent nor more than six percent volume of concrete.

F. Admixtures:
   1. May be used at contractors option to provide workability at low slumps, increased compressive strength, retardation or acceleration of the concrete.
   2. Chemical Admixtures: ASTM C494. Mineral Admixtures: ASTM C618.
   3. The cement factor shall not be reduced and changes shall be made in the other mix proportions to ensure the minimum strength requirements.
   4. Use of admixtures approved in writing by Architect. No additional expense to the Owner will be allowed.
   5. No calcium chloride shall be used.
   6. Before any admixture is accepted for use, the Contractor shall submit certified laboratory reports on each additive material to the architectural consultant. The report shall show the following:
      a. Confirmation of compliance with the applicable ASTM Standard.
      b. Evaluation of the effects of the admixture on the properties of the concrete to be made on the job, including consideration of the anticipated ambient conditions on the job, and proposed construction procedures.
      c. Determination of within-lot uniformity of product proposed for use.

2.03 CONCRETE MIXES

A. Concrete Proportions:
   1. Concrete shall be homogenous, and when hardened, shall have the required strength, resistance to deterioration, durability, water tightness and the properties as specified.

2. Minimum concrete strength at 28 days shall be;
   a. 3,000 psi for walks, terraces, curbs and gutters.
   b. 4,000 psi for concrete pavement and pads.
3. Slump of concrete:
   a. Pavement: 2-1/2 inch minimum to 4 inch maximum.
   b. Ramps and sloping surfaces: Not more than 3 inches.

B. Ready-Mix Concrete:
1. Ready-mix concrete shall conform to ASTM C94. The mixing agitation shall begin within 30 minutes, and the concrete shall be discharged from the truck within one hour after the water has been added to the concrete mix.
2. Delivery tickets are to accompany each concrete truck and shall be kept in the job superintendent's file. Delivery tickets must indicate the following information or be subject to rejection:
   a. Name of project.
   b. Supplier of concrete.
   c. Truck identity and ticket serial number.
   d. Date of delivery.
   e. Brand of cement.
   f. Cement content.
   g. Strength classification.
   h. Batching time.
   i. Point of deposit.
   j. Total amount of water.
   k. Weight of aggregate.
   l. Daily temperature.
   m. Number of cubic yards in load.
   n. Admixture content.
   o. Name of Contractor.
   p. Name of driver.
   q. Time loaded and first mixing of concrete.
   r. Reading of revolution counter.
3. Quantity of water used for each batch shall be accurately measured.

2.04 JOINT MATERIALS

A. Sealed expansion and contraction joints: Filler of nonbituminous rubber or cork conforming to ASTM D1752.
B. Non-sealed joints:
1. Non-sealed Joints: Subject to compliance with requirements, manufacturers offering specified items which may be incorporated in the work include the following.
   a. "Flexcell" - Celotex Corp., Tampa, FL (813) 873-1700.
   b. "Seal Tight Fiber Expansion Joint" - W.R. Meadows, Inc., Hampshire, IL (800) 342-5976.
2. Filler premolded bituminous type conforming to ASTM D1751.
3. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

C. Noncompressive Filler:
1. Noncompressive Filler: Subject to compliance with requirements, manufacturers offering specified items which may be incorporated in the work include the following.
   a. "Styrofoam SM" - Dow Chemical Co., Midland, MI (517) 636-0754.
   b. "Foamular" - Owens Corning, Toledo, OH (800) 828-7155.
2. 2 inch or 1 inch thick sheets.

3. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

D. Compressive Filler:
1. Compressive Filler: Subject to compliance with requirements, manufacturers offering specified items which may be incorporated in the work include the following.
   a. "Ethafoam" - Dow Chemical Co., Midland, MI (800) 322-8723.
   b. "Rodofoam No. 423" - Sternson Group, Brampton, ON (800) 265-8417.
2. 2 inch or 1 inch thick sheets, compression modulus within the range of 15 to 25 pounds per square inch per inch.
3. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

E. Filler Adhesive for Noncompressive Filler and Compressive Filler:
1. Filler Adhesive: Subject to compliance with requirements, manufacturers offering specified items which may be incorporated in the work include the following.
   a. "General Purpose Mastic No. 11" - Dow Chemical Co., Midland, MI (800) 322-8723.
   b. "Rodofast" - Sternson Group, Brampton, ON (800) 265-8417.
2. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

F. Slab-on-grade Construction Joints: Provide a full slab depth 24 gauge metal preshaped key, approximate depth of key to be 1/4 slab thickness and a key width of about 1/10 slab thickness.

G. Joint Sealants: ASTM C920. Non-priming, pourable, self-leveling polyurethane. Subject to compliance with project requirements manufacturers offering joint sealants which may be incorporated in the Work include, but are not limited to the following:
1. Sonolastic Paving Joint Sealant, by Sonneborn, Shakopee, MN (800) 433-9517.
2. Sonomeric CT 1 Sealant, by Sonneborn, Shakopee, MN (800) 433-9517.
3. Sonomeric CT 2 Sealant, by Sonneborn, Shakopee, MN (800) 433-9517.
4. Vulkem 45, by Mameco, Cleveland, OH (800) 321-6412.
5. Chem-Caulk, by Bostik, Middleton, MA (800) 726-7845.
6. "THC-900" - Tremco, Beachwood, OH (800) 562-2728.
7. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

2.05 CURING MATERIALS

A. Sealers:
1. Sealers: Subject to compliance with requirements, manufacturers offering specified items which may be incorporated in the work include the following.
   a. "Polyseal" - W.R. Meadows, Inc., Hampshire, IL (800) 342-5976.
   b. "Kure-N-Seal" - Sonneborn, Shakopee, MN (800) 433-9517.
   c. "Cure-Hard" - W.R. Meadows, Inc., Elgin, IL (312) 683-4500.
2. ASTM C156 and ASTM C309, Type I. Material shall become integral part of concrete and leave slab free of residue or film.
3. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

B. Membrane: Opaque-white polyethylene sheet, 0.006 inch thick, meeting requirements of ASTM C171.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution Requirements: Verification of existing conditions before starting work.
B. Verification of Conditions: Verify that field measurements, surfaces, substrates and conditions are as required, and ready to for earthwork operations to begin.
   1. Verify gradients and elevations of base are correct, and base is dry.
C. Report in writing to Contracting Officer prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.
D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the United States Postal Service.

3.02 SUBGRADE PREPARATION

A. Conform with the requirements specified in Section 312000 .
B. Thoroughly wet subgrade and then compact with two passes of a 500 pound roller.
C. Pumping: Where concrete paving or sidewalks, and curbs are to be placed, yielding material deflecting more than 1/2 inch under a 500 lb. roller shall be removed to a depth of not less than 4 inches below subgrade elevation and replaced with an approved granular material which shall then be compacted as described above.
D. The subgrade shall be in a moist condition when the concrete is placed. In cold weather the subgrade shall be prepared and protected so as to provide a subgrade free from frost when the concrete is deposited.

3.03 FORM CONSTRUCTION

A. Comply with the requirements of Section 033000. Install sufficient quantity of forms to allow continuous progress of the work and so that forms can remain in place at least 24 hours after concrete placement.
B. Check complete formwork for grade and alignment to the following tolerances:
   1. Top of form: Not more than 1/8 inch in 10 feet.
   2. Vertical face: Longitudinal axis not more than 1/4 inch in 10 feet.

3.04 PLACING REINFORCEMENT

A. Support reinforcing and wire securely together to prevent displacement by construction loads and traffic, or the placing of concrete. For slabs on grade, supporting pieces of concrete blocks or bricks may be used.
B. Place wire mesh reinforcing two inches above bottom of slab unless otherwise indicated.
C. Reinforcement shall be kept clean from oil, dirt and loose mill scale or other coatings which might destroy the concrete bond. Remove tags and markings prior to concrete placement.
D. Do not place concrete until reinforcement has been inspected and approved by local authorities, if required.

3.05 CONCRETE PLACEMENT AND FINISHING

A. Tamp and consolidate concrete with a suitable wood or metal tamping bar and the surface shall be finished to grade with a wood float.

B. Finished surfaces shall not vary more than 3/16 inch from the testing edge of a 10 foot straightedge.
C. Curb Expansion Joints: Fill joints with 1/2 inch thick joint filler strips conforming to ASTM D1751 or ASTM D1752.
D. Contraction Joints: Divide the surface of paving, walks and terraces into rectangular areas not to exceed 5 feet 0 inches each way.
   1. Cut a groove in the top portion of the slab to a depth of at least one-fourth of the slab thickness using a jointer or by sawing a groove in the hardened concrete with a power-driven saw.
   2. Membrane-cured surface damaged during the sawing operations shall be resprayed as soon as the surface becomes dry.
E. Slab Finishes: ACI 301, paragraph 11.7 and as follows:
   1. Broom Finish: On stair treads with abrasive nosings and on walks, unless other finishes have been indicated or specified.
   2. Broom or Belt Finish: On level walks. Broom in direction perpendicular to travel and approved sample panel. Submit joint pattern layout prior to starting work.

3.06 TOLERANCES

A. Horizontal slabs: Finished surfaces true with no deviation in excess of 1/8 inch when tested with a 10 foot straightedge, non-accumulative. No coarse aggregate showing.
B. Steps:
   1. Variation in steps within a flight of stairs:
      a. Rise: 1/8 inch.
      b. Tread: 1/4 inch.
   2. Variation in consecutive steps:
      a. Rise: 1/16 inch.
      b. Tread: 1/8 inch.

3.07 EXPANSION JOINTS

A. Install transverse expansion joints at returns and 15 feet on center.
B. Install longitudinal expansion joints where curbs and paved areas abut each other, buildings, other concrete slabs and pads or vertical restraints.
C. Place joint filler with top edge 1/4 inch below the surface and shall be held in place with steel pins or other devices to prevent warping of the filler during floating and finishing.
D. Immediately after finishing operations are completed, round joint edges with edging tool having a radius of 1/8 inch. Remove concrete over the joint filler.
E. At the end of the curing period, clean and fill expansion joints with joint sealer. Fill joints flush with concrete surface. Dummy groove joints shall not be sealed.

3.08 CURING

A. Immediately after the finishing operations, the exposed concrete surface shall be cured for 7 days by the mat, impervious sheet, or membrane-curing method.

3.09 BACKFILLING

A. After curing, remove debris and backfill the adjoining areas, grade and compact to conform to the surrounding area in accordance with the lines and grades indicated.

3.10 PROTECTION

A. Protect the completed work from damage. Repair damaged concrete and clean concrete discolored during construction. Remove work that is damaged and reconstruct to entire length between regularly scheduled joints. Refinishing damaged portion is not acceptable.

B. Prevent cars and trucks from driving on new pavement for a minimum of 14 days.

END OF SECTION 32 13 13

SECTION 32 17 23 - PAVEMENT MARKINGS

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Painted pavement markings.
   2. Painted curbs.

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other Documents.

C. Related Sections:
   1. Section 321216 - Asphalt Paving: Asphalt paving substrate for marking application.
   2. Section 321313 - Concrete Paving: Concrete paving substrate for marking application.
   3. Section 099100 – Painting: Painting exterior bollards.

1.02 SUBMITTALS

A. Section 013300 - Submittal Procedures: Procedures for submittals.
   1. Product Data: Technical data sheets indicating manufacturer's catalog number, paint type description, and VOC content for each paint type specified.
   2. Assurance/Control Submittals:
      a. Certificates: Manufacturer certificate that Products meet or exceed specified requirements.
      b. Test Reports: Manufacturer Material Safety Data Sheets (MSDS) for each paint type specified.

1.03 QUALITY ASSURANCE

A. Regulatory Requirements: Provide paint materials that conform to Federal, State, and local restrictions for Volatile Organic Compounds (VOC) and lead-free content.

1.04 DELIVERY, STORAGE AND HANDLING

A. Section 016000 - Product Requirements: Transport, handle, store, and protect products.

B. Deliver paint materials in sealed original labeled containers, bearing manufacturer's name, type of paint, brand name, lot number, brand code, coverage, surface preparation, drying time, cleanup requirements, color designation, and instructions for mixing and/or reducing.

C. Store paint materials at minimum ambient temperature of 45 degrees F and a maximum of 90 degrees F, in ventilated area, and as required by manufacturer's published instructions.

1.05 PROJECT CONDITIONS OR SITE CONDITIONS

A. Maintain access for vehicular and pedestrian traffic as required for other construction activities. Utilize flagmen, barricades, warning signs and warning lights as required.

1.06 MAINTENANCE

A. Section 017704 – Closeout Procedures and Training: Requirements for Closeout Submittals.
   1. Extra Materials:
      a. Provide 1 gallon of each color to Contracting Officer.
      b. Label each container with color and type, in addition to manufacturer's label.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Subject to compliance with project requirements, manufacturers offering specified Products which may be incorporated into the Work include the following:
   1. Sherwin-Williams Company, Cleveland, OH (800) 321-8194.
   2. Benjamin Moore and Company, Montvale, NJ (201) 573-9600.
   3. Pittsburgh Paints (PPG), Pittsburgh, PA (800) 441-9695.

B. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

2.02 MATERIALS

A. Ready-mixed; pigments fully ground maintaining a soft paste consistency, capable of readily and uniformly dispersing to a complete homogeneous mixture providing good flowing and brushing properties capable of drying or curing free of streaks or sags. Dry to traffic and touch in 2 hours.

B. Traffic Paint: Flat, Water Base, Acrylic, complying with Federal Specifications TT-P 1952D
   1. 1st Coat:
      a. Sherwin-Williams: Pro-Park Waterborne Traffic Marking Paint, B97 Series MDF 9 mils.
      b. Benjamin Moore: SuperSpec HP Safety & Zone Marking Paint P58, MDF 9 mils
      c. PPG Zoneline Traffic & Marking Paint, 11-50 Series, MDF 9 mils.
   2. 2nd Coat:
      a. Sherwin-Williams: Pro-Park Waterborne Traffic Marking Paint, B97 Series MDF 9 mils.
      b. Benjamin Moore: SuperSpec HP Safety & Zone Marking Paint P58, MDF 9 mils
      c. PPG Zoneline Traffic & Marking Paint, 11-50 Series, MDF 9 mils.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.

B. Verification of Conditions: Verify that field measurements, surfaces, substrates and conditions are as required, and ready to receive Work.

C. Report in writing to Contracting Officer prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.

D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the United States Postal Service.

3.02 PREPARATION

A. Sweep pavement and surfaces to receive paint markings clean of dust and dirt. Allow pavement to cure a minimum of 60 days prior to application of paint markings.

B. Clean surfaces free of glaze and grease, road film, and other foreign materials.

C. Where existing pavement markings are indicated on Drawings to be removed or would interfere with the adhesion of new paint, use a motorized abrasive device to remove existing markings.
   1. Use equipment that will not damage existing paving or create surface hazardous to vehicle
   2. or pedestrian traffic.

3. Use marking removal methods approved by governing authority having jurisdiction in areas within public rights-of-way.

3.03 APPLICATION

A. Apply paint products in accordance with manufacturer's published instructions using application procedures approved for the particular application and substrate to the specified Minimum Dry Film Thickness (MDF). Apply each coat to uniform finish.
B. Do not apply paint markings on surfaces that are not dry and if rain is expected within 24 hours.
C. Do not apply paint markings when surface or air temperature is below 50 degrees F.
D. Apply 2 coats at manufacturer recommended rate without addition of thinner. Apply with mechanical equipment to produce uniform straight edges. At sidewalk curbs and crosswalks, use straightedge to provide uniform, clean, and straight stripe.

3.04 PAINT MARKING SCHEDULE

A. Paint the following items with colors indicated below:
   1. Pedestrian Crosswalks: White
   2. Fire Lanes: Red or per local code.
   3. Lane Striping Where Separating Traffic in Opposite Directions: Yellow.
   4. Lane Striping Where Separating Traffic in Same Direction: White.
   5. Handicap Symbols: Per local code.
   6. Parking Stall Striping: White

END OF SECTION 32 17 23

SECTION 32 31 13 - CHAIN LINK FENCES AND GATES

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Chain link fence framework, fabric, and accessories.
   2. Excavation for post bases, concrete footings for posts, and center drop for gates.
   3. Chain link manual and motorized gates and related hardware.

B. Related Sections:
   1. Section 033000 - Cast-In-Place Concrete: Post footings.
   2. Section 281304 – Physical Access Control System
   3. Section 270500 - Common Work Results for Communications.
   4. Section 271500 – Communications Horizontal Cabling.

1.02 REFERENCES

A. American Society for Testing and Materials (ASTM):
   1. ASTM A 90 - Tests for Weight of Coating on Zinc-Coated (Galvanized) Iron or Steel Articles.
   2. ASTM A 116 - Specification for Zinc-Coated (Galvanized) Steel Woven Wire Fence Fabric.
   3. ASTM F 1184 - Specification for Industrial and Commercial Horizontal Slide Gates, Type II, Class 2.
   4. ASTM A 123 - Specification for Zinc (Hot- Dip Galvanized) Coatings on Iron and Steel Products.
   5. ASTM A 392 - Specification for Zinc-Coated Steel Chain-Link Fence Fabric.
   6. ASTM F 567 - Standard Practice for Installation of Chain Link Fence.
   7. ASTM A 824 - Specification for Metallic-Coated Steel Marcelled Tension Wire Use with Chain Link Fence.
   8. ASTM F 1043 - Specification for Strength and Protective Coatings on Metal Industrial Chain Link Fence Framework.
   9. ASTM F 668 - Specification for Poly (Vinyl Chloride) (PVC) Coated Steel Chain Link Fence Fabric.
   10. ASTM F 900 - Specification for Industrial and Commercial Swing Gates.
   11. ASTM F 1083 - Specification for Pipe, Steel, Hot-Dipped Zinc Coated (Galvanized) Welded, For Fence Structures.
   12. ASTM F 2200 – Specification for gates to be automated.

B. Underwriter's Laboratories (UL):
   1. UL325, Door, Drapery, Gate, Louver, Window Operators, and Systems.

C. Chain Link Fence Manufacturer's Institute (CLFMI):
   1. CLF-PM0610 (July 2011) - Product Manual.

1.03 SUBMITTALS

A. Procedures for submittals.
   1. Product Data: Submit product data for fabric, posts, accessories, fittings, and hardware.
   2. Shop Drawings: Include plan layout, grid, spacing of components, accessories, fittings, hardware, anchorage's, and schedule of components.
   3. Assurance/Control Submittals:
      a. Certificates: Manufacturer's certificate certifying that Products meet or exceed specified requirements.

b. Qualification Documentation: Submit documentation of experience indicating compliance with specified qualification requirements.

1.04 QUALITY ASSURANCE

A. Perform Work in accordance with CLFMI PM.
B. Installer Qualifications: Company specializing in performing the Work of this Section with minimum 5 years documented experience.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Deliver products to site, store, and protect products.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Subject to compliance with project requirements, manufacturers offering Products which may be incorporated in the Work include the following:
   1. Allied Tube & Conduit, Harvey, IL (800) 882-5543.
   2. Anchor Fence Division, Master-Halco, Incorporated, Baltimore, MD (800) 229-5615.
   3. Merchant's Metals, Houston, TX (800) 254-0080.
   4. The Tymetal Corporation, Fort Miller, NY (518) 695-9000.
   5. HySecurity, Kent, Washington, (800) 321-9947.

B. Product Requirements: Product options and substitutions. Substitutions: Permitted.

2.02 MATERIALS

A. Conform to CLFMI Product Manual.
B. Steel Framing:
   1. Type I: ASTM F 1083 Schedule 40, standard weight galvanized steel pipe, welded construction, minimum yield strength of 25 ksi; coating conforming to ASTM F 1043 Group IA on pipe exterior and interior.
   2. Type II: ASTM F 1043, cold-formed and welded galvanized steel pipe with minimum yield strength of 50 ksi; coating conforming to ASTM F 1043 Group IC on pipe exterior and Group ID on pipe interior.
   3. Type III (Roll-formed "C" sections): ASTM F 1043, cold-formed galvanized steel post cold-formed and welded galvanized steel pipe with minimum yield strength of 45 ksi ; coating conforming to ASTM F 1043 on post exterior and interior.

C. Fabric: ASTM A 392; Class 2: 2 ounce zinc 9 gage (0.148 inch diameter) galvanized steel wire, 2 inch diamond mesh interwoven wire, top and bottom selvages knuckled or knuckled and twist.
D. Fabric: ASTM A 668-2b; 9 gage (0.148 inch diameter) polyolefin ASTM F668-2b coated galvanized steel wire, 2 inch diamond mesh interwoven wire, top and bottom selvages knuckled or knuckled and twist.

2.03 MI6 ES

A. Footing Concrete: 3,000 psi Portland cement concrete.
B. Grout: Premixed, factory-packaged, non-staining, non-corrosive grout. Provide type formulated for exterior application.

2.04 COMPONENTS

A. End, Corner, and Pull Posts: Minimum sizes and weights as follows:

1. Up to 6 Foot Fabric Height:
    a. Type I Posts: 2.375 inch outside diameter pipe, 3.65 pounds per lineal foot.
    b. Type II Posts: 2.375 inch outside diameter pipe, 3.12 pounds per lineal foot.
2. Over 6 Foot to 13 Foot Fabric Height:
    a. Type I Posts: 2.875 inch outside diameter pipe, 5.79 pounds per lineal foot.
    b. Type II Posts: 2.875 inch outside diameter pipe, 4.64 pounds per lineal foot.
3. 13 Foot and Over Fabric Height (If required):
    a. Type I Posts: Round; 4.0 inch outside diameter pipe, 9.10 pounds per lineal foot.
    b. Type II Posts: 4.0 inch outside diameter pipe, 6.56 pounds per lineal foot.

B. Line (Intermediate) Posts: Minimum sizes and weights as follows:
1. Up to 6 Foot Fabric Height:
    a. Type I Posts: Round; 1.90 inch outside diameter pipe, 2.72 pounds per lineal foot.
    b. Type II Posts: 1.90 inch outside diameter pipe, 2.28 pounds per lineal foot.
    c. Type III Posts: 1.875 inch x 1.625 inch, 2.28 pounds per linear foot.

C. Swinging Gate Posts: For leaf widths, as follows:
1. Up to 4 Feet Width:
    a. Type I Posts: 2.875 inch outside diameter pipe, 5.79 pounds per lineal foot.
    b. Type II Posts: 2.875 inch outside diameter pipe, 4.64 pounds per lineal foot.
2. Between 4 Feet and 10 Feet Width:
    a. Type I Posts: Round; 4.00 inch outside diameter pipe, 9.10 pounds per lineal foot.
    b. Type II Posts: 4.00 inch outside diameter pipe, 6.56 pounds per lineal foot.
3. Between 10 Feet and 15 Feet Width:
    a. Type I Posts: 6.625 inch outside diameter pipe, 8.97 pounds per lineal foot.

D. Sliding Gate Posts:
1. All leaf widths:
    a. Type I Posts: Round; 4.00 inch outside diameter pipe, 9.10 pounds per lineal foot.
    b. Type II Posts: 4.00 inch outside diameter pipe, 6.56 pounds per lineal foot.

E. Bottom Rail and Intermediate Rails: Manufacturer's longest lengths.
1. Typical:
    a. Type I: Round; 1.66 inch outside diameter pipe, 2.27pounds per lineal foot.
    b. Type II: 1.66 inch outside diameter pipe 1.83 pounds per lineal foot.
    c. Type III Posts: 1.625 inch x 1.25 inch, 1.37 pounds per linear foot.
2. Couplings: Expansion type, approximately 6 inches long.
3. Attaching Devices: Means of attaching bottom rail securely to each gate, corner, pull, and end post.

F. Swinging Gate Hardware:
1. Hinges: Size and material to suit gate size; offset to permit 180 degree gate opening. Provide 1-1/2 pair of hinges for each leaf over 6 foot 0 inch nominal height.
2. Latch: Forked type or plunger-bar type to permit operation from both sides of gate, with padlock eye.
3. Double Gate Hardware: In addition to the above, provide gate stops for double gates, consisting of mushroom type flush plate with anchors set in concrete to engage center drop rod or plunger bar. Configure for use of one padlock to lock both gate leaves.

G. Sliding Gate Hardware:
1. Provide manufacturer's standard heavy duty track, ball-bearing hanger sheaves, overhead framing and supports, guides, stays, bracing, and accessories required. Tymetal, Fortress type gate is basis of design. Gates may not exceed 30 feet in width.
2. For 10 feet to 30 feet opening:
    a. Frame shall be fabricated from 6063-T6 aluminum alloy extrusion. The top member shall be 3 inch x 5 inch aluminum structural channel/tube extrusion weighing not less

than 3.9 lbs/lf. The top member shall be “keyed” to interlock with the “keyed” track member. The bottom member shall be a single horizontal aluminum structural tube weighing not less than 2.0 lbs/lf or a spliced 2 inch x 5 inch aluminum structural channel weighing not less than 2.65 lbs/lf. The two horizontal sections may be spliced in the field.

b. SPLICING: A 1/4 inch x 5 inch x 24 inch galvanized steel splice plate shall be used to secure the two 5 inch channel bottom members together utilizing eight 3/8 inch x 1 1/2 inch plated carriage bolts with lock nuts. The top members shall be spliced together on the side opposite the track member using a 1/4 inch x 2 inch x 24 inch aluminum splice plate secured with six 1/4 inch x 1/2 inch drive rivets on one side and welded to the top member on the other side. On the track side, the track is to be overlapped 24 inch onto the opposing section, interlocked with the top member and vertically secured in place using six 1/4 inch x 1/2 inch drive rivets and horizontally secured in place using six 5/16 inch x 1 inch plated hex head cap screws. The respective splice end vertical member shall be 1 inch x 2 inch, weighing not less than 0.82 lbs/lf. The 1 inch x 2 inch members will be joined utilizing 5/16 inch x 2 3/4 inch plated hex head cap screws, quantity varying by height of gate.

c. The vertical members shall alternate between 2 inch x 2 inch and 1 inch x 2 inch in cross section weighing not less 1.1 lbs/lf and 0.82 lbs/lf respectively. The spacing for the vertical intermediates shall be no greater than half the height of the gate.

d. The gate frame shall have a separate semi-enclosed “keyed” track, extruded from 6105-T5 aluminum alloy, weighing not less than 2.9 lbs/lf. Track member to be located on only one side of the top member. When interlocked with the “keyed” top member and welded to it, it forms a composite structure with the top of the gate frame. Welds to be placed alternately along the top and side of the track at 9 inch centers and a minimum of 2 inches long.

e. The gate frame is to be supported from the track by two swivel type, self aligning, 4-wheeled, sealed lubricant, ball-bearing truck assemblies.

f. Diagonal “X” bracing of 3/16 inch minimum diameter stainless steel aircraft cable shall be installed to brace the gate panels and to provide a ready means of vertical alignment.

2.05 ACCESSORIES

A. Sleeves: Galvanized steel pipe with inside diameter not less than 1/2 inch greater than outside diameter of fence posts. Provide steel plate closure welded to bottom of sleeves of width and length not less than 1 inch greater than outside diameter of sleeve.
   1. Up to 6 Foot Fabric Height: Provide sleeve not less than 12 inches long.

B. Tension Wire: 7 gage steel, metallic-coated coil spring wire, in accordance with ASTM A 824, located at the top of fence fabric.

C. Wire Ties: 11 gage galvanized steel.

D. Post Brace Assembly: Manufacturer's standard adjustable brace at end and gate posts and at both sides of corner and pull posts, with horizontal brace located at mid-height of fabric. Use same materials as top rail for brace, and truss to line posts with 0.375 inch diameter rod and adjustable tightener.

E. Post Tops: Galvanized steel, weather tight closure cap for tubular posts, one cap for each post. Furnish cap with openings to permit passage of top rail.

F. Stretcher Bars: Galvanized steel, one piece lengths equal to full height of fabric; with minimum cross section of 3/16 inch x 3/4 inch. Provide one stretcher bar for each gate and end post, one for each bottom rail, and two for each corner and pull post.

G. Stretcher Bar Bands: Manufacturer's standard.

H. Gate Cross-Bracing: 3/8 inch diameter galvanized steel adjustable length truss rods.

2.0X FABRICATION

A. Fabricate swing gate perimeter frames of 1.90 inch outside diameter galvanized steel pipe. Provide horizontal and vertical members to ensure proper gate operation and for attachment of fabric, hardware, and accessories. Space frame members maximum 8 feet apart.
B. Assemble gate frames rigidly by welding or with special fittings and rivets. Use same fabric as for fence. Install fabric with stretcher bars at vertical edges. Bars may also be used at top and bottom edges. Attach stretchers to frame at not more than 15 inches on center. Install diagonal cross-bracing on gates as required to ensure frame rigidity without sag or twist.
C. Attach hardware to provide security against removal or breakage.

2.07 FINISHES

A. All fence posts, fabric, and accessories shall be galvanized.

2.08 GATE OPERATORS

A. Gate operators must conform to UL325, Standards for Safety. The operator must be tested by an independent testing laboratory such as UL or ETL and found to conform to these standards. The completed installation shall conform to applicable ASTM and UL requirements.
B. Basis of Design: HySecurity Slide Driver Unit.
C. All electrical work is to be done by qualified electricians and is to conform to all applicable local, state, and federal codes.
D. General Operation
   1. The operator must be designed for high-cycle applications and low maintenance. The operator shall be capable of actuating gates up to 30 feet in overall length. The gate operator must be able to operate gates up to 150 per cent of weight of actual gate at 2.2feet (66 cm) per second.
   2. All fasteners, except structural bolts, are to be stainless steel, or other non-corrosive material.
   3. The operator is to provide wear compensating, spring-loaded, friction-feed type drive mechanism. The drive mechanism is to consist of two drive wheels that can be manually disconnected by a toggle style disconnect. This disconnect is to instantly disengage the drive wheels for manual operation. The operator, upon returning to automatic operation by engaging the drive mechanism, shall function properly without regard to the gate's actual position.
E. Housing Construction
   1. The housing cover must swing open to allow access to the internal components.
   2. The housing cover must be lockable.
   3. All operator cover locks are to be keyed alike.
   4. The housing, chassis and cover to be galvanized for corrosion resistance per ASTM 123 M.
F. Electric Motor
   1. The electric motor used in the gate operators must have a continuous-duty rating of two horsepower with a service factor of 1.15 or greater and shall be available in all voltages and phases to suit the installation requirements of the site.
   2. The electric motors must have built-in overload protection and resettable with a sealed pushbutton reset.
G. Hydraulic System

1. The hydraulic system must be self-contained and contain pump, reservoir, two position control valve, hydraulic hoses, fittings and hydraulic motors.
2. All hydraulic hoses shall have a minimum burst pressure of 12,000 pounds per square inches.
3. The hydraulic motors must be automatically locked when the control valve is in the de-energized to prevent slippage of the drive wheels.
4. The hydraulic system must be soft-start and soft-stop to minimize shock loads transmitted to the gate system including a reverse delay to maximize gate hardware life.

H. Electrical
1. Built in "warn before operate" system.
2. Anti-tailgate mode.
3. 26 programmable user relay output options.
4. Built-in power surge/lightning strike protection.
5. Control circuit: 24VDC.
6. Electrical enclosure: Oversized, metal, with hinged lid gasketed for protection from intrusion of foreign objects, and providing ample space for the addition of accessories.
7. Menu configuration, event logging and system diagnostics easily accessible with integral touchpad or a PC and free START software.
8. Limit switch shall feature a built in LED "tripped" indicator light. The limit switch must readily accessible, adjustable and replaceable with normal hand tools.
9. The limit switches are to provide the ability to remote monitor the gate position when in the fully closed and fully open positions.
10. Provide individual surge protection at both ends of all power and low voltage controls conductors serving the gate operator(s). Refer to Section 281304.

I. Accessories
1. Through beam type photo eyes.
2. Heater with thermostat for cold damp climates.
3. Snow brushes and blades for cold snowy climates
4. Strobe or other similar visual beacon to operate simultaneously with standard gate operator "warn before operate" audible beacon.

J. Inductive vehicle loop detectors
1. Inside and outside obstruction loops are to be installed to prevent the gate from closing when vehicle traffic is present. Anti-Tailgating logic is to be applied to entrance lane gates.
2. Free exit loops are to be installed for exit lane gates.
3. Loops for gates with heavy truck traffic will have no side of the loop less than 6'.
4. Loop wire to be stranded Thhn or XLPE, crosslink poly-ethelene jacketed type acceptable for direct burial.
5. Refer to detail drawings for specific loop placement or refer to manufacturers recommendations.

PART 3 - E6 ECUTION

3.01 E6 AMINATION

A. Execution Requirements: Verification of existing conditions before starting work.
B. Verification of Conditions: Verify that field measurements, surfaces, substrates and conditions are as required, and ready to receive Work.
C. Report in writing to Contracting Officer prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.

D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the United States Postal Service.

3.02 INSTALLATION

A. Install fence in accordance with ASTM F 567 and manufacturer's published instructions.
B. Install gates in accordance with ASTM F 900, ASTM F2200 or ASTM 1184 as applicable and to manufacturer's published instructions.
C. Space line posts 10 feet 0 inches on center maximum, unless otherwise indicated on Drawings.
D. Grade-set Posts:
   1. Drill or hand excavate.
   2. Excavate each post hole to 12 inch diameter, or not less than four times diameter of post.
   3. Excavate approximately 3 inches lower than post bottom; set post bottom not less than 36 inches below finish grade.
   4. Hold post in position while placing, consolidating, and finishing concrete.
E. Sleeve-set Posts: Anchor posts in concrete by means of pipe sleeves preset and anchored into concrete. After posts have been inserted into sleeves, fill annular space between post and sleeve solid with grout, mixed and placed to manufacturer's recommendations.
F. Rails: Run rail between post, bending smoothly for curved runs located at the bottom of the fence fabric. Provide expansion couplings as recommended by fencing manufacturer.
G. Center Rails: Provide center rails where indicated. Install in one piece between posts and flush with post on fabric side, using offset fittings where necessary.
H. Brace Assemblies: Install braces so posts are plumb with rod in tension.
I. Tension Wire: Install tension wires through post cap loops before stretching fabric and tie to each post cap with not less than 6 gage galvanized wire. Fasten fabric to tension wire using 11 gage galvanized steel hog rings spaces 24 inches on center.
J. Fabric: The fence fabric must be installed within 2 inches between finish grade and bottom selvage. Pull fabric taut and tie to posts, rails, and tension wires. Install fabric on the exterior side of the fence, and anchor to framework so fabric remains in tension after pulling force is released.
K. Stretcher Bars: To secure end, corner, pull, and gate posts, thread through or clamp to fabric 4 inches on center and secure to posts with metal bands spaced 15 inches on center.
L. Tie Wires:
   1. Use U-shaped wire conforming with diameter of pipe to which attached, clasping pipe and fabric firmly with ends twisted two full turns. Bend wire ends to minimize hazards to persons or clothing.
   2. Tie fabric to line posts with wire ties spaced 12 inches on center. Tie fabric to rails and braces with wire ties spaced 24 inches on center. Manufacturer's standard procedure will be accepted if of equal strength and durability.
M. Fasteners: Install nuts for tension bands and hardware bolts on side of fence opposite fabric side. Peen ends of bolts or score threads to prevent removal of nuts.
N. Gates: Install gates plumb, level, and secure for full opening without interference. Install ground-set items in concrete for anchorage. Adjust hardware for smooth operation.

3.03 CONSTRUCTION

A. Site Tolerances:
   1. Maximum Variation from Plumb: 1/4 inch.
   2. Maximum Offset from True Position: 1 inch.

3. Locate fencing components completely within site boundaries. Do not infringe adjacent property lines.
4. Maximum Fence Distance from Ground: 1 1/2 inches.
5. Maximum Gate Distance from Ground: 4 inches.

B. Gate Operator:
1. Coordinate conduit runs and electrical connections with Access Control Section 281304, Common Work Results for Communications Section 270500, and Communications Horizontal Cabling Section 271500.

3.04 FIELD QUALITY CONTROL

A. Test gate operator through ten full cycles and adjust for operation without binding, scraping or uneven motion. Test limit switches for proper "at rest" gate position.
B. All anchor bolts shall be fully concealed in the finished installation. Owner, or owner's representative, shall complete "punch list" with installing contractor prior to final acceptance of the installation and submit completed warranty documentation to manufacturers where applicable.

3.05 CONTINUED SERVICE AND DOCUMENTATION

A. Train owner's personnel on how to safely shut off electrical power, release and manually operate the gates. Additionally, demonstrate the general maintenance of the gate operator and accessories and provide one copy of "Installation and Reference" manual for the owner's use (a second manual is available upon request). Manuals will identify parts of the equipment for future procurement. Direct maintenance personnel to HySecurity's website, specifically the technical support sections.

END OF SECTION 32 31 13

SECTION 32 71 13 - PRECAST CONCRETE JERSEY BARRIERS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:

1. Precast concrete Jersey barriers with chain link fence attachment, and installation, as shown on the drawings, and as herein specified.

B. Related Documents: The Contract Documents, as defined in Division 1 Section - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other Documents.

C. Related Sections include the following:

1. Division 32 Section: Chain Link Fences

1.03 SUBMITTALS

A. Product Data: Submit manufacturer's technical data and installation instructions for concrete crash barriers, options and accessories.

B. Shop Drawings: Provide system details, plan layout and spacing of components including posts and panels.

1.04 QUALITY ASSURANCE

A. Provide concrete barrier as manufactured by a single fabricator, including necessary accessories.

B. Installation by company specializing in commercial/institutional quality fencing with five years experience.

PART 2 - PRODUCTS

2.01 GENERAL

A. All construction to conform to local building department codes.
B. The constractor shall verify all conditions at the job site prior to beginning the work.

2.02 MATERIALS

A. Precast Concrete Crash Barrier: Provide 36" tall by 12' long concrete crash barrier with chain link fence attachment option, as manufactured by Waskey Custom Precast Concrete Products, Baton Rouge, LA (225) 926-8752, Dura-Crete (801) 972-8686, or equal as approved by Architect. Provide galvanized concrete pins to interlock barrier sections and anchor as required. Paint safety yellow. Add reflective material if required by facility.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Securing of barriers: Drill holes for securing of barriers to pavement.
B. Fencing: Install fencing so posts are plumb when attached to barriers. Install posts at 6' intervals.

END OF SECTION 34 71 13

SECTION 32 71 13 - PRECAST CONCRETE JERSEY BARRIERS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

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   1. Division 32 Section: Chain Link Fences

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PART 3 - EXECUTION

3.01 INSTALLATION

A. Securing of barriers: Drill holes for securing of barriers to pavement.
B. Fencing: Install fencing so posts are plumb when attached to barriers. Install posts at 6' intervals.

END OF SECTION 34 71 13

SECTION 32 92 00 - TURF AND GRASSES

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:
   1. Seed.
   2. Sod.
   3. Sprigs.
   4. Mulches.
   5. Asphalt Adhesive.
   6. Water.
   7. Erosion Control Material.

B. Related Documents: The Contract Documents, as defined in Section 011000 - Summary of Work, apply to the Work of this Section. Additional requirements and information necessary to complete the Work of this Section may be found in other documents.

C. Related Sections:
   1. Section 311000 - Site Clearing: Mulch from recycled site debris.
   2. Section 312000 - Earth Moving: Topsoil material.
   3. Section 313200 - Soil Stabilization: Stabilization materials and procedures.
   4. Section 312500 - Erosion and Sedimentation Controls: Slope and erosion protection materials.
   5. Section 329200 - Plants: Planting materials.
   6. Section 092900 - Gypsum Board: Soil amendment from recycled scrap gypsum.

1.02 REFERENCES

A. American Society For Testing and Materials (ASTM):
   1. ASTM C 602 - Specification for Agricultural Liming Materials.
   2. ASTM D 977 - Specification for Emulsified Asphalt.

B. American Sod Producers Association (ASPA):
   1. ASPA STSMT - Specification for Turfgrass Sod Materials and Transplanting/Installing.

1.03 SUBMITTALS

A. Section 013300 - Submittal Procedures: Procedures for submittals.
   1. Assurance/Control Submittals:
      a. Certificates:
         1) Submit certificate from seed supplier for each grass-seed mixture stating the botanical and common name and percentage by weight of each species and variety, and percentage of purity, germination, and weed seed. Include the year of production and date of packaging.
         2) Submit certificate from sod supplier for each seed mixture, identifying sod source, including name and telephone number of supplier.

B. Section 017704 - Closeout Procedures and Training: Procedures for closeout submittals.
   1. Operation and Maintenance Data: Include maintenance instructions, cutting method and maximum grass height, types of application frequency, and recommended coverage of fertilizer for one full growing cycle.

1.04 QUALITY ASSURANCE

A. Regulatory Requirements: Conform to applicable requirements of the Local and State Department of Agriculture Extension Service of the state in which the project is located.

1.05 ENVIRONMENTAL REQUIREMENTS

A. Resource Management:
   1. Renewable Resources: Plants specified are indigenous, low maintenance varieties, tolerant of site's existing soils and climate without supplemental irrigation or fertilization once established.
      a. Soil amendments: No chemical fertilizers; use organic/natural matter to support establishment of indigenous plants; use inorganic materials such as sand or gypsum to improve workability and drainage of soil as appropriate to indigenous plants.
      b. Mulch: Provide organic mulch products.
   2. Recycled Content:
      a. Wood fiber mulch: Provide products manufactured from 100 % post-consumer paper content and yard trimming composts.
      b. Mulch from recycled site debris: Coordinate with Section 311000 - Site Clearing to identify and prepare suitable organic debris for use as mulch on site.
      c. Soil amendment from recycled scrap gypsum: Coordinate with Section 092900 - Gypsum Board to prepare scrap gypsum board for use as soil amendment.

PART 2 - PRODUCTS

2.01 SEED

A. Classification:
   1. State Approved of latest season's crop delivered in original sealed packages bearing producer's guaranteed analysis for percentages of mixtures, purity, germination, weedseed content, and inert material.
   2. Label in conformance with applicable state seed laws.
   3. Wet, moldy, or damaged seed will be rejected.

2.02 SOD

A. Classification:
   1. Certified as classified in ASPA STSMT.
   2. Machine cut sod at a uniform thickness of 3/4 inch with a tolerance of 1/4 inch, excluding top growth and thatch. Each individual sod piece capable of supporting its own weight when lifted by ends.
   3. Broken pads, irregularly shaped pieces, torn or uneven ends will be rejected.
   4. Wood pegs and wire staples for anchorage as recommended by sod supplier.

2.03 SPRIGS

A. Healthy living stems, stolons, or rhizomes and attached roots of locally adapted grass without adhering soil, including two to three nodes, from 4 to 6 inches long, obtained from heavy and dense sod.
   1. Provide sprigs which have been grown under climatic conditions similar to those in locality of Project Site.
   2. Coordinate harvesting and planting operations to prevent exposure of sprigs to sun for more than 30 minutes before covering and moistening.
   3. Sprigs containing weeds or other detrimental material or that are heat damaged will be rejected.

2.04 MULCHES

A. Provide mulch free from noxious weeds, mold, and other deleterious materials.
B. Straw: Stalks from oats, wheat, rye, barley, or rice. Air-dry condition of proper consistency for placing with commercial mulch blowing equipment.
C. Hay: Use only marsh hay for lawn areas. Air-dry condition of proper consistency for placing with commercial mulch blowing equipment.
D. Wood Cellulose Fiber:
   1. Processed to contain no growth or germination-inhibiting factors, dyed with non toxic, biodegradable dye to an appropriate color to facilitate visual metering of materials application.
   2. Composition on air-dry weight basis: 9 to 15 percent moisture, pH range from 3.5 to 5.0
   3. Use with hydraulic application of grass seed and fertilizer.
   4. Provide organic mulch products manufactured from 100 percent post-consumer paper content and yard trimming composts.
   5. Manufacturers:
      a. National Fiber, Belcher, MA, (800) 282-7711 or (413) 283-8747.
      b. Wood Recycling Inc., Woburn, MA, (800) 982-8732 or (617) 937-0855.
      c. Section 016000 - Product Requirements: Product options and substitutions. Substitutions: Permitted.

2.05 TABILIZING MATERIALS

A. Specified in Section 313200.
B. Asphalt Adhesive: ASTM D 977, Grade RS-1. Use with straw or hay mulch
C. Cellulose Fiber: Use for anchoring straw. Fiber binding shall be applied at a net dry weight of 750 pounds per acre. Cellulose fiber may be mixed with water. Mixture shall contain maximum of 50 pounds of cellulose fiber per 100 gallons of water.
D. Mulch Netting: Stake light weight plastic netting over the mulch according to manufacturer's recommendations. Stakes shall be driven to ground level.

2.06 WATER

A. Suitable quality for irrigation.

2.07 EROSION CONTROL MATERIAL

A. Specified in Section 312500.
B. Net: Heavy, twisted jute mesh, plastic mesh, biodegradable paper fabric with knitted yarns, or standard weave burlap.
C. Blanket: Fiber

2.08 TOPSOIL

A. Topsoil:
   1. Containing organic matter as needed to support establishment of plants. Maximum particle size, 3/4 inch, with maximum 3 percent retained on 1/4 inch screen.
   2. Component Percentages:

a. Silt:	25 to 50
b. Clay:	10 to 30
c. Sand:	20 to 30
d. pH:	5.5 to 7.0
e. Soluble Salts:	600 ppm maximum

f. pH: 5.5 to 7.0.

2.09 PH ADJUSTERS

A. Lime:
   1. Material: ASTM C 602, Class T, agricultural commercial grade ground limestone containing not less than 50 percent of total oxides.
   2. Gradation: Minimum 75 percent passing 100 mesh sieve and 100 percent passing 20 mesh sieve.

B. Ferrous Sulfate: Commercial Grade.

2.10 FERTILIZER

A. Bonemeal: Commercial, raw, finely ground; minimum 4 percent nitrogen and 20 percent phosphoric acid.
B. Superphosphate: Commercial-Grade complete fertilizer of neutral character consisting of fast-and-slow-release nitrogen, 50 percent derived from natural organic sources of urea-form, phosphorous, and potassium in following composition:
   1. Composition: 1 pound per 1000 square feet of actual nitrogen, 4 percent phosphorous, and 2 percent potassium, by weight.
   2. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in soil reports from a qualified soil-testing agency.

C. Slow-Release Fertilizer: Granular fertilizer consisting of 50 percent water-insoluble nitrogen, phosphorous, and potassium in the following composition:
   1. Composition: 20 percent nitrogen, 10 percent phosphorous, and 10 percent potassium, by weight.
   2. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in soil reports from qualified soil-testing agency.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Section 017300 - Execution: Verification of existing conditions before starting work.
B. Verification of Conditions: Verify that field measurements, surfaces, and conditions are as required, and ready to receive Work.
C. Report in writing to Contracting Officer prevailing conditions that will adversely affect satisfactory execution of the Work of this Section. Do not proceed with Work until unsatisfactory conditions have been corrected.
D. By beginning Work, Contractor accepts conditions and assumes responsibility for correcting unsuitable conditions encountered at no additional cost to the United States Postal Service.

3.02 PREPARATION OF SUBSOIL

A. Prepare subsoil to eliminate uneven areas and low spots. Maintain lines, levels, profiles, and contours. Make changes in grade gradual. Blend slopes into level areas.
B. Remove foreign materials, weeds, and undesirable plants and their roots. Remove contaminated subsoil.
C. Scarify subsoil to a depth of 3 inches where topsoil is to be placed. Repeat cultivation in areas where equipment, used for hauling and spreading topsoil, has compacted subsoil.
D. Place topsoil as specified in Section 312000.

3.03 FERTILIZING

A. Apply fertilizer in accordance with manufacturer's published instructions.
B. Apply after smooth after smooth raking of topsoil and prior to roller compaction.
C. Do not apply fertilizer at same time or with same machine as will be used to apply seed.
D. Mix thoroughly into upper 2 inches of topsoil.
E. Lightly water to aid the dissipation of fertilizer.
F. No chemical fertilizers.

3.04 SEEDING

A. Sow one-half of seed in one direction and remainder at right angles to first sowing.
B. Cover seed to average depth of 1/2 inch by means of spike-tooth harrow, cultipacker, or other recommended device.
C. Drill Seeding:
   1. Use cultipacker seeders.
   2. Drill seed uniformly to an average depth of ½ inch and at a rate of 8 pounds per 1,000 square feet.
D. Hydroseeding:
   1. Mix seed, fertilizer, and wood cellulose fiber in required amount of water to product a homogeneous slurry. Add wood cellulose fiber after seed, water, and fertilizer have been thoroughly mixed and apply at the rate of 200 pounds per acre dry weight.
   2. Hydraulically spray material on ground to form a blotter-like cover impregnated uniformly with grass seed.
   3. Immediately following application of slurry mix, make separate application of wood cellulose mulch at the rate of 1,000 pounds, dry weight, per acre.
   4. Apply cover so that rainfall or applied water will percolate to underlying soil.
E. Mulch:
   1. Spread evenly at rate of 4 tons per acre.
   2. Anchor by crimping mulch with serrated disc, or by spraying asphalt emulsion on mulched surface.
   3. Take precautionary measures to prevent asphalt materials from marking of defacing structures, pavements, utilities, or plantings.
F. Rolling:
   1. Immediately after seeding, firm entire area except for slopes in excess of 3 to 1 with a roller not exceeding 90 pounds for each foot of roller width.
   2. If seeding is performed with cultipacker-type seeder or hydroseeding, rolling may be eliminated.
G. Erosion Control Material: Install in accordance with manufacturer's instructions.

3.05 SODDING

A. Placing:
   1. Place a maximum of 20 hours after initial harvesting, in accordance with ASPA GSS as modified herein.
   2. Thoroughly moisten areas to be sodded immediately prior to placing.
B. Spot Sodding:
   1. Cut sod into plugs 2 inches square or 2 inches in diameter. Place individual pieces on 4 inch centers and press firmly into soil by foot pressure or by tamping.
   2. Overseed for erosion control on all spot sodded areas.
   3. Place seed, as specified above, at the rate of 1 pounds per 1,000 square feet.

C. Slopes and Ditches:
   1. For slopes 2:1 and greater, lay with long edge parallel to slope.
   2. V-ditches and flat bottomed ditches, lay with long edge perpendicular to flow of water.
   3. Anchor each piece of sod with wood pegs or wire staples maximum 2 feet on center.
   4. On slope areas, start sodding at bottom of slope.

D. Finishing: After completing sodding, blend edges of sodded area smoothly into surrounding area.

E. Watering: Start immediately after completing each day's sodding. Apply at a rate sufficient to ensure thorough wetting of soil to minimum depth of 4 inches.

3.06 CLEANING AND PROTECTION

A. Remove soil and debris created by lawn work from paved areas. Clean wheels of vehicles before leaving site to avoid tracking soil onto surface of roads, walks, or other paved areas.

B. Immediately after seeding, sodding or sprigging, protect the area against traffic or other use.

C. Restore existing lawn and grass areas which have been damaged during execution of this work to original condition.

D. Keep one paved pedestrian access route and one paved vehicular access route to each building clean at all time. Clean other paving when work in adjacent areas is complete.

3.07 ESTABLISHMENT PERIOD

A. Definitions:
   1. Lawns and grasses establishment period will be in effect until lawns and grasses have been mowed 3 times.
   2. Stand of lawn and grass is 95 percent ground cover of established species.

B. Maintenance During Establishment Period:
   1. Mow lawns and grassed areas to an average height of 2 inches whenever average height of grass becomes 4 inches.
   2. Promotion of growth: Mow, remove excess clippings, eradicate weeds, water, fertilize, overseed, and perform other operations necessary to promote growth.
   3. Post-fertilize areas with commercial grade 7 days after planting and at intervals of 4 weeks thereafter until accepted. Apply fertilizer uniformly at the rate of 1 pounds per 1,000 square feet.

3.08 FINAL INSPECTION AND ACCEPTANCE

A. Final Inspection and Acceptance:
   1. Final inspection will be made upon written request from the Contractor at least 10 days prior to last day of lawn and grasses establishment period.
   2. Final acceptance will be based upon a satisfactory stand of lawns and grasses as defined in the paragraph entitled, "Establishment Period”

B. Replanting: Replant areas which do not have a satisfactory stand of lawns and grasses.

C. Contractor is to maintain lawns and grasses for one year from completion.

END OF SECTION 32 92 00